
High-Performance Thermoplastic Composite Electromagnetic Shielding for Avionics

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ABSTRACT

As the aviation industry moves towards lighter and more sustainable aircraft, carbon fibre reinforced thermoplastic composites are increasingly being considered as replacements for conventional aluminium structures. However, as these materials take on a greater structural role, their ability to shield critical avionics from electromagnetic interference becomes an important and largely unresolved question. This thesis investigates the electromagnetic shielding performance of two thermoplastic composite systems across a frequency range directly relevant to 5G networks and modern radar systems.

In this study, two types of materials are explored, carbon fibre polyphenylene sulphide (CF/PPS) and carbon fibre polyamide-12 (CF/PA12). Five laminate configurations were manufactured, inspected, and tested across a broad range of properties. The results show that CF/PPS laminates were more difficult to manufacture cleanly, exhibiting higher levels of internal porosity, while CF/PA12 consolidated more reliably under the same conditions.

Despite this, both material systems demonstrated strong mechanical performance and competitive shielding effectiveness, with the best-performing configurations achieving shielding levels comparable to existing carbon fibre epoxy systems reported in literature. The study finds that how the fibres are oriented within the laminate has a far greater influence on shielding performance than the choice of thermoplastic matrix itself. Ultimately, this thesis demonstrates that thermoplastic composites are credible candidates for electromagnetic shielding in aviation, and that with appropriate laminate design and manufacturing control, they can meet the demands of next-generation aircraft.

1.0 Introduction

Electromagnetic compatibility (EMC) is not only a core consideration, but a necessity in modern aviation. In Europe, aircraft are now required to comply with Regulation (EU) 2018/1139, which requires manufacturers to demonstrate that critical systems remain tolerant to electromagnetic disturbances throughout the aircraft's operational envelope (1). This shift towards the application-specific EMC requirements for EMC reflects the hidden threat to critical avionics onboard aircraft as electrification continues to drive today's highly congested radio-frequency environment that is becoming increasingly complex and dense.

To understand the extent of this problem, it's important to recognise that EMC refers to the capability for devices to operate in an electromagnetic environment within a degree of safety, whilst avoiding disruption of other devices by electromagnetic interference (EMI) (2). EMC is concerned with not just the *immunity* against EMI from surrounding devices, but also the *suppression* of operationally disruptive EMI emissions from the device itself, both of which can be propagated by *radiation* and *conduction* coupling mechanisms (2). The critical solution to suppressing these mechanisms of propagation involves enclosing the source within a conductive enclosure with a combination of panels, meshes, and gaskets to form a 'final defense' against EMI, also known as *shielding*. This same principle is used in conventional microwave ovens that prevent high-power microwaves from escaping (3).

With most applications, the 'gold standard' to shield electronics has always been conventional metal alloys. As detailed in an comprehensive report by NASA in 1987, aluminium alloys are preferred for avionics due to their balance of high electrical conductivity, lower density, and affordability (4). However, as outlined in the Paris Agreement, the need to limit global warming to less than 1.5°C above pre-industry levels is imperative (5). Despite aviation contributing only two percent of global greenhouse gas (GHG) emissions, demand for air travel will continue to grow (6). As such, the aviation industry has moved away from aluminium alloys towards high-performance polymer composites to achieve lighter aircraft.

Polymer composites can be categorised as thermosets (TS) or thermoplastics (TP). Unlike TS composites that rely on permanent cross-linking between polymer chains, TP composites have proven to be more sustainable and offer several advantages over their TS counterparts that include longer shelf-life, easier re-processability, lower turnover time, improved fracture toughness, greater impact resistance and higher fatigue resistance (7). The shifting adoption for thermoplastic (TP) polymers can also be clearly seen in the European polymer composites market. The AVK Federation for Reinforced Plastics has highlighted that thermoplastic (TP) composites market share in Europe grew from 54 percent in 2015 to 58 percent in 2025, with a growing 62 percent contribution from the transportation industry (8).

Building on this momentum, the aerospace industry has increasingly gravitated towards carbon fibre reinforced thermoplastic polymer composites (CFRTPs). Commercial applications include the Airbus H160 main rotor hub seen in Figure 1(a) and the Rolls-Royce Trent XWB acoustic liner seen in Figure 1(b). The most significant catalyst however, has been the decade-long Airbus Multifunctional Fuselage Demonstrator (MFFD) project, seen in Figure 1(c). As key part of the European Clean Sky 2 initiative, MFFD saw extensive research into high-performance uni-directional (UD) CFRTP laminates, achieving a fuselage mass reduction of one tonne compared to aluminium and up to 67 percent increase in manufacturing rate (9). Ultimately, MFFD demonstrated technical and economic benefits for CFRTPs to be more widely adopted in future aircraft.

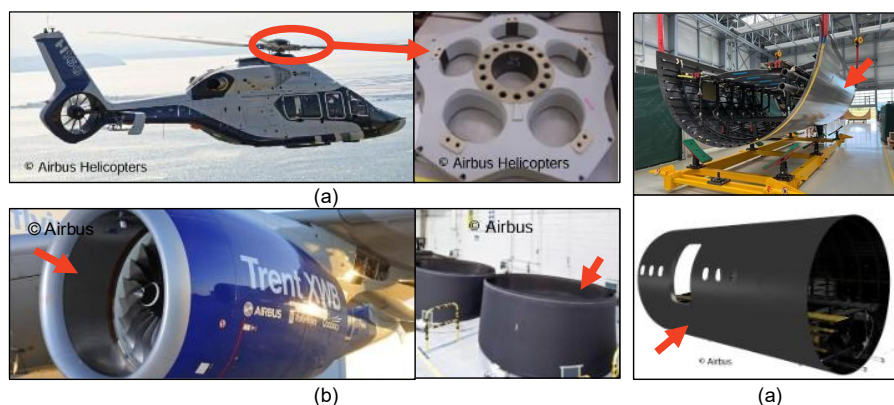


Figure 1. Commercial applications of CFRTPs. (a) Airbus H160, (b) Rolls-Royce Trent XWB, (c) Airbus MFFD; modified from (9,10).

2.0 Background and Literature Review

2.1 Fundamentals to Electromagnetic Shielding Effectiveness

The core principles of electromagnetic shielding can be traced to the early developments of electrostatics from the early 19th century. Followed by the rise of the theory of electromagnetism and electrodynamics, core principles from this period have since played an important role in the development of antennas, the design of shielding enclosures, and physics behind electric motors today. Amongst the many influential individuals, Michael Faraday's contributions to electromagnetism have been widely regarded to be most significant. Founding the law of induction in 1831, Faraday also introduced the idea of electromagnetic shielding in 1836 by constructing a hollow metallic cube out of a light wooden frame and covering it in metal foil for insulation. Later popularised as the Faraday cage, the structure demonstrated charged particles in the structure responding to an applied electrical field. The phenomena involved equal charge of opposite signs induced on neighbouring particles within the structure, effectively cancelling out the applied field (11).

Inspired by Faraday's seminal work, James Clerk Maxwell later presented a unified approach in his 1873 treatise (4,12). Unifying the key findings of Ampère, Gauss and Faraday, the Maxwell equations, as seen in Eq. (1-4), describe the interactions between electric and magnetic fields, as well as propagation behaviours, where ∇ represents the vector differential operator, $\mathbf{E}$ is the electric field vector, $\mathbf{B}$ is the magnetic field vector, ρ is the electric charge density, ϵ_0 is the permittivity of free space, μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ H/m), and $\mathbf{J}$ represents the electric current density vector.

$$(i) \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (\text{Gauss's Law for Electricity}) \quad (1)$$

$$(ii) \quad \nabla \cdot \mathbf{B} = 0 \quad (\text{Gauss's Law for Magnetism}) \quad (2)$$

$$(iii) \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{Faraday's Law of Electromagnetism}) \quad (3)$$

$$(iv) \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (\text{Ampère's Law with Maxwell's correction}) \quad (4)$$

Using Maxwell's equations, Sergei Alexander Schelkunoff later applied the equations to derive what later became known as the Schelkunoff's theory during his time at Bell Labs (13). Schelkunoff's derivation expressed total shielding effectiveness (SE_T) as the ratio of the incident EM field power (P_0) to transmitted power (P_t) and defined three distinct components: absorption (SE_A), reflection (SE_R), and multiple reflection (SE_{MR}). The final formulations are shown in Eq. (5-8), excluding the multiple reflection correction term which is generally neglected when SE_A is greater than 10 dB (13), although some articles report up to 15 dB.

$$SE_T = SE_A + SE_R + SE_{MR} \quad (5)$$

$$SE_T = -10 \log_{10} \left(\frac{P_t}{P_0} \right) = -10 \log_{10} (|S_{11}|^2) \quad (6)$$

$$SE_A = -10 \log_{10} \left(\frac{|S_{21}|^2}{1 - |S_{11}|^2} \right) \quad (7)$$

$$SE_R = -10 \log_{10} (1 - |S_{11}|^2) \quad (8)$$

Represented here are the input reflection coefficient, S_{11} and the forward transmission coefficient, S_{21} . Generally referred to as S-parameters, they can be obtained experimentally from two-port vector network analysers. However, while these formulations are relatively intuitive, reliance on experimentally derived values is not practical. To support this, an alternative formulation for homogeneous materials used by Rajan et al (14) can be written as Eq. (9) for highly conductive materials.

$$SE_T = \underbrace{-8.686t \left(\frac{\sigma \omega \mu_r}{2} \right)^{0.5}}_{SE_A} \underbrace{-10 \log_{10} \left(\frac{\sigma}{16\omega \epsilon_0 \mu_r} \right)}_{SE_R} \quad (9)$$

Where σ is electric bulk conductivity of the specimen, ϵ_0 is the absolute permittivity of air ($\epsilon_0 \approx 8.85 \times 10^{-12}$ F/m), ω here represents the angular frequency ($\omega = 2\pi f$), and μ_r is magnetic relative permeability which is approximately equal to 1 for non-ferromagnetic materials. This approach has been proven to be effective for highly conductive homogeneous polymer materials such as carbon-based filler polymer composites (15) and nonwoven laminated polymer composites (14). However, there remains a challenge with unidirectional laminates due to preferential conductivity pathways from anisotropy and multiple internal reflections (16).

2.1.1 Electromagnetic Shielding Spectrum

Throughout the development of avionics, avionics have played a major role in navigating the skies. In this process, a wide range of frequencies have since arised. As shown in Figure 2(a), there are three main sources of EMI which can impact the health of the avionics. They may arise through the form of *transients* from un-predictable sources like lightning, *radio frequencies* from radar, or power generation sources in the form of high-power electronics. These frequencies can be as low as 400 Hz and as high as 300 GHz, as such adequate understanding of the shielding performance of TP composite laminates across the whole spectrum is of high-importance.

As noted in a recent policy review by Forge and Vu (17), the spectrum of most importance of the future was highlighted to be in the mid-band (1-6 GHz) which is most preferred by modern network carriers due to its balance of data speed and latency (17). The 1-6 GHz frequency range lies across the L-band, S-band, and partially of the C-band, as seen in Figure 2(b). The two of most importance here is the S-band which is used for mainly weather surveillance in radar but is also the range for which technologies such as Bluetooth (2.40-2.48 GHz) (18). In addition, Europe is planning to deploy a significant range of its future 5G network within the S-band, specifically 1.4 GHz, 2.0 Ghz, and 3.4-3.8 GHz (17). Overall, as aircraft increasingly operate in environments saturated with 5G signals in the 1.4–3.8 GHz range, ensuring the smooth adoption of composite airframe structures provide adequate shielding for critical avionics within this band is critical for the safety of the next generation of aircraft.

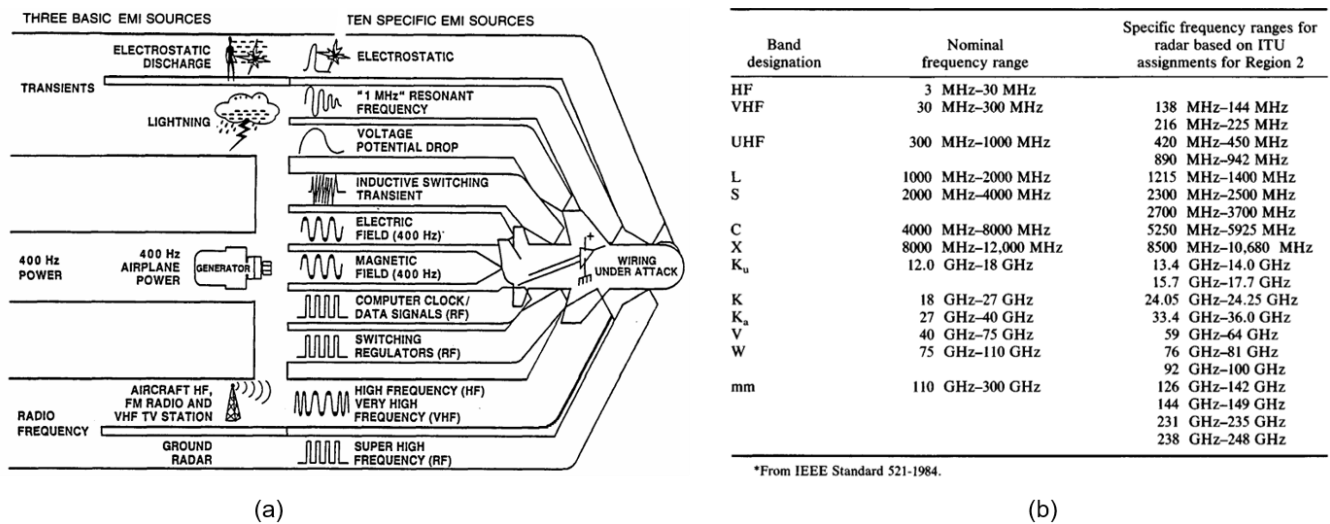


Figure 2. Historical shielding spectrums used in avionics and radar.

(a) EMI sources of risk to avionics (4); (b) microwave frequency bands by International Telecommunications Union (ITU) (18).

2.1.2 Electromagnetic Shielding Experimental Setups

There are numerous approaches in measuring the electromagnetic shielding effectiveness. Depending on the field of interest, different approaches may be adopted as appropriate. When measuring near field shielding effectiveness where field lines are more spherical, one approach is to measure the results using a dual transverse electromagnetic (DTEM) cell method. For far-field planar waves, these may be measured via *shielded enclosures*, *transmission line*, or *free space*, as shown in Table 1.

Shielded enclosures are widely used for comparing measurements between different shielding materials (19). Comprised of an outer metal box with two antennas situated on either side of the box, the laminate is placed in between the two antennas, as shown in Figure 3(a). Once clamped and sealed, the measurements from the receiving antenna on the inside are then used to derive the shielding effectiveness. As specified in IEEE 299-2006 standard, while the test standard has a wide range between 9 kHz to 18 GHz, it can be extended up to 100 GHz, reducing the need for multiple tests to obtain the full spectrum. However, coupled with seal issues between the specimen and the fixture, the large specimen sizes and significant test equipment ultimately make the process expensive to operate. There are some studies that have attempted this with thin metals or layered polymer composites (20–22), using specimen sizes up to 500 mm x 500 mm.

The *transmission line* method (TLM) is one of the most popular categories. The reason most studies have gravitated to TLM methods is because results are generally more comparable in between laboratories (19). For studies looking to use the coaxial TLM variant shown in Figure 3(b), there currently exists the ASTM D4935 method for testing these materials; similarly, for waveguide TLM, the test methodology also has the ASTM D5568 standard. An advantage of this method is the ability to SE_A and SE_R components using a vector network analyser (VNA). There is also significantly smaller specimens that can be achieved, with coaxial TLM methods achieving as low as 13 mm in diameter (23), although these are not in accordance to standards due to the different frequency and fixture. The downside to the TLM method is the relatively higher precision in cutting methods to achieve the right specimen geometries required due to smaller specimen sizes. In addition, due to the frequency cutoff, careful selection of the right fixture is necessary.

Lastly, *free space* method (FSM) can be used to measure shielding effectiveness. The setup consists of spot-focusing horn lens antennas with the specimen placed in the focal point between the two specimens. There are some studies that have explored this method (24–26). The advantage of this approach is the flexibility over frequencies as the waves are no longer constrained. In addition, there is clear line-of-sight of the specimens, and S-parameters can still be recorded in this approach. The downside is precision over the beam focus, there also may be wave diffraction near the edges requiring large samples or use of an anechoic chamber (27)

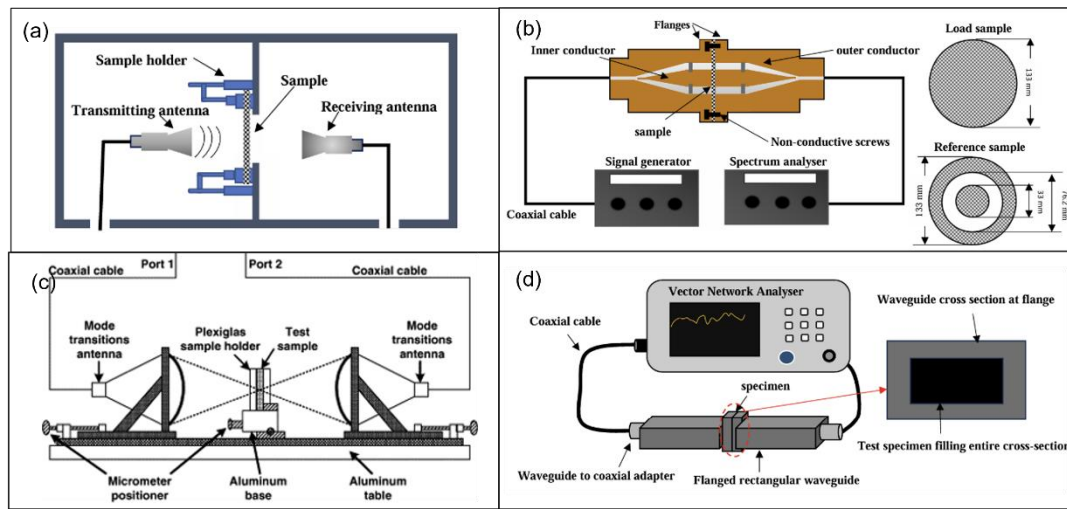


Figure 3. Four experimental setups for measuring shielding effectiveness.

(a) IEEE 299 shielded enclosure, (b) ASTM D4935 coaxial TLM (27), (c) free-space method (26), (d) ASTM D5568 waveguide TLM (27).

Table 1. Select range of experimental methodologies used for measuring shielding effectiveness.

Experiment				Sample			
Type	Standard & Model	Frequency (GHz)	Analyser	Type	Size (mm)	Thickness (mm)	Ref.
-	-	-	-	-	-	-	-
Coaxial	ASTM D4935; Custom	0.03 – 1.5	Keysight E5062A	Round	Ø 133.1	1.0 – 3.0	(28)
Coaxial	Non-standard; EpsiMu Multiwave PE13	0.05 – 8.0	Anritsu Shockline MS 46522B	Round	Ø 13.0	0.4 – 3.0	(23)
Coaxial	Non-standard; Elgal SET-19A	80.0 – 110.0	Hewlett-Packard 8510C	Round	Ø 150.0	60.0	(29)
Coaxial	Modified ASTM D4935; KEYCOM S-39D-D4935	0.3 – 3.0	Agilent 8753-ES	Round	Ø 49.0	0.025 – 0.15	(30)
Waveguide	Modified ASTM D5568	8.2 – 12.4	Keysight N5230A	Rectangular	10.16 x 22.86	1.0	(31)
Waveguide	Modified ASTM D5568	12.4 – 18.0	Keysight N5234B	Rectangular	7.90 x 15.80	1.6 – 3.6	(32)
Waveguide	Modified ASTM D5568; Pasternack PE-W90S001-6 1701, PE-W62S001-6 1650	8.2 – 18.0	Anritsu Shockline MS 46522B	Rectangular	10.16 x 22.86; 7.90 x 15.80	0.4 – 3.0	(23)
Abs. Box	Modified IEEE P2715	0.7 – 18	Keysight P9375A	Rectangular	300 x 500	2.0	(33)
Free Space	Non-standard; Custom	8.2 – 12.4	Hewlett-Packard HP 8722ES	Rectangular	150 x 150	0.9 – 1.9	(26)

* Abs. Box stands for absorber box.

2.2 Development of Electromagnetic Shielding for Unidirectional Composite Laminates

2.2.1 Shielding Mechanisms of Unidirectional Composite Laminates

One of the properties of using waveguide TLM methods is that the operating frequencies change based on the frequency, wave speed, and the size of the waveguide. As long as the frequencies stay within the operating frequencies, generally transverse electric field (TE) mode stays active. Figure 4(a) and 4(b) illustrates unidirectional fibres being aligned to the y-direction (90°) and x-direction (0°) respectively. In the former, plies are oriented parallel to the electric field, achieving an increase in almost increases the shielding effectiveness by about five times. The reason for this is that electrons oscillating along this plane of motion move in the direction parallel to the direction of the fibres, which is the direction of least resistance and are conducted away effectively. As such, when plies are oriented such that they are in the 90° laminate orientation, where fibres are parallel to the short-side of the waveguide, this theoretically inhibits the motion of the electrons, which are now forced to move in the perpendicular direction to the plies. These fields can be observed in Figure 4(c) and 4Figure 4(d).

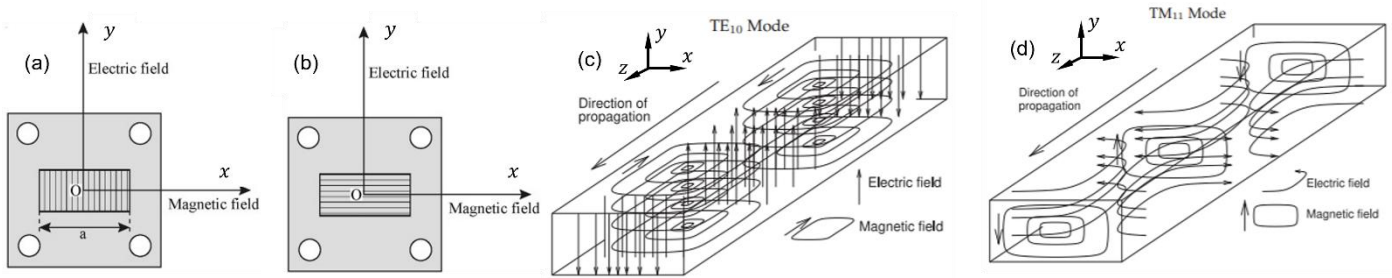


Figure 4. Electric and magnetic fields within rectangular waveguides. (a) UD fibres oriented parallel to electric field (34), (b) UD fibres oriented perpendicular to electric field (34), (c) TE₁₀ mode (35) (d) TM₁₁ mode (35).

Compared to homogeneous materials, as illustrated in Figure 5, the laminated structure of carbon fibre laminates can influence the shielding effectiveness of the material. Shielding arises from electromagnetic waves encountering an interface between two layers with differing fibre orientations. Hong et al (36) found that in their experimentation, the $[0/90/0/90]$ layup with carbon fibre/epoxy unidirectional laminates performed best, followed by $[0/90/0/0]$, $[0/0/90/90]$, and $[0/0/0/0]$. This phenomenon was also observed in the approach carried out by Kim et al by increasing ply angles in 15 degree angle increments (37). The results observed that shielding effectiveness increased in a relatively stable manner from 20 dB up to 90 dB for 6-ply laminates of carbon fibre epoxy laminates. The highest shielding effectiveness was achieved by $[0/90]_3$ and was followed by $[0/45]_3$, $[0/30]_3$, $[0/15]_3$, and $[0/0]_3$. With this in mind, a cross-ply (CP) layup generally achieves shielding effectiveness higher than quasi-isotropic (QI) layups (15,16).

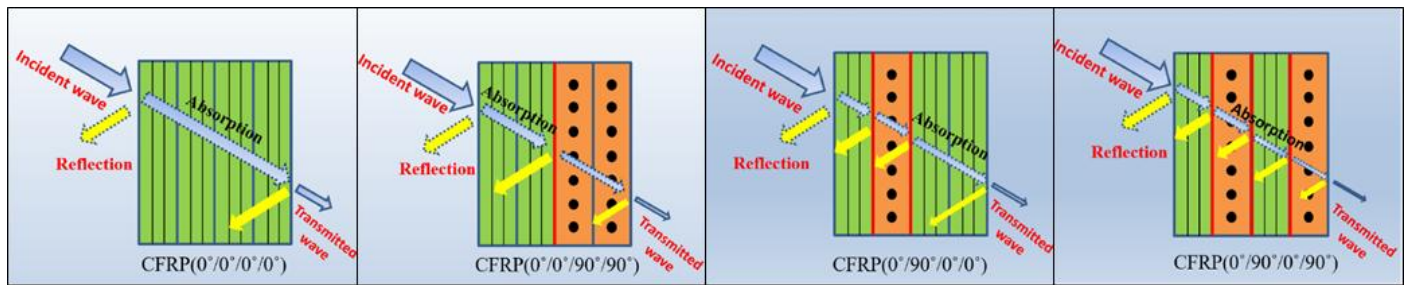


Figure 5. Mechanisms in shielding effectiveness of unidirectional composite laminates. Modified from (36).

Table 2 details four sets of results from unidirectional polymer composite laminates compiled from literature, specifically focused on CP and QI layups. Inspecting the results here, Kukdo T300/YD-128 laminates in general appear to be hold the best shielding performance performance in comparison to other laminate systems here, requiring much less laminates compared to the Syensqo 34-700/LTM45 system and NGF XN80/EP system. Conversely, despite using the same fibre system, the Syensqo LTM45 system with layup $[0/90]_2$ appears to perform much better than the other UNITIKA PA6 system with a layup of $[0/90]_3$. This may have been discrepancies from measurement methodology used given the two were not carried out with the same setup. In the same study it was observed that pitch-based carbon fibres appear to have low conductivity values, the study found that XN80 carbon fibres have a conductivity of 11 kS/m compared to almost two times higher at 19 kS/m for T700 carbon fibres.

Table 2. Summary of selected UD polymer composite laminates.

Matrix / Process	Fibre	Layup	Thickness (mm)	SE _T (dB)	Frequency	Setup	Ref.
-	-	-	(mm)	(dB)	-	-	-
EP (Kukdo YD-128) vacuum bag moulding	PAN-based (Toray T300)	[0/90]	~0.3	64.16	0.5-3 GHz	Coaxial, Mod. ASTM D4935, S. G. & Receiver**	(37)
		[0/90] ₂	~0.6	78.49			
		[0/90] ₃	~0.9	90.89			
PA6 (UNITIKA Nylon 6), prepreg, autoclave	PAN-based (Grafil 34-700)	[0/90]	0.476	44-45.6	0.1-1.0 GHz	Coaxial, Elgal SET-19A, HP-8752C VNA	(38)
		[0/90/0]	0.743	48.4-56.4			
		[0/90] ₂	0.957	61.6-67.2			
EP (Syensqo LTM45) prepreg, autoclave	PAN-based (Grafil 34-700)	[0/90] ₄	1.0	84.5	0.03-1.5 GHz	Coaxial, ASTM D4935 R&S ZNLE3 VNA	(16)
		[0/90] ₈	2.0	93.3			
		[45/0/-45/90] ₂	1.0	83.8			
		[45/0/-45/90] ₄	2.0	92.2			
EP (NGF -) prepreg, (-)	Pitch-based (NGF GRANOC XN80, 125 gsm)	[0/90/0/90]	0.55	~28.5-22.6 dB	5-15 GHz	Coaxial, Custom GPIB, N/A	(36)
		[0/90/0/90] ₂	~1.10	~22.5-28.9 dB			

* Value not reported, estimated values used instead.

** Signal generator and receiver; EP stands for Epoxy

2.3 Conductivity

Conductivity is an important consideration when it comes to evaluating shielding effectiveness and there are two types of conductivity measurements when it comes to composite laminates. There is surface electrical conductivity, and bulk electrical conductivity measurements approaches. Surface conductivity is often used for longitudinal directions and can be applied to the transverse direction too. Bulk conductivity adopts Ohm's equations and is often used in through-thickness calculations. The orientations of the conductivity directions, in-plane parallel ($\sigma_{\parallel}$), in-plane transverse ($\sigma_{\perp}$), and through-thickness (σ_{tt}) are as shown in Figure 6.

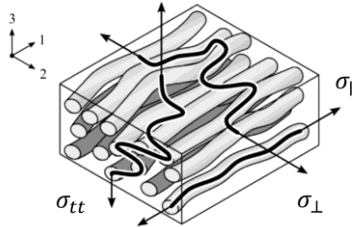


Figure 6. Conductivity directions in unidirectional carbon fibres. (39)

Most studies generally carry out the four-point probe method, due to its effectiveness in removing contact resistance. However, Buser et al carried out an investigation over 14 methods across 1975-2023, and pointed out there is a limitation of the approach for UD plies which lies in the single set of potentiometric probes, which results in inconsistency in the results measured. To measure the wide range of thermoplastic tapes, a six-point method was developed, as shown in Figure 7(a). Surface longitudinal electrical conductivity, $\sigma_{\parallel}$ of the high-temperature thermoplastic tapes (AS4/PEEK, AS4/LMPAEK, T700/PEEK) was found to be 36.6-39.4 kS/m. While this approach is attractive, obtaining the conductivity calculations is less intuitive. For this reason, Mikinka and Siwak pivoted to the two-probe method shown in Figure 7(c). Their approach involved applying pressure onto the surfaces of interest to decrease the number of asperities. The electrical conductivity value $\sigma_{\parallel}$ was found to be 13.72-14.82 kS/m for CP layups, and 7.81-8.52 kS/m for QI layups for LTM45/34-700 laminates. Patadia et al's work is shown in Figure 7(b), their analysis show that for IM7/APC2 CF/PEEK material found that $\sigma_{\perp}$ of 0.108 S/m and $\sigma_{\parallel}$ of 36.4% from 447 S/m.

Considerations prior to making these measurements have been detailed in many studies, most significantly by Li et al (40), whereby abrasion was found to decrease surface resistivity from 2-3 Ω/sq to 0.00103 Ω/sq . There has also been reported the influence of silver paint on results (41)

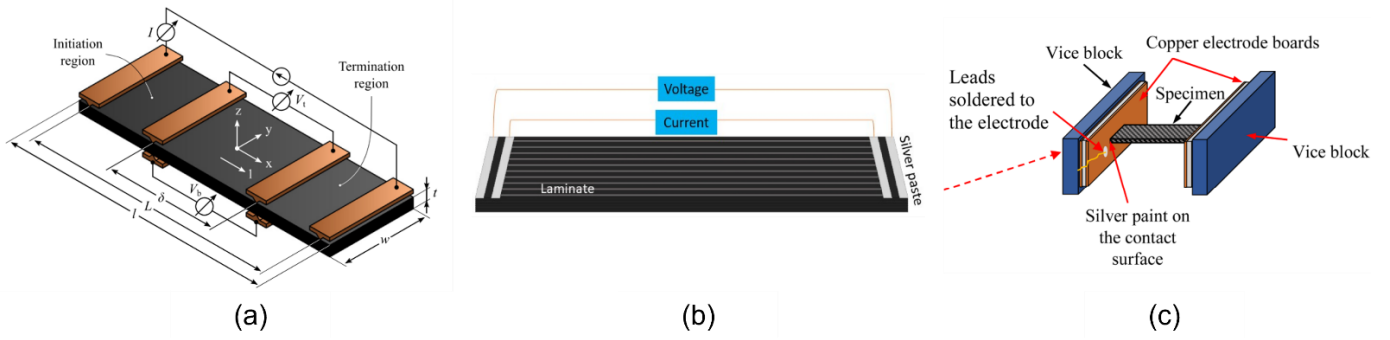


Figure 7. In-plane conductivity measurement methodologies.

(a) six-point probe method (39), (b) four-point probe method (42), (c) two-point probe method (16).

2.3.1 Enhancement of Shielding Effectiveness

There has been active work to enhance the conductivity of unidirectional laminates, particularly in the through-thickness direction. Mikinka and Siwak (43) carried out an in-depth study on this topic recently, and developed a comprehensive database covering a range of improvement methods. While metal is a popular shielding enhancement material, there exists the risk of galvanic corrosion. There are solutions such as nickel or the addition of glass fibre layer, however these require additional processing steps. With this in mind, the studies of interest in this discussion will be in relation to carbon fibre polymer composite laminates shielding effectiveness that have been reported using the X-band waveguides TLM method measuring the 8-12 GHz range.

An approach towards enhancing shielding effectiveness is interleaving or addition of conductive material in between plies to reduce the resistance of the interplay resin layer. In one example, Patadia et al also explored the incorporation of interlayer carbon nanotubes (CNTs) ‘sock’ lining in between carbon fibre polyether ether ketone (PEEK) plies to construct a hybrid laminate composite with layup $[0]_8$ and nominal thickness of 2 mm. Their study found that $\sigma_{\perp}$ increased by 23,359% from 0.108 S/m up to 25.3 S/m, and that $\sigma_{\parallel}$ increased by 36.4% from 447 S/m up to 610 S/m. They saw improvements in shielding effectiveness measurements where y-direction SE_T increased from 77.8 dB to 88.7 dB; similarly the x-direction SE_T at 10 GHz increased from 11.3 dB to 52.1 dB (42).

In another study, Gong et al explored a different method by agitating the surface of each ply with CNTs via dry spray deposition. Using a layup of $[0^\circ/45^\circ/-45^\circ/90^\circ]_2$, and a thickness ~ 0.5 mm; the final the enhancement with CNTs resulted in an increase from 45-62 dB up to 53-74 dB (44).

Another approach is to directly coat carbon fibres using conductive materials, improving the interfacial interaction between fibers and the matrix, resulting in improved conductivity of the composite overall. Kong et al presented surface functionalised polyaniline (PANI) onto woven Toray T300 (3K) plain weave carbon fibre. The laminate was fabricated with an epoxy resin via hot press method with a thickness of 610 μm , resulting in which were shown to improve the response from 53-54 dB to as high as 74-82 dB when combined with 4% reduced graphene oxide/Fe₃O₄ (RGF) (45).

Finally, there has also been explorations of adding films or layers on top of the laminate to increase surface conductivity but also EM shielding performance. Zhu et al reported development of nylon filter/nickel film that was hot pressed onto UD T300 carbon fibre/epoxy laminates. Manufacturing the laminates as $[0/45/90/-45]_s$ with a final laminate thickness of 0.19 mm, the study saw an increase an increased SE_T from 39.77 dB to 63.1 dB (46).

Overall, the enhancement of conductivity and shielding effectiveness can be achieved by addition of interlaminar methods such as fibre surface functionalisation, interlayer veils, or conductive coatings have proven to be effective, increasing the conductivity by around 10 dB. There remains active development within this space, however, there remains less development within this space on UD CFRTP laminates.

2.4 Motivation, Aims and Objectives

As seen in the literature review, there exists many opportunities for development of unidirectional composite laminates. CFRTP composite laminates are gaining traction and adoption is growing. The wider aviation industry is currently in search for a more sustainable future where materials can be reprocessed for other applications. There are now explorations of the incorporation of composites integration into avionics systems (27), however, there remains several unsolved challenges.

First, beside the work of Chung and Eddib (38), remarkably little existing information is available regarding the electromagnetic shielding effectiveness (SE) of CFRTP composite laminates. This anisotropy, combined with the influence of the thermoplastic matrix on fibre-matrix interfacial conductivity, represents a largely unexplored area. Second, there exists a significant deficiency in the literature with respect to the statistical rigour applied to electromagnetic shielding measurements of composite laminates. Many, if not all, of the studies reviewed reported results from only a single laminate specimen, with no repetition of measurements across multiple samples of the same configuration. Third, there appears to be a notable lack of discussion on the shielding performance of materials in the S-Band 2-4 GHz range, given the fact that shielding performance changes significantly based on frequency.

Looking into the next decade, Airbus has already demonstrated the industrial viability of carbon fibre polyphenylene sulphide (CF/PPS) at scale, with the material currently being used in the manufacture of fairings for the A320neo, derived from experience gained through the A380 pylon cowls [X]. Jean-Fulcrand et al. further demonstrated that CF/PPS exhibits significant promise in terms of long-term environmental durability, showing minimal surface damage following weathering, lower hygroscopicity, and negligible UV degradation when compared to PEI and PEEK (47). Even under aggressive high-moisture and corrosive environments, CF/PPS demonstrated markedly superior performance when subjected to simulated sea condition exposure (48). The established industrial precedent combined with compelling environmental durability data positions CF/PPS as an ideal candidate for further investigation.

There is also the shift observed in the offshore industry with Strohm (49). The industry is currently seeing a pivot towards TP composites due to their high impact resistance, better fatigue, and corrosion resistance. There exists demand for not just high-durability composites, but low-melting point materials such as carbon fibre polyamide-12 (CF/PA12).

The aim of this project is to experimentally develop and characterise high-performance electromagnetic shielding for avionics manufactured from thermoplastic unidirectional laminates in the S-band region. The objectives of this study are to:

- Manufacture carbon fibre thermoplastic composite laminates and evaluate the manufacturing quality of all laminates.
- Characterise the bulk electrical conductivities and flexural performance of the selected thermoplastic laminates.
- Measure the shielding effectiveness of thermoplastic laminates in the S-band region.
- Investigate the relationship between manufacturing quality, bulk electrical conductivity, and shielding performance.
- Compare the shielding effectiveness and mechanical performance to the state-of-the-art, aluminium.

3.0 Experimental Methodology

3.1 Introduction

This chapter presents the experimental approach employed to characterise the electromagnetic shielding performance of CF/PPS and CF/PA12 thermoplastic composite laminates.

The workflow followed in this study is summarised in Figure 8, which provides a visual overview of the sequential decision-making process from initial manufacturing through to the completion of characterisation. For each laminate configuration, the procedure began by defining the key processing parameters; namely the layup sequence, consolidation temperature, and applied pressure. This was informed by DSC results and a review of relevant literature to ensure manufacturing was carried out in line with established best practices. A standard operating procedure was developed to ensure repeatability across press cycles, and images alongside process notes were recorded throughout for documentation purposes.

Following consolidation, preliminary quality assessments were performed by visually inspecting the laminate surface finish and conducting acoustic tap tests to identify regions of potential delamination or porosity. Ultrasonic C-scans were subsequently carried out to provide a more systematic, full-panel assessment of internal quality. Where laminates were found to be unsatisfactory, the processing parameters including applied pressure, mould preparation, and thermal cycle were reviewed and the manufacturing process repeated until acceptable consolidation quality was achieved.

Once satisfactory laminate quality was confirmed, a structured characterisation programme was undertaken. To minimise material consumption, electromagnetic shielding effectiveness measurements were prioritised and conducted first, as these required intact full-panel specimens. To reduce the risk of methodological error, pilot studies on one or two specimens were carried out prior to full batch testing, with results reviewed against literature values. Where measurements were found to be inconsistent or the methodology ineffective, the approach was refined accordingly. For example, three distinct electrode configurations were evaluated for through-thickness conductivity measurements before the most repeatable method was selected. Once a satisfactory methodology was established, the full specimen batch was tested.

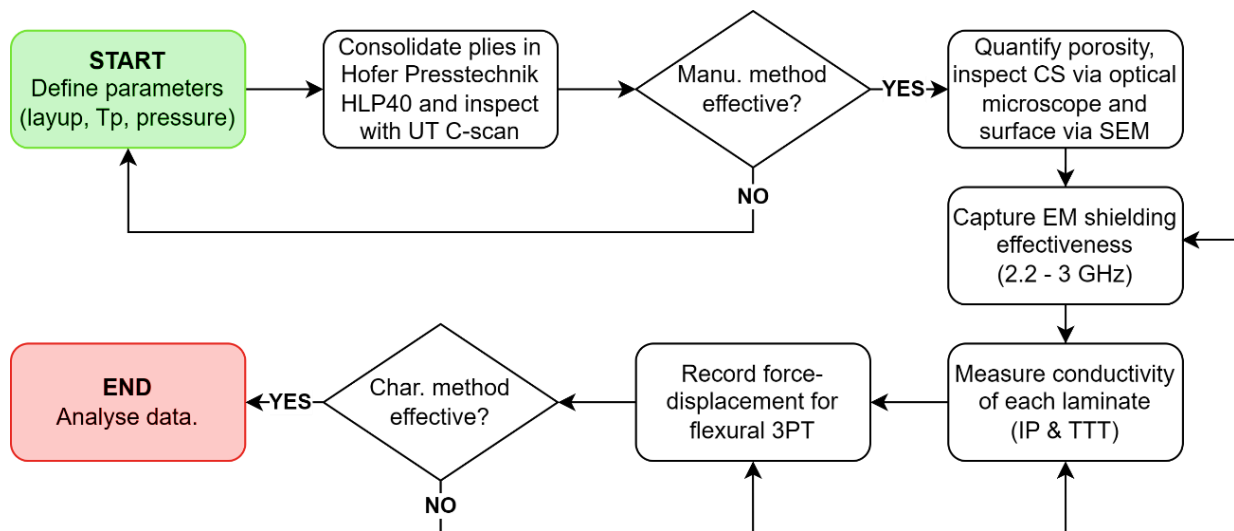


Figure 8. Overview of experimental procedure carried out in this project.

3.2 Materials

Carbon fibre/polyphenylene sulfide (CF/PPS) planar laminates were manufactured using unidirectional (UD) prepreg *Cetex® AS-4/TC1100* from Toray Advanced Composites, USA. The prepreg had a density of $\rho = 1.60 \text{ g/cm}^3$, a nominal ply thickness of $t = 0.16 \text{ mm}$, and a tape width of $w = 101 \text{ mm}$, consisting of PAN-based AS4A (Hexcel Corporation, USA) standard modulus carbon fibres at a fibre volume fraction $v_f = 59\%$ and a fibre areal weight (FAW) of 150 g/m^2 . The *Cetex® TC1100 PPS* resin system used has maximum moisture absorption of 0.02% (50).

Carbon fibre/polyamide-12 (CF/PA12) planar laminates were fabricated using a modified *VESTAPE® PA12-CF45* UD prepreg tape supplied by Evonik Operations GmbH, Germany. This prepreg system had a density of $\rho = 1.36 \text{ g/cm}^3$, a nominal ply thickness of $t = 0.20 \text{ mm}$, and a tape width of $w = 104 \text{ mm}$, consisting of PAN-based unknown standard modulus carbon fibres at a fibre volume fraction $v_f = 45\%$ with an FAW of 343 g/m^2 . The *Evonik PA12* resin system used has maximum moisture absorption of 0.7% (51).

Table 3. Mechanical and physical properties of prepreg from manufacturer data sheets.

Symbol	Property	CetexAS-4/TC1100 (50)	VESTAPE PA12/CF (51)
σ_t	Tensile strength (0°), RTD	2160 MPa	1750 MPa
E_t	Tensile modulus (0°), RTD	124 GPa	100 GPa
σ_c	Compressive strength (0°), RTD	1100 MPa	350 MPa
E_c	Compressive modulus (0°), RTD	117 GPa	-
σ_B	Flexural strength (0°), RTD	-	900 MPa
E_B	Flexural modulus (0°), RTD	-	95 GPa
v_f	Fibre volume fraction	59%	45%
ρ	Prepreg density	1.60 g/cm ³	1.36 g/cm ³

3.2.1 Differential Scanning Calorimetry (DSC)

Prior to starting experimentation, differential scanning calorimetry (DSC) was carried out using the Discovery DSC (TA Instruments, USA). This was carried out to confirm the identity of each prepreg and established the most optimal processing temperature to be used for manufacturing.

Three specimens were prepared for each material with each specimen containing two squares of prepreg (3mm x 3mm) for a weight of 2-3 mg. Specimens were enclosed inside Tzero Black aluminium pans (TA Instruments, USA) and sealed with a press. A new empty reference pan was prepared as required. Next, a three-step program for CF/PPS specimens was specified, starting with equilibrating to 20 °C and ramping up to 400 °C at 10 °C/min, before dropping back down to 20 °C at the same rate. Similarly, CF/PA12 specimens were equilibrated at 20 °C and ramped to 220 °C at 10 °C/min, before temperature was dropped down to 20 °C at the same rate. Between each step, the temperature was programmed to be held isothermally for four minutes to allow the stabilisation of the system.

Following this, specimens were placed into the DSC, and runs were programmed for each specimen. Each specimen underwent three runs of each procedure. The first run would allow any thermal history to be removed in addition to the release of any absorbed gases, allowing acquisition of the actual second and third runs.

Upon completion, analysis of the melting, glass-transition, and crystallisation temperatures of each material was done in TRIOS software (TA Instruments, USA). Values from the second and third runs were recorded for each specimen, giving a total of six readings for both CF/PPS and CF/PA12.

3.3 Manufacturing

3.3.1 Layup Configuration and Ply Cutting Scheme

Three layup configurations ($[0]_{14}$, $[(45/-45/0/90/45/-45/0)_s]$, $[(0/90)_3/0]_s$) were investigated for the CF/PPS laminates, and two configurations ($[0]_{10}$, $[45/-45/0/90/0]_s$) were investigated for the CF/PA12 laminates. All laminates were fabricated to a nominal thickness of $t = 2$ mm with an approximate planform size of $100 \text{ mm} \times 150 \text{ mm}$ (2:3 aspect ratio).

The reason to fabricate panels with these dimensions were primarily to reduce manual labour involved in preparing plies, however interestingly, while separate to the project's objectives, it was found that combined with the cutting scheme laid out in Figure 9, this study's approach resulted in very minimal material waste. In fact, cutting ± 45 -degree plies with the approach here actually achieved a utilisation factor of $U = 98.99\%$. When compared to even the most optimised cutting scheme reported by Mikinka (26), using the 1:1 aspect ratio resulted in only $U = 70.8\%$ utilisation. Further discussion of the cutting scheme and analysis conducted can be found in Appendix A.

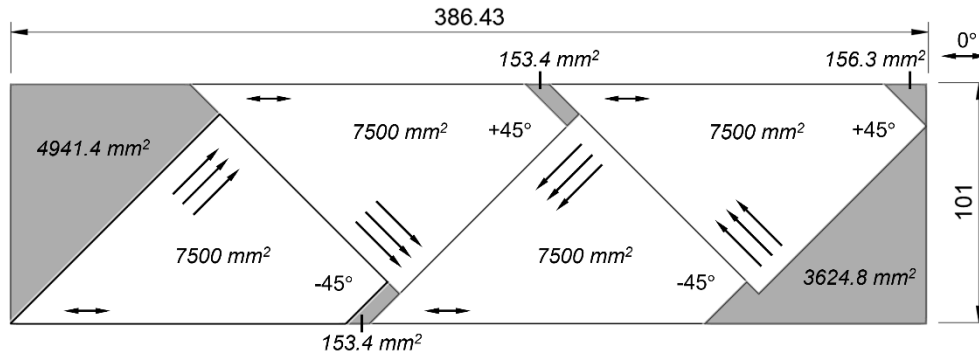


Figure 9. Prepreg cutting template design for CF/PPS ± 45 -degree plies. Similar template was designed for CF/PA12.

3.3.2 Ply Preparation and Layup Procedure

Prepreg rolls for CF/PPS and CF/PA12 were stored in a controlled environment at room temperature (25°C). Corresponding ply templates (made from cardstock) were then secured in the orientation and location specified in the cutting scheme with non-residue blue flash tape. Prepreg plies were then carefully cut individually from each roll while minimising longitudinal fibre splitting using sharp RS PRO 230 mm stainless steel scissors (RS Components, UK). Once cut, plies were categorised by ply orientation and sealed in labelled bags.

During layup, unlike TS prepregs, the non-tacky nature of TP prepreg required a more radical approach similar to the method used by Bandaru et al (52). To ensure ply orientation was maintained, spot welding was used. This was done in the order shown in Figure 10 using the T245-A chisel tip and CDB soldering station with temperature control (JBC Soldering, Spain). Welding temperatures used were set slightly above melting temperature; CF/PPS plies carried out using a temperature of 340°C and CF/PA12 plies was carried out using a temperature of 240°C . Once at temperature, a small amount of pressure was applied at each location and held for approximately one to two seconds. For alignment, this whole procedure was done inside a steel 'picture-frame', as detailed in the next section.

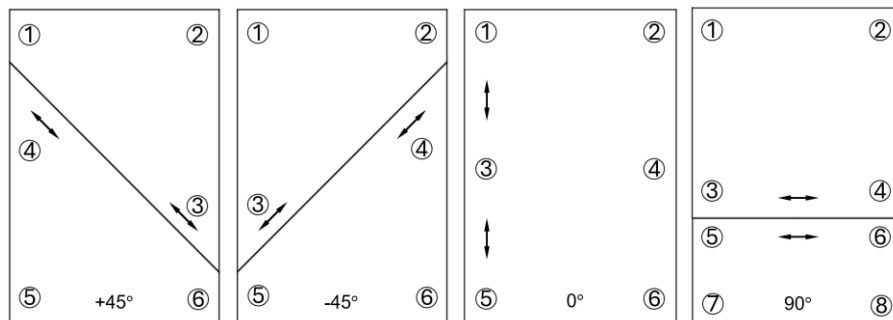


Figure 10. Spot weld location and order used when securing plies to form laminate. Numbers indicate order of weld.

3.3.3 High-Temperature Mould Preparation

Laminates were consolidated using a three-element high-temperature mould comprising of a top caul plate, bottom caul plate, and ‘picture-frame’. Picture-frames were waterjet-cut from mild-steel sheets ($t = 2$ mm) with internal dimensions 104 mm x 154 mm and outer dimensions 144 mm x 194 mm as detailed in (Appendix A). The frames were then sanded using 360-grit hand pads (Mirka, Finland) and P1200 silicon-carbide abrasive paper (MetPrep, UK) to minimise the risk of mechanical lock. After sanding, the moulds were cleaned with a degreaser. To ensure reliable release with minimal residue, three to five thin layers of *Chemlease® 41-90 EZ* high-temperature liquid mould release agent (Chem-Trend, USA; rated up to 450 °C) were applied using Kimtech® 7644 lint-free process wipes (Kimberly-Clark, USA), with a 15-minute wait time between layers. Following application, the frames were left to dry and fully cure for at least 30 minutes.

Polishing and release-agent application were repeated for each frame before every CF/PPS press cycle. This was necessary due to persistent PPS residue, which required light scraping with a sharpened chisel for removal. In contrast, CF/PA12 panels did not exhibit the same adhesion behaviour; therefore, only a single re-application layer was required between cycles.

3.3.4 High Temperature Press Consolidation Procedure

According to manufacturer guidance and standard practice, all laminates were manufactured using the stack illustrated in Figure 11 and processed in an HLP40 hydraulic laboratory press (Hofer Presstechnik GmbH, Germany).

Three laminates were manufactured simultaneously for each press cycle. The procedure began by wiping clean the bottom aluminium caul plate (500 mm × 500 mm × 10 mm), after which a layer of porous brown polytetrafluoroethylene (PTFE) release film was placed on top. Three steel picture frames and three laminate stacks were then positioned in the orientation shown in Figure 4. A high-temperature thermocouple was placed adjacent to one of the frames and secured using 3M 363 high-temperature aluminium-foil/glass-cloth tape (3M, USA). A second layer of PTFE release film was applied, and the top caul plate (400 mm × 400 mm × 10 mm) was carefully positioned. Once the stack was complete, the press-close position was recorded and input parameters calculated for each material’s press cycle. This was adjusted accordingly as detailed in Appendix A.

Two unique press cycles were used for the two materials in this study. A three-stage press cycle was used for the CF/PPS laminates and conversely, a two-stage press cycle was used for the CF/PA12 laminates. For CF/PPS laminates, the platens and mould were heated under a pressure of 500 kPa to 325 °C at a ramp rate of 4 °C/min. Upon reaching this temperature, the pressure was increased first to 1000 kPa and held for 15 minutes, this was then increased to a final pressure of 1200 kPa and held for another 15 minutes. Finally, platen heating was switched off, and the mould was allowed to air cool at the final pressure of 1200 kPa overnight. For CF/PA12 laminates, the platens and mould were heated under a pressure of 500 kPa to 225 °C at a ramp rate of 2 °C/min. Once this temperature was reached, the pressure was increased to 1000 kPa and held for 5 minutes. Upon heating was switched off and the mould cooled under final pressure of 1000 kPa overnight. Laminates were then released by hand and edges sanded using a disk sander to remove any excess remaining after release.

The temperature and pressure of the overall process were recorded over time and exported as a .TXT file and further analysed in MATLAB.

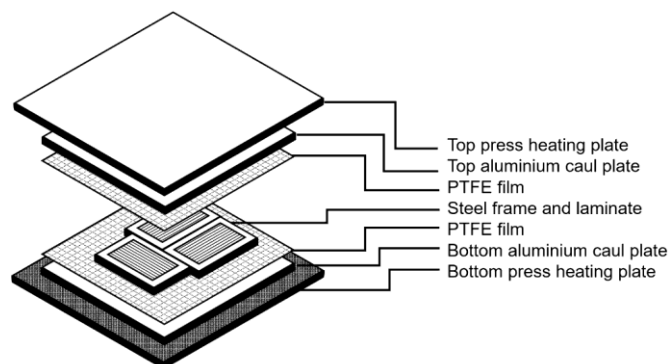


Figure 11. Stack used in heated press consolidation to manufacture laminates.

3.4 Quality Inspection Methodologies

3.4.1 Overview

Four methodologies were employed to investigate the quality of the laminates manufactured, each employed to analyse a specific area of interest, as seen in Figure 12. First, ultrasonic (UT) C-scans were done for rapid, full-panel mapping of defects. Next, this was complemented by micro-computed tomography (μ CT) scans to correlate UT C-scans to internal morphology. This was followed by optical microscopy for detailed inspection of the laminate cross-sections which helped provide detailed information on ply orientation, fibre-alignment, voids, and resin-rich regions. Finally, selective electron microscopy (SEM) was done to image the surface morphology of the edges after waterjet-cutting and their contributions to properties such as in-plane conductivity.

3.4.2 Ultrasonic (UT) C-Scan and Micro-Computed Tomography (μ CT) Scan

UT C-scans are a widely used non-destructive technique. These were carried out on each laminate using the Triton 1700 TT immersion tank equipped with a 10 MHz single-element transducer, which offers a balance of defect detection and penetration depth (Tecnitest, Spain). Scanning speed was set to 100 m/s with a scanning index of 0.5 mm. Acquisition was then carried out in pulse-echo mode using AOSUT software (Tecnitest NDT, Spain) and analysed with VisualScan v3.0 (Tecnitest NDT, Spain). To ensure the same scale was applied across different laminates, it was found to be necessary to include laminates from the previous scan.

After UT C-scans were acquired, a highly-porous small region (20 mm x 10 mm x 2 mm) was inspected using the Xradia 510 Versa CT scanner (Zeiss, Germany). Some studies use a voxel size of less than 2 μ m to achieve at least three voxels across each fibre (53,54). However, the focus here was to capture the macrovoids located in this region so acquisition was carried out at a larger voxel resolution of 6.2 μ m. Using the Scout-and-Scan Control system v16.2 (Zeiss, Germany), parameters used here included setting source to 80 kV (7 W), objective to 0.4X, bin size to 1, and exposure time to 2 seconds. 1601 projections were acquired and model was reconstructed using Dragonfly Pro (Comet Technologies, Canada).

3.4.3 Optical Microscopy and Scanning Electron Microscopy (SEM)

Four specimens (20 mm x 20 mm) were cut and stood up using spring clips, then set with KLEER-SET Type FF polyester casting resin and hardener (MetPrep, UK). Polishing was then done incrementally using P320, P600, P1200 and P2500 silicon carbide abrasive paper, followed by 6 μ m diamond suspension with planocloth, and then finishing with 0.3 μ m alumina suspension and 0.03 μ m alumina suspension using Multicloth soft nap cloth (MetPrep, UK). These were done with standard programmes using the Saphir 550 semi-automatic polisher with Rubin 520 attachment (ATM-M, Germany). Once complete, all samples were cleaned carefully under running water and then dried with a microfibre cloth. Optical microscopy was then captured using the 3-axis HR-2500 microscopy system with light diffusion adapter (Hirox, Japan). Further analysis of images was done in Fiji v2.16.0 (ImageJ2).

SEM micrographs were taken using the Hitachi S-3700N SEM (Hitachi High-Tech, Japan). Specimens were secured onto metal stages with Loctite® Double Bubble (Henkel Loctite, USA) mixed with carbon black for electrical continuity. Thereafter, specimens were placed into the platinum coater for batch sputtering platinum ($t = 4$ nm) onto the surface of the samples. The stage was then attached to a support stand, placed into the SEM and closed. Compressed air was used between each step to clear dust from the surface. To ensure spatial resolution, images were captured in slow-mode at a working distance of 5 mm and accelerating voltage of 5 kV.

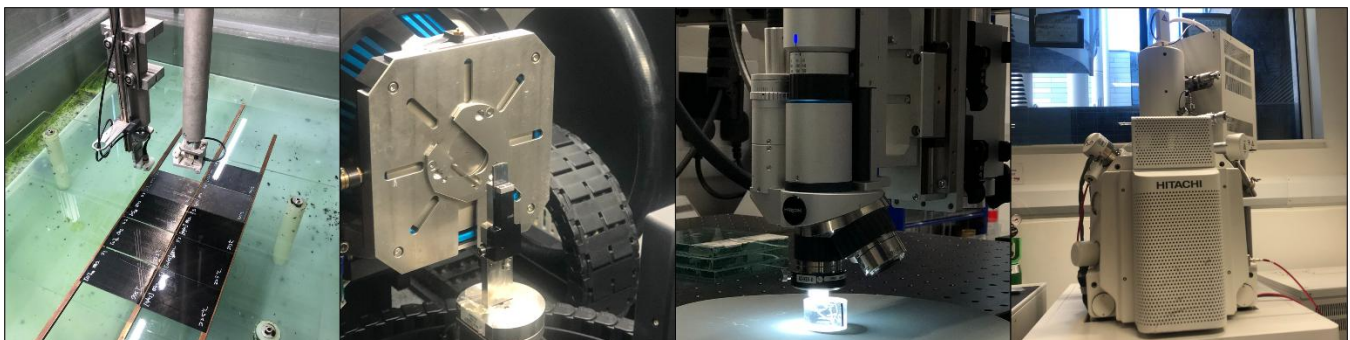


Figure 12. Quality inspection methodologies. From left to right, UT C-scan of CF/PA12 laminates; μ CT scan of CF/PPS laminate; optical microscopy; Hitachi S-3700N SEM.

3.5 Characterisation Methodologies

3.5.1 Density, Porosity, and Flexural Properties

Prior to flexural testing, density and void percentage (porosity) measurements were performed. Owing to the regular geometry of the laminates, physical measurements were taken in accordance with ASTM D2734. Four measurements of width and length were recorded using digital vernier calipers, followed by thickness measurements using a digital micrometer. Mass was measured to four decimals using Sartorius ENTRIS 124i-1S (Sartorius, Germany) as seen in Figure 13. Porosity was subsequently calculated using Eq. (10) using the measured density ($\rho_{meas.}$) and manufacturer density ($\rho_{manu.}$) from Table 3.

$$v_{void} = \frac{\rho_{manu.} - \rho_{meas.}}{\rho_{manu.}} \quad (10)$$

It should be noted that density of CF/PPS laminates were measured with the same laminates (100 mm x 150 mm) that were eventually used for electromagnetic shielding experiments, whereas for CF/PA12 laminates, these were measured using IP conductivity specimens (20 mm x 10 mm). The effects of this decision will be discussed in the next chapter.



Figure 13. Measurement of laminate mass for void percentage calculations.

To characterise the flexural strength and flexural stiffness of the laminates manufactured, the established ASTM D790 three-point-bending (3PT) method for reinforced plastics was used. However, due to the focus of this study being on the electromagnetic shielding properties, a smaller test campaign carried out for this area. This test campaign included three specimens for each CF/PPS configuration, and two specimens for each CF/PA12 configuration. As such, caution was practiced when interpreting results.

Specimens for each configuration were manufactured by waterjet with nominal width of $w = 25$ mm, thickness, $t = 2$ mm, and length, $l = 150$ mm. The decision to use these dimensions was necessary due to the anisotropy of unidirectional laminates, which requires high span-to-thickness ratios. The intended span for the flexural test was 120 mm. During testing of the CF/PPS specimens, however, the span was inadvertently set to $L = 100$ mm due to a setup error. The CF/PA12 specimens were subsequently tested using the intended $L = 120$ mm span. The effect of on the calculated flexural properties was considered in the results.

Test procedure involved taking three measurements of the width and thickness using digital vernier calipers and averaged. Subsequently, The test programme was then set up with crosshead speed was set to 12 mm/s on Instron 50 kN 5969 testing machine (Instron, USA) with data acquisition done through the testing machine and recorded using Bluehill Universal (Instron, USA). Analysis was carried out with the corrected 3PT bending flexural equations.

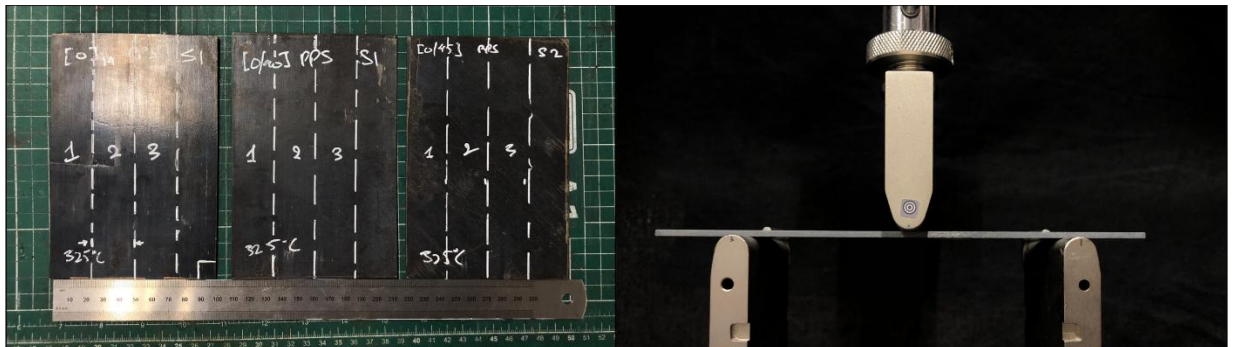


Figure 14. Specimen preparation and flexural testing of laminates. From left to right, specimens cut from one panel; setup used.

3.5.2 In-Plane and Through-Thickness Bulk Electrical Conductivity

Two types of bulk electrical conductivity were measured: in-plane (IP) bulk conductivity & through-thickness (TTT) bulk conductivity. IP specimens were 20 mm x 10 mm x 2 mm, whereas TT specimens were 12.5 mm x 12.5 mm x 2 mm. These specimens were cut by waterjet and dried at room temperature for at least 24 hours.

Using P1200 silicon-carbide abrasive paper (MetPrep, UK), TTT specimens were carefully sanded by approximately 10 μm on both sides (top and bottom) to expose the carbon fibres under the thin polymer layer. While significantly lower than the thickness mentioned by Li et al (40), this was sufficient after evaluating the optical micrographs of the cross-section. The sides for IP specimens were also carefully sanded using the same abrasive paper. Specimens were then cleaned using isopropyl alcohol (IPA) and dried with Kimtech® 7644 lint-free process wipes (Kimberly-Clark, USA). With gloves, four measurements were made for each specimen dimension and averaged. For thickness, this was measured with a digital micrometer, whereas the length and width were measured with digital vernier calipers. Finally, TTT specimens received an additional precise layer of 3M 1170 conductive aluminium tape (3M, USA) which was trimmed using a scalpel.

Five readings were recorded for each specimen using the Keysight 34465A Digit Multimeter (Keysight Technologies, USA) in 4-wire DC resistance mode which is more appropriate for low resistance readings less than 1 k Ω (55). The methodology used here is seen in Figure 15, which is a modified ASTM D4496 two-probe method adapted from the work of Mikinka (27). This approach provided the most repeatable and accurate readings after comparing it to the two other methods that were carried out, which are detailed in Appendix A. Here, both IP and TTT specimens were clamped to a fixed position in between two conductive aluminium surfaces (electrodes) with insulation from the vice polymer pads. These were connected to gold-plated kelvin clips to their respective terminals.

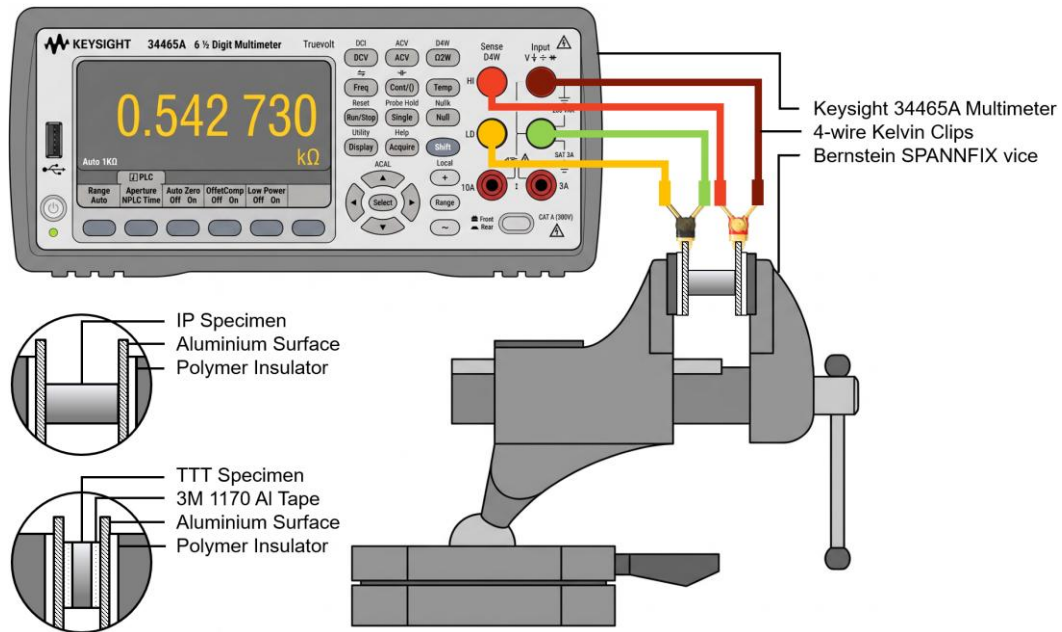


Figure 15. Diagram of electrical conductivity measurement setup.

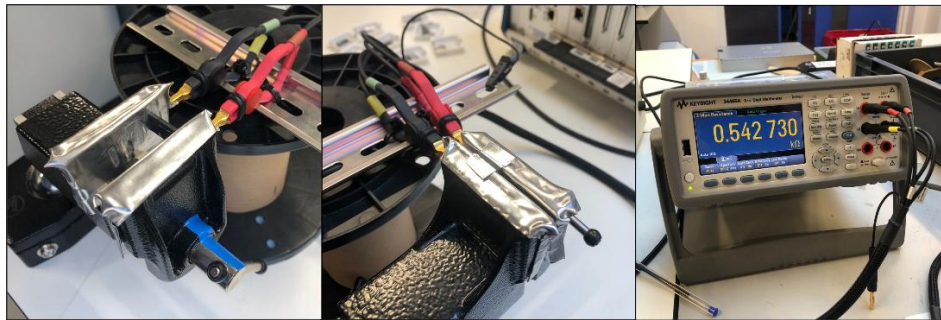


Figure 16. Final setups used for electrical conductivity measurements. From left to right, IP measurement; TTT measurement; Keysight 34465A.

3.5.3 Electromagnetic Shielding Effectiveness (SE)

A modified ASTM D5568 methodology was used due to its popularity and low-cost of starting up, as previously discussed. The reason the latter was an important consideration was due to the lack of existing apparatus, as such, a similar method to work of Sorgucu (56) was used in this study. As seen in Figure 17, the system developed used WR340 rectangular waveguide adapters in the FDP26 flange style (A-INFO, USA), which supported the 2.2 GHz to 3 GHz range. S-parameters were measured using the R&S ZVL3 2-port vector network analyser (Rohde & Schwarz, Germany), which supported the 9 kHz to 3 GHz range. Adapters were connected using SMA male-to-female connector cables and fixated with Gorilla Tape heavy-duty cloth-backed adhesive tape (Gorilla Glue Company, USA),

Three specimens were prepared for each configuration previously mentioned in Table 4. An aluminium reference was also fabricated with the same dimensions. For each specimen, holes were carefully drilled ($\varnothing = 6.3$ mm) into the laminate (100 mm x 150 mm x 2 mm) on a bench drill press with a template, as seen in Appendix C. Measures were taken to minimise delamination by applying backing tape onto both the entrance and exit side of the material, which helped hold the fibres together. Each specimen would be sandwiched between the two waveguides with M6 x 40 steel hexagonal bolts and tightened using a pair of RS PRO 10 mm spanners (RS Components, UK), as seen in Figure 6.

The procedure involved in testing started with full Through-Open-Short-Match (TOSM) two-port calibration. This was done prior to starting the experiment and was repeated for every measurement if there was visible ‘drifting’ of S11 parameter linear magnitude away from baseline. Throughout this process, 0.9 Nm torque wrenches (Agilent, USA) were used to avoid overtightening of the connections for consistency. After connecting the whole system, the resulting linear-magnitude of S-parameters (S11, S21, S12, S22) were each captured in .CSV files. Post-processing of the signals was done with a custom MATLAB algorithm using Eq. (5-7) which can be found in Appendix C. The algorithm filters signals to reduce noise and analysed in more detail in the next chapter.

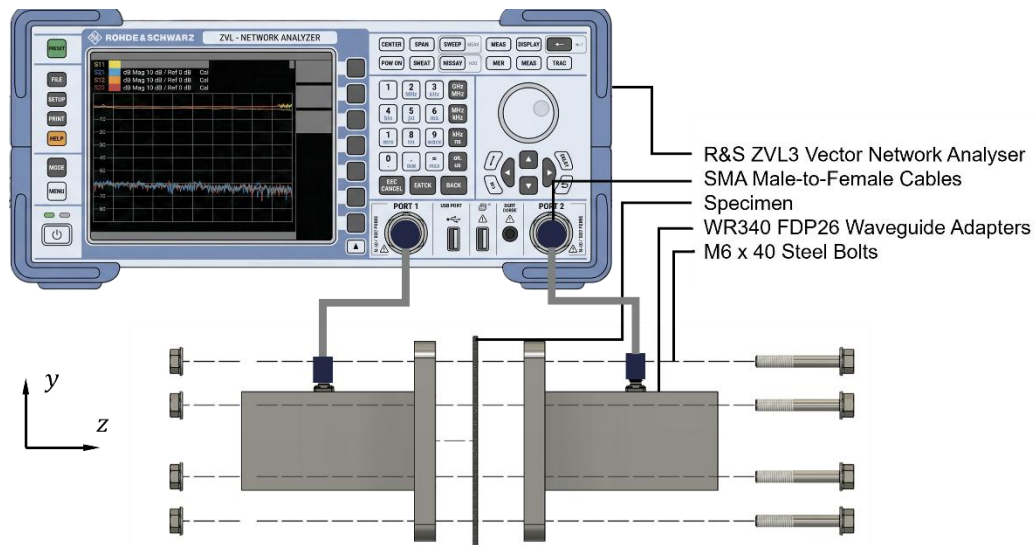


Figure 17. Diagram of electromagnetic shielding effectiveness measurement setup.

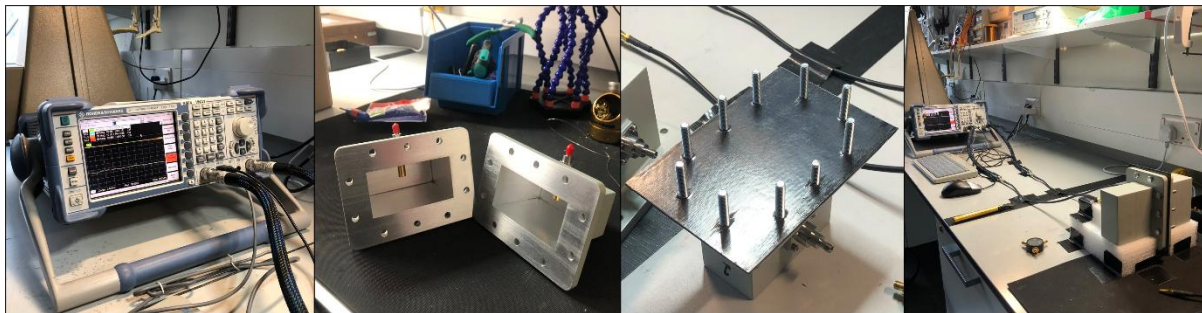


Figure 18. Setup used for electromagnetic shielding effectiveness measurements. From left to right, R&S ZVL3 vector network analyser; WR340 FDP26 waveguide adapters; mounting specimens in progress; overview of setup.

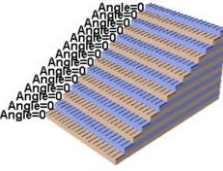
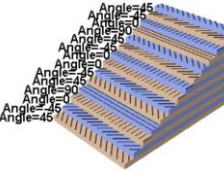
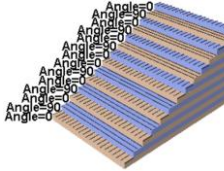
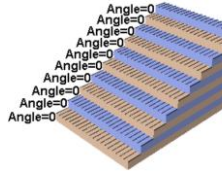
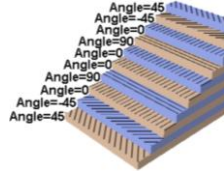
4.0 Results and Discussion

4.1 Introduction

This chapter presents and evaluates the performance of CF/PPS and CF/PA12 laminates using the methodology outlined. To assess the performance of the two laminates in their electromagnetic shielding applications, the findings are presented across four distinct sections. In Section 4.2, this study will first assess prepreg thermal characteristics, the process involved in manufacturing the laminates, and their associated quality using first non-destructive methods before microscopy. In Section 4.3, the physical, mechanical performance, and electrical conductivity of the laminates will be reported. Finally, in Section 4.4, analysis of the results from electromagnetic shielding and discussion of their connection to electrical conductivity and manufacturing quality will be explored, the discussion will then shift to a broader assessment on density-normalised multifunctional performance.

As seen in Table 4, each specimen in this chapter has been given a specimen identification number. To streamline analysis, the same specimen numbers have been standardised across all characterisation tests. However, it should be noted that the specimens may not always be prepared from the same laminate identification number and will be specified where applicable. For instance, flexural tests on specimen PPS-000-S1, PPS-000-S2, and PPS-000-S3 were all derived from one laminate (PPS-000-S4), as such interpretation of defects from other laminates (PPS-000-S1, PPS-000-S2, PPS-000-S3) are not directly applicable.

Table 4. CFRTP laminate configurations investigated in this study.

Config.	PPS-000-S#	PPS-045-S#	PPS-090-S#	PA12-000-S#	PPS-045-S#
					
Material	CF/PPS	CF/PPS	CF/PPS	CF/PA12	CF/PA12
Layup	[0] ₁₄	[(45/-45/0/90/45/-45/0)] ₅	[(0/90)3/0] ₅	[0] ₁₀	[45/-45/0/90/0] ₅

4.1.1 Terminology

Defined in Table 5 is a range of properties on results presented in the following section. Included in this table are the property's corresponding symbols, units and standards, as mentioned in the methodology section.

Table 5. Range of key properties investigated in this study and their corresponding standards.

Symbol	Property	Unit	Standard
T_g	Glass-transition temperature	[°C]	-
T_m	Melting temperature	[°C]	-
T_c	Crystallisation temperature	[°C]	-
ρ	Density	[g/cm ³]	-
v_v	Void Fraction	[%]	ASTM D2734
E_B	Flexural Stiffness	[GPa]	ASTM D790
σ_B	Flexural Strength	[MPa]	ASTM D790
$\sigma_{ }$	In-Plane Conductivity (Longitudinal)	[S/m]	ASTM D4496
$\sigma_{\perp}$	In-Plane Conductivity (Transverse)	[S/m]	ASTM D4496
σ_{tt}	Through-thickness Conductivity	[S/m]	ASTM D4496
SE_T	Shielding Effectiveness (Total)	[dB]	ASTM D5568
SE_A	Shielding Effectiveness (Absorption)	[dB]	ASTM D5568
SE_R	Shielding Effectiveness (Reflection)	[dB]	ASTM D5568

4.2 Manufacturing, Quality, and Microstructure

4.2.1 Analysis on Differential Scanning Calorimetry (DSC)

As semi-crystalline polymers, both prepreg materials show distinct T_c , T_m , and T_g values. According to each manufacturer's data sheet, Toray (50) specifies their PPS polymer to have $T_g = 90$ °C and $T_m = 280$ °C; whereas Evonik (51) specifies their PA12 polymer to have $T_g = 40$ °C and $T_m = 178$ °C. Inspecting the thermal properties of each prepreg material, as presented in Table 6, it can be observed that other than T_g measured here being 10 °C higher, all thermal properties here are in line with both manufacturer's specifications, as such the processing temperatures can be based on manufacturer's recommended guidelines.

The coefficient of variance (COV) was significantly higher for T_g compared to T_m and T_c . One contributor of variance may be related to the manual selection of the T_g region for analysis in TRIOS software by eye, where the start and end points of the inflection region were selected and mid-point generated by TRIOS. Manual selection could have resulted in error as the locations selected would not have been the same across graphs. Though, the challenge of selecting this region was accentuated by the difficulty of distinguishing the T_g slope due to the heating and cooling rates used. One solution to this could have been increasing heating and cooling rates could have improved accuracy, but this would have artificially shifted towards the right, as described by Knappe (57), resulting in inaccurate thermal properties. The most effective solution here would be to have investigated this parameter in more detail using dynamic mechanical thermal analysis (DMTA) using the storage modulus and loss factor to identify T_g . Alternatively, a modulated DSC program could have also been carried out for improved results.

Table 6. Thermal properties CF/PPS and CF/PA12 prepreg.

	Cetex AS-4/TC1100			VESTAPE PA12/CF		
Symbol	Value	SD	COV	Value	SD	COV
T_g	89.3	4.7	5.27%	50.2	4.0	7.92%
T_m	279.5	0.7	0.25%	176.5	0.1	0.03%
T_c	219.6	3.6	1.64%	160.0	0.1	0.06%

Four DSC thermograms from two specimens of CF/PPS and CF/PA12 are shown in Figure 19, each selected from Run 1 and Run 3. Focusing on CF/PPS, two observations of note is the disappearance of the exothermic peak around 120 °C from heating, which is likely due to cold crystallisation from rapid cooling during the manufacturing process (58,59). There is also a shift and shape change of the T_c peak, where crystallisation is delayed in Run 3 and the peak has become broader with a slight shoulder. One possible explanation for this phenomenon is heterogeneous nucleation sites such as the embedded carbon fibres (60,61). However, the same peak shape can be observed in pure PPS polymer from the NETZH database, although the report noted an endothermic peak near 159 °C (62) which was not observed. The effect of repeated thermal cycles on T_c can shift results as described by Ren et al (63) due to the self-nucleation characteristic of PPS polymer from residual crystalline fragments, however this likely cannot be applied here as PPS polymer reverts to fully random coil conformation when heated up to 400 °C (63). For CF/PA12 prepreg, there was a notable appearance of a double peak for T_m that was not visible in the first run. This is common indicator of distinct crystal structures (α -crystal and γ -crystal) in PA12 polymer, resulting in distinct melting peaks as described in the seminal paper of Hiramatsu et al in 1983 (64). However, a similar phenomenon was observed when analysing similar polymers when reinforced with carbon fibre, Liang et al (65) found the addition of carbon fibres played a role as a nucleating agent, resulting in the creation of double T_m peaks in the carbon fibre reinforced polyamide-6 (CF/PA6) in their study.

Overall, these observations were relatively consistent across all specimens of CF/PPS and CF/PA12, although CF/PPS did face unusual anomalies in PPS-S1-R1 and PPS-S2-R1 (Appendix B), before stabilising in subsequent runs. From the analysis conducted, that cooling rate is critical and can result in incomplete crystallisation if rapid cooling processes like quenching are used, as such slow cooling rates should be adopted. The author also acknowledges that T_c and T_m peaks are influenced by additives and number of thermal cycles, particularly with carbon fibre reinforced polymers. As such, while not within the scope of this study, further investigation on the neat polymers to achieve certainty on the DSC thermogram characteristics of the two prepreg materials investigated here.

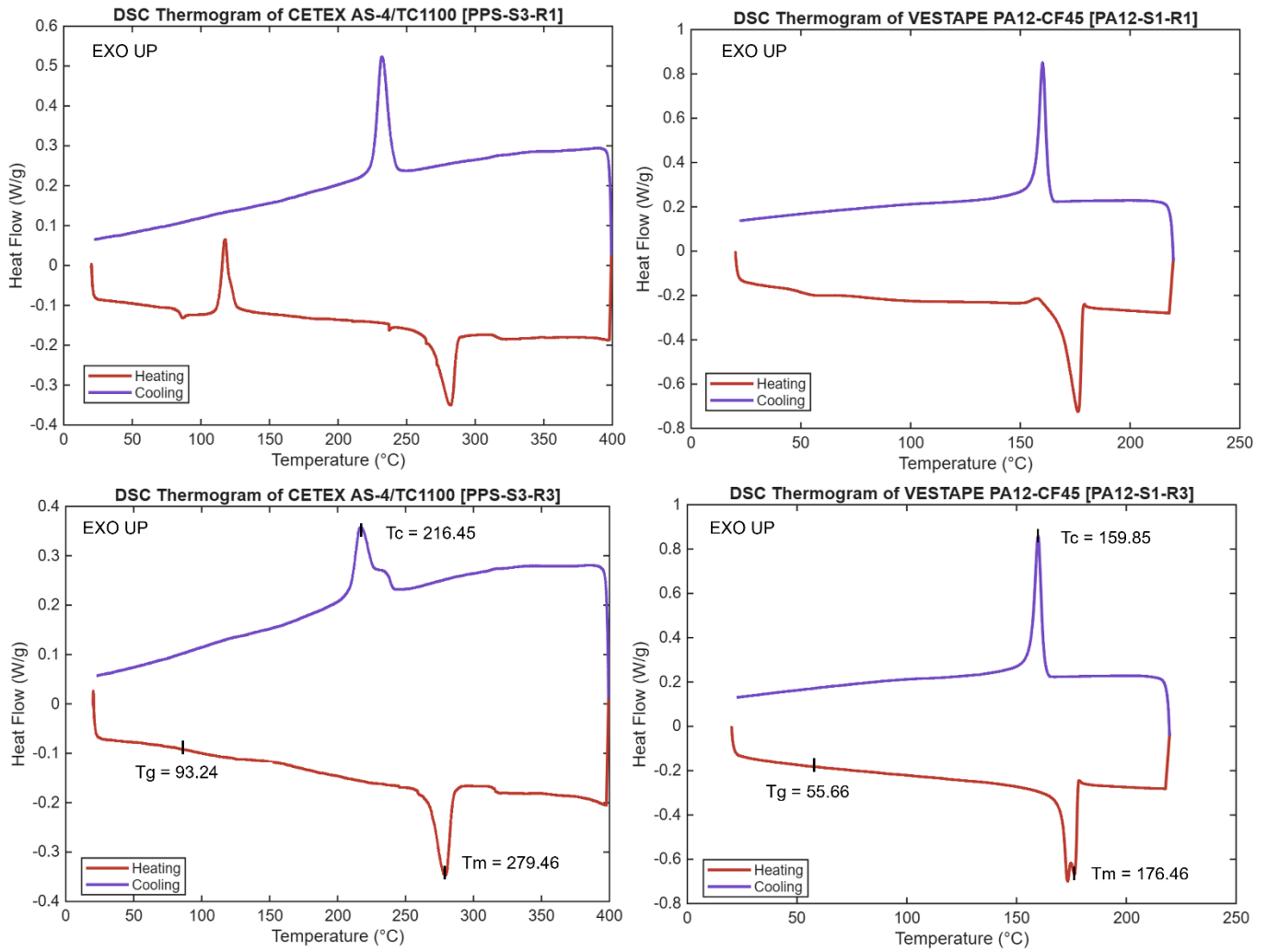


Figure 19. Differential scanning calorimetry thermograms of as-received prepreg material and after two runs.

4.2.2 Analysis on Heated Press Consolidation

Using the information obtained on the thermal properties of each material, press consolidation was carried out with processing temperature cycle as specified in Section 3.3.4. The final manufactured laminates for CF/PPS and CF/PA12 are shown in Figure 20 and Figure 21 respectively. Across both laminates, a notable amount of flash/residue was produced at the laminate edges after consolidation, likely due to the greater difference between the nominal stack height (2.24 mm) and the final consolidated thickness (1.98 mm), resulting in more material being displaced laterally as flash. On the other hand, CF/PA12 laminates had a nominal stack height of 2.00 mm, with the final laminates measuring an average thickness of $t = 1.94 \pm 0.02$ mm. The residue from CF/PPS had an off-white appearance and was highly porous yet stiff making removal difficult. On the contrary, the residue from CF/PA12 had a darker, semi-transparent appearance and was more ductile, an indicator of lower crystallinity which allowed removal without tools.

During manufacturing, one interesting observation made was the difficult in releasing the laminates from the frames, particularly PPS-045 and PPS-090, and PA12-045 laminates. The metal frame contracted significantly during cooling due to its relatively high thermal coefficient of expansion compared to the carbon fibre-dominated laminates, which exhibit near-zero in-plane CTE. This differential contraction caused the frame to clamp around the laminate, making release difficult. Despite this, by supporting the laminate from below and applying localised pressure at the corners to overcome the frictional clamping force.

After the edges were trimmed and sanded flush, there was generally little variance in geometry between each laminate. None of the laminates exhibited curvature, as expected given the symmetric layup configurations used. However, there was a random variation in the surface quality in laminates despite the being manufactured with the same procedure and materials. The type of surface defects observed can be split into two categories, namely: local variations in surface smoothness, and small indents.



Figure 20. Press-consolidated CF/PPS laminates.



Figure 21. Press-consolidated CF/PA12 laminates.

There are several reasons surface quality could have been affected. The most probable reason for this could have been from PTFE film being reused across each manufacturing session. Repeated use of PTFE release film likely introduced microscale surface damage and contamination, which transferred to the laminate surface during consolidation under high temperature and pressure. Despite this, the surface defects appeared to have negligible effect on laminate geometry and mechanical integrity, though their influence on ultrasonic C-scan signal quality will be discussed in Section 4.2.3.

Analysing one of the temperature and pressure datasets logged, we can identify two key pieces of information from Figure 22. First, there were two spikes in temperature with the magnitude of the second spike (317°C) being lower than the first (328°C), suggesting the control system progressively stabilised over the course of the cycle. This characteristic was not independent to this process cycle, and the same phenomenon was also observed when inspecting the recorded temperatures of the press heating plates, suggesting this is a characteristic of the press control system, whereby a delay in deactivating the heating elements causes a transient temperature overshoot coinciding with pressure application. Second, the temperature of the laminate converged to a temperature of 315 °C, giving a difference of 10 °C from the programmed temperature. The reason for this could be a combination of the PTFE films acting as insulating layers, in addition to the high thermal capacity of the thick caul plates used. This combined effect may have induced a delay in achieving 325 °C. Overall, the setup achieved a satisfactory overall cooling rate of 0.243 °C/min, which is very low but ideal for polymer crystallisation as noted in Section 4.2.1 (66).

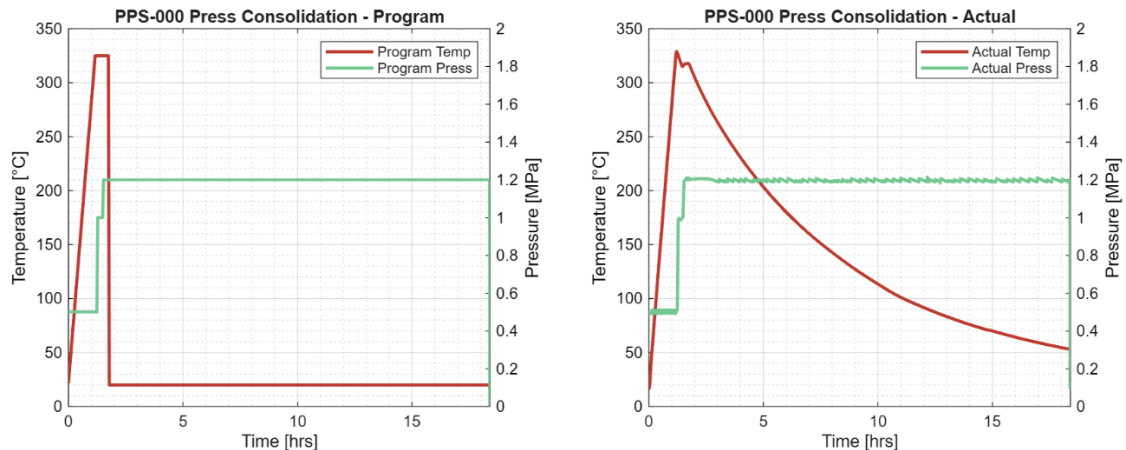


Figure 22. Press consolidation program compared to actual readings from one process cycle (PPS-000).

4.2.3 Analysis on Ultrasonic (UT) C-Scans

Turning our attention to UT C-scans, Figure 23 presents the final attenuation plots which are a function of the incident amplitude (A_i) and back-wall amplitude (A_f) signals. The signals were emitted and recorded using a single transducer in pulse-echo mode. Using this gating system and after setting a back-wall amplitude threshold to the signal noise floor, defects such as porosity and delamination can be detected. In general, higher porosity regions correspond to low A_f values; macro-voids and delamination on the other hand correspond to $A_f \approx 0$ dB, as described by Stone and Clark in 1975 (67). This will be confirmed using μ CT scans.

With this in mind, there has emerged two groups of approaches. One approach is to correlate porosity to attenuation as seen in the analysis conducted by Shi et al (68) on quasi-isotropic balanced layups of carbon fibre epoxy composite (CF/EP). The study investigated the relationship of porosity range from 0-2.5% using 5 MHz UT signals and fit a third-order polynomial to correlate the two variables with Ying & Truell's wave scattering theory. On the other hand, many researchers (69,70) have imposed a strict 6 dB threshold for laminate quality, whereby attenuation greater than 6 dB refers to regions with high porosity. The studies assume that attenuation coefficient stays constant between different types of carbon fibre composite laminates. Ultimately, as demonstrated by a recent study in 2026 by Ono (71), attenuation values can be influenced by parameters including fibre type, matrix type, layup, laminate thickness, and pore morphology, as such, careful interpretation of attenuation values is necessary.

Overall, a nuanced approach to evaluating quality is optimal when evaluating porosity. However, for this study, a qualitative approach will be used in the upcoming discussion, defining macrovoids and delaminations as 0-4 dB (black), microvoids or low-porosity regions as 4-18 dB (blue/dark blue), whereas high-porosity regions as 18-40 dB (teal-yellow-white). Figure 24

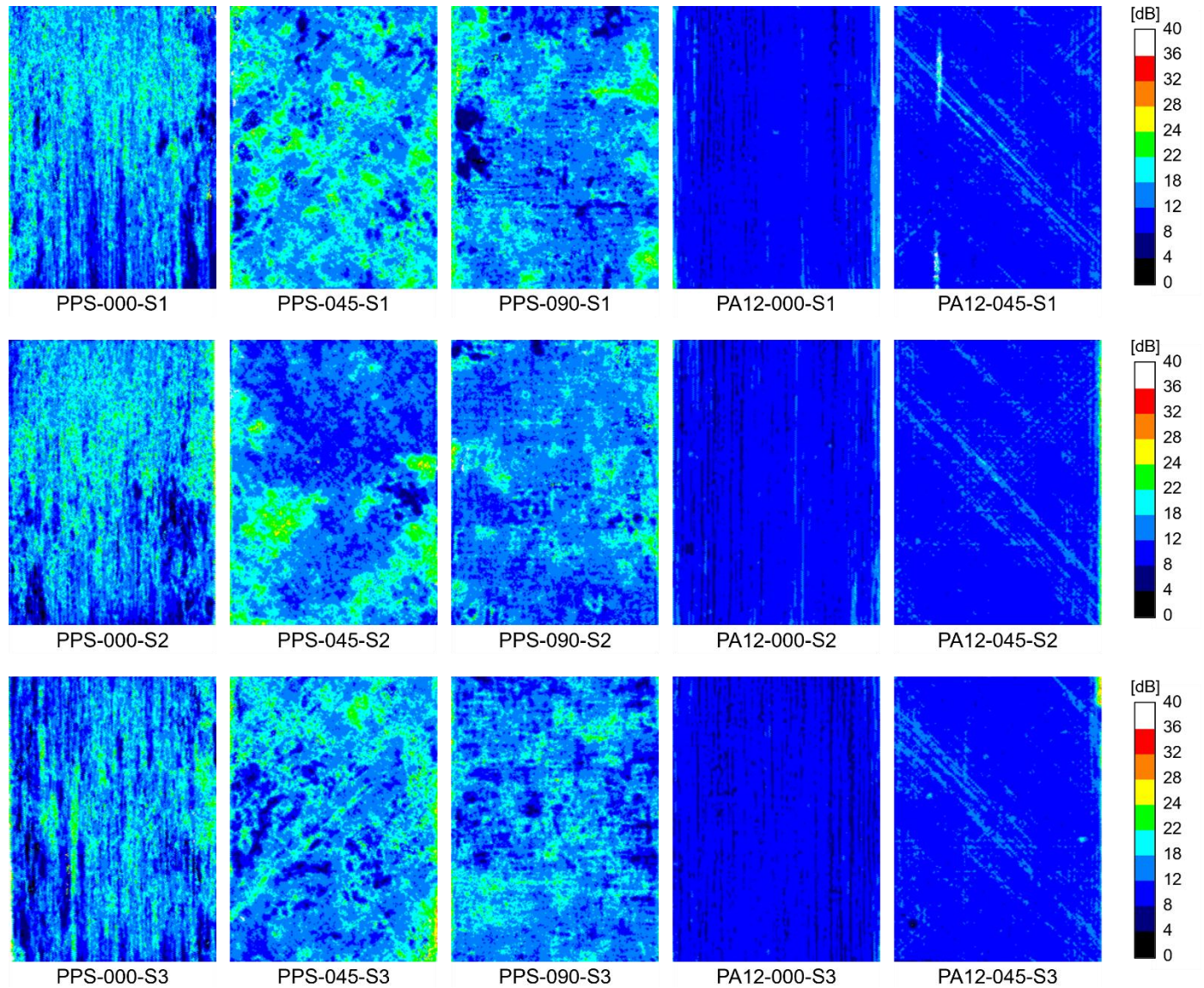


Figure 23. Ultrasonic C-scan attenuation plots of selected specimens in their corresponding laminate configuration.

Observing the UT C-scans, all three CF/PPS laminate configurations manufactured appeared to demonstrate presence of macrovoids and high-porosity regions, with notable differences in morphologies. The scans on PPS-000 specimens generally indicated columnar behaviour of macrovoids, or ‘channelling’, located near laminate edges with a mix of high-porosity and low-porosity regions scattered throughout the laminate. This was quite different compared to the PPS-045 specimens which saw large zonal behaviour of these high-porosity regions, notably accumulating along the boundary between ply edges due to methodology in Section 3.3.2. A similar phenomenon was observed with the PPS-090 specimens with zonal behaviour of high-porosity regions, although high-porosity zones were generally smaller in these specimens. Presence of macrovoids could be detected in PPS-045 and PPS-090 specimens which were randomly within the laminate. Conversely, CF/PA12 laminate configurations appeared to demonstrate low percentage of porosity across the two laminate configurations investigated, with most of the region showing 8-12 dB. Initial inspection of the scans of PA12-000 specimens showed small vertical striations that occasionally decreased to 4-8 dB or increased to 12-16 dB. Comparatively, PA12-045 specimens exhibited a similar effect of diagonally-oriented striations and minimal macrovoids could be observed.

Mechanisms that minimise the formation of voids in the polymer come from: void compression, dissolution and diffusion of gas molecules from voids, as well as in-plane dissolution, as confirmed in the findings of Swamy et al (72). These mechanisms were not as effective for CF/PPS laminates possibly due to material properties, including the high fibre volume fraction of the prepreg, the high viscosity of PPS polymer, and prepreg age as prepreg was manufactured in 2013. Due to a phenomenon related to the tortuosity of fibres (54), the two factors could have elevated the effect of local tortuosity on impeding melt polymer flow leading to entrapment of air, and subsequent creation of voids. The effect observed by Gomasasca et al (54) was found when examining another high-temperature prepreg tape in CF/PEEK and the findings found a correlation between tortuosity and void fraction. As will be shown in the CT analysis in the following section, there is evidence to suggest this mechanism was a significant contributor.

Other points of consideration include the higher number of laminates in the CF/PPS configurations compared to CF/PA12 laminates. This would have introduced more opportunities for entrapped air between plies. Next, there is also the possibility of uneven heating as seen in Figure 24 with the top and bottom heating plate each exhibiting a difference in temperature of 32 °C and 53.7 °C respectively. Colder regions would have higher viscosity restricting flow, while hotter regions would exhibit reduced viscosity, potentially causing preferential resin flow and localised resin-starved zones. Given the lower maximum temperature than expected of 315 °C recorded by the thermocouple, the temperature could have been quite low in some local regions. The contribution to the differences in manufacturing however is difficult to quantify without knowing what the temperatures were like, especially given CF/PA12 did not face these issues and that the thick caul plates used likely distributed the uneven heating.

To conclude this section, we have examined the approach used here to acquire C-scans of thermoplastic laminates, analysed the laminate quality, identified potential issues and understood what mechanisms could have driven these issues. The primary driver of defects in CF/PPS laminates was deduced to be material-specific rather than process-setup related, as supported by the successful consolidation of low-porosity CF/PA12 laminates under identical conditions. However, quantifying this requires further research by employing a systematic statistical approach (i.e. Taguchi method) as seen in the optimisation of other high-temperature TP laminates such as CF/PEEK (73).

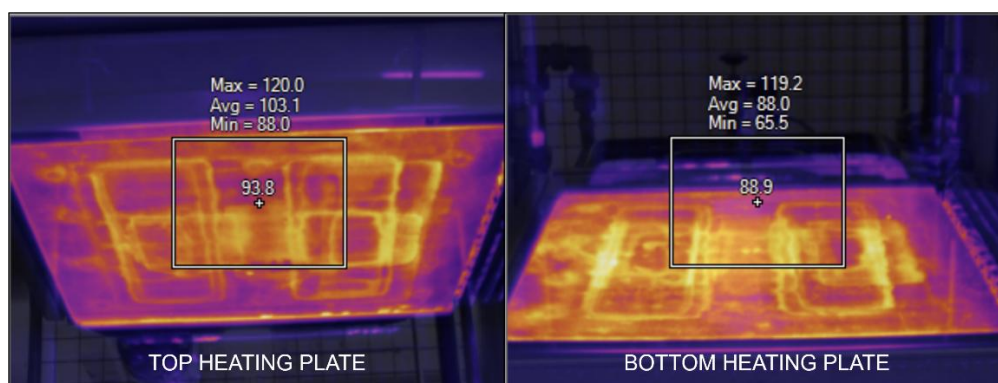


Figure 24. Hofer HLP40 hydraulic laboratory press temperature distribution.

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4.2.4 Analysis of Micro-Computed Tomography (μ CT) Scans

Through processing of computed tomography (μ CT) scans, reconstruction and thresholding allowed identification of the porous geometry within as shown in Figure 25. A section of the bottom-right corner of PPS-000-S1 laminate that indicated 0-4 dB was cut and used for inspection. Measuring 20mm (X) x 10 mm (Y) x 2 mm (Z), the top half of this specimen was imaged after attaching it to the specimen holder. Using Dragonfly Pro, appropriate thresholds were then applied to denoise and isolate the features of interest, the cross-section of the plane normal to the x -axis was set up and identified as XX. This was followed by the cutting planes normal to the y -axis (YY) and z -axis (ZZ) were then placed, allowing capture of the following images.

Examining the annotated section XX, we can first observe clear presence of a macrovoid measuring of approximately 1 mm in width with sharp linear features extending to both the top and bottom surfaces, consistent with through-thickness void channelling. There is also presence of a smaller void located towards the right edge of the laminate with smaller dimensions, extending towards the right edge. This cross-section also appears to highlight what seems to be microvoids scattered on the left side, although given the imaging resolution and the geometric unsharpness introduced by the specimen's proximity to the source, these may be imaging artefacts rather than true microstructural features (74). Geometric unsharpness is a core consideration when conducting μ CT scans at high-resolution of these specimens and during capture, this was not minimised as the specimen was placed close to the source instead of the detector. As a result, significant denoising and thresholding may have created these micro artifacts. In most practices, it is ideal to minimise object-to-detector distance for feature clarity.

Moving to the inspection of section ZZ, we can now clearly observe that the macrovoid extends through the whole length of the specimen, showing evidence of the columnar behaviour of the voids observed in UT C-scans. There is also clear fibre-misalignment away from the y -axis shown in section YY, where one of the plies appears to be displaced out-of-plane, with voids present above and below. This is consistent with ply-level waviness or tortuosity, and the co-location of voids and ply displacement suggests a coupled mechanism whereby fiber tortuosity impedes melt flow, promoting void entrapment, which in turn removes lateral support and allows further ply displacement.

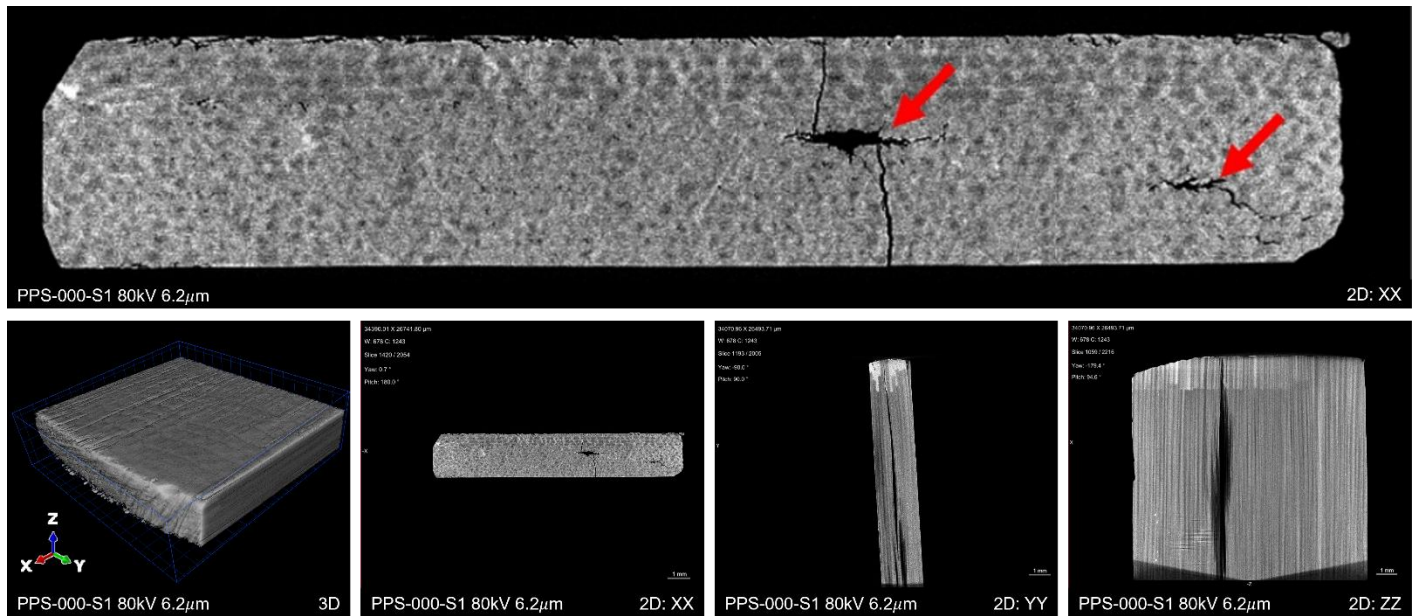


Figure 25. Computed tomography scans of PPS-000-S1 sample of 0-4dB region.

4.2.5 Analysis on Optical Microscopy

Optical microscopy is often carried out to characterise the microstructure of UD composite laminates. This technique complements the previous analysis by providing detailed high-resolution imaging of the cross-section and enables observation of fibre distribution, resin-rich regions, and interfaces. A key advantage of optical microscopy is its cost-effectiveness; it requires less labour-intensive sample preparation than SEM and significantly shorter acquisition times than μ CT scans, making it well-suited for screening both CF/PPS and CF/PA12 configurations.

Figure 26 presents the optical micrographs of each laminate configuration, with annotations identifying key microstructural features: resin-rich regions indicated by *arrows*, voids by *dotted outlines*, and fibre-dense regions by *dashed outlines*. Examination of the cross-sectional morphologies confirms that the fibre orientations are consistent with the intended layup configurations: 0° fibres appear as circular cross-sections, $\pm 45^\circ$ fibres as ellipses, and 90° fibres as elongated horizontal features, collectively providing confidence that the laminates were manufactured to the correct fibre orientations.

Voids are observed in both PPS-000-S1 and PPS-090-S2, though they differ notably in their size, morphology, and spatial distribution. In PPS-000-S1, voids exhibit irregular and sometimes elongated geometries, with equivalent diameters ranging from approximately 100 to 250 μm , as indicated in the higher-magnification images in the bottom right. These larger voids are not uniformly distributed throughout the laminate but concentrated within a localised horizontal band spanning multiple plies, suggesting that void formation in this configuration is associated with a specific region of inadequate consolidation, similar to the phenomenon mentioned by entrapped gas during processing. In contrast, voids observed in PPS-090-S2 are considerably smaller, approximately 50 μm in diameter, and are more randomly distributed across multiple plies rather than confined to a discrete zone. This difference in void characteristics between the two configurations may reflect the influence of fibre orientation on resin flow behaviour during consolidation, where the 90° fibre architecture offers different resistance to matrix flow and gas evacuation compared to the 0° configuration. This likely also explains the zonal characteristics of porosity mentioned earlier.

Resin-rich regions are also observed predominantly at ply interfaces and between fibre bundles, appearing as flat matrix-dominant zones devoid of fibres. These are most prominent in PPS-045-S2 configuration, where they form a systematic arrangement, visible in the particularly visible within the 0° plies as localised pockets, with fibre-dense regions in between. These fibre-dense regions can be identified in the 0° cross-sections of both PPS and PA12 laminates, appearing as darker regions of tightly packed clusters of fibre cross-sections with minimal matrix present. The formation of such regions may be attributed to slow cooling rates used. Similarly, Emon et al (75) used ambient cooling during their manufacturing of glass-fibre polypropylene TP laminates and observed the same behaviour, suggesting that fibres may have slowly moved closer together during the slow cooling phase. Fibre-dense regions appear more prevalent in the PA12 laminates, suggesting that despite its superior consolidation behaviour relative to PPS, localised fibre bundling remains a manufacturing challenge.

Overall, there are three key conclusions from this analysis. Through imaging the cross-sections, visual evidence of severe porosity was observed; resin-rich regions were identified to be present in all samples which were more pronounced in PPS laminates; and fibre-dense regions were identified. While defects like voids and resin-rich regions do not directly reduce fibre volume fraction at the macro-scale, they may represent reduced local stiffness and result in preferential sites for matrix cracking and interlaminar delamination under mechanical loading.

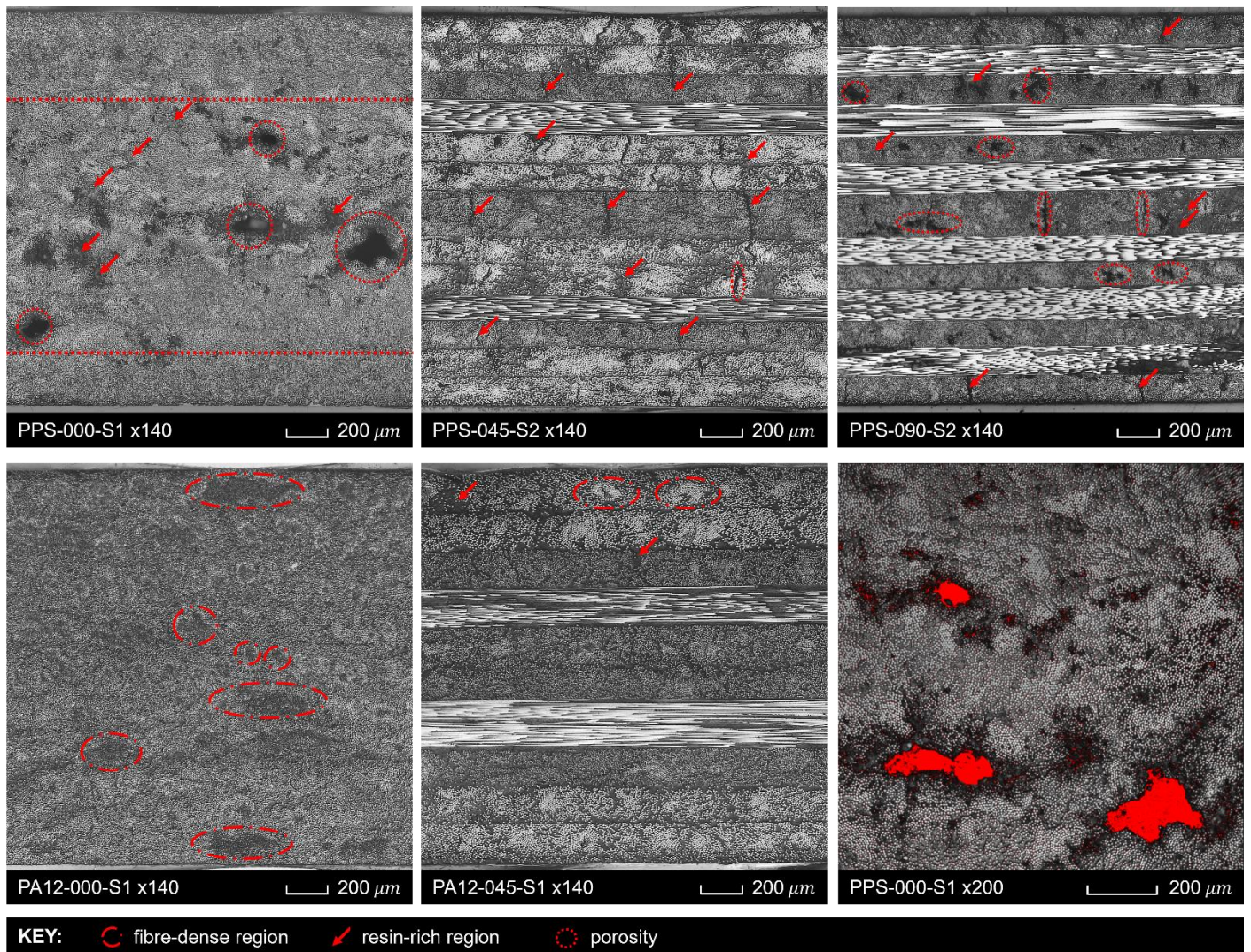


Figure 26. Comparison of optical micrographs of cross-section with annotations on key observations.

4.2.6 Analysis on Scanning Electron Microscopy (SEM)

Scanning electron microscopy (SEM) was carried out to characterise the surface morphology of thermoplastic composite laminates at high resolution. This technique complements optical microscopy and μ CT work by providing microscale topographical detail of the laminate surface, enabling direct observation of fibre exposure, matrix coverage, and surface texture as a function of surface preparation method. This is useful as it informs subsequent measurements on bulk electrical conductivity and results collected from flexural testing.

As shown in the SEM micrographs in Figure 27, a clear distinction is evident between the uncut and waterjet-cut surfaces. The uncut CF/PPS surface presents a smooth, consolidated morphology in which the carbon fibres are largely obscured beneath a continuous PPS polymer layer, with only minor surface defects such as small voids evident. In contrast, the waterjet-cut surface exhibits a markedly disrupted topography, characterised by severely fragmented PPS matrix material in the form of displaced polymer flakes, some of which appears to have reattached to the exposed fibre-ends. The waterjet-cut surface also presents significant exposure of the underlying carbon fibres, and a notably greater surface roughness as indicated by the depth of surface relief (vertical distance between the highest and lowest parts visible in the micrograph).

Overall, the uncut surface presents a relatively smooth, resin-dominated morphology with minimal fibre exposure, whereas the waterjet-cut surface exhibits significant mechanical disruption, with fragmented matrix material alongside exposed carbon fibres. This difference in surface condition is of relevance to in-plane bulk electrical conductivity measurements, as the degree of fibre exposure at the measurement interface directly governs the quality of electrical contact established and therefore electrical conductivity measurements. Flexural properties may be influenced by these results, as some specimens were tested with uncut surfaces due to material limitations.

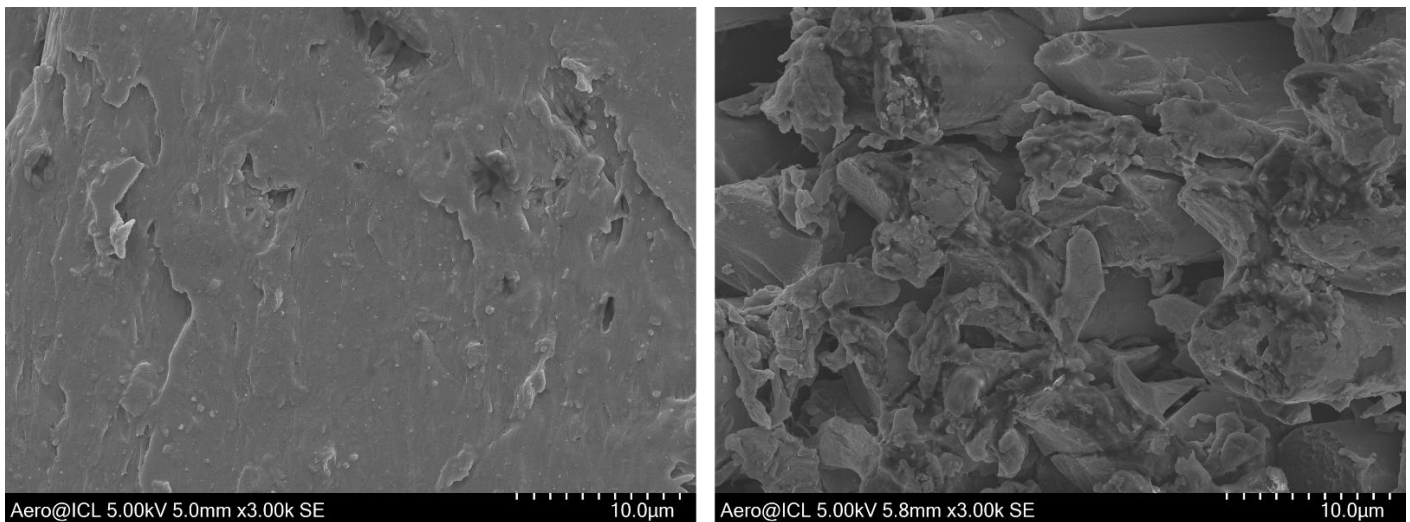


Figure 27. Selective electron microscopy images of CF/PPS laminate side-profile, before and after waterjet-cutting.

4.3 Physical, Mechanical, and Electrical Properties

4.3.1 Density and Porosity

The average, standard deviation (SD), and coefficient of variance (COV) were calculated for density and averaged across three specimens. Overall, PPS-090 emerges as the most consistent and well-consolidated laminate configuration, while the high variance observed in the PA12 data warrants cautious interpretation and further investigation. The results presented are broadly consistent with the observations derived from the UT C-scans, though the contribution of macrovoids was found to be somewhat overestimated in the case of PPS-000. While more advanced methodologies, such as acid digestion and detailed μ CT volume scans, are employed in research for porosity characterisation, the approach presented here was adopted as it provided a practical and cost-effective alternative for estimating the void content within the laminate configurations studied.

presents the density and porosity measurements for the five laminate configurations studied, quantifying notable differences observed between CF/PPS and CF/PA12 laminates. The CF/PPS laminates demonstrated consistently higher densities, 1.5334 to 1.5423 g/cm³, compared to the CF/PA12 laminates, which recorded lower densities between 1.3386 and 1.3496 g/cm³ respectively. Among the PPS variants, PPS-090 exhibited the highest density and the lowest coefficient of variation (COV) of 0.13%, matching observations of the significant low porosity zones observed in UT C-scans. In terms of porosity, PPS-045 recorded the highest void volume fraction with a COV of 14.71%, suggesting that the high porosity regions observed in UT C-scans played a major role, overall contributing to a total porosity higher than PPS-000 specimens despite the significant macrovoids observed. The PA12 laminates, although exhibiting exceptionally low absolute void fractions of around 1.5737% and 0.7619%, presented exceptionally high COV values of 78.00% and 93.62%, respectively, rendering these measurements statistically unreliable for direct comparison.

Table 7. Experimental density and porosity of each laminate configuration.

Material	ρ	SD	COV	v_{void}	SD	COV
-	(g/cm ³)	(g/cm ³)	(%)	(%)	(g/cm ³)	(%)
PPS-000	1.5405	0.0040	0.26%	3.7203	0.2528	6.80%
PPS-045	1.5334	0.0098	0.64%	4.1641	0.6124	14.71%
PPS-090	1.5423	0.0020	0.13%	3.6050	0.1263	3.50%
PA12-000*	1.3386	0.0167	1.25%	1.5737	1.2275	78.00%
PA12-045*	1.3496	0.0097	0.72%	0.7619	0.7133	93.62%

* Measurements made with IP conductivity specimens.

It should be noted that PA12 measurements were obtained using small in-plane (IP) conductivity specimens, which may have introduced systematic bias into both the density and porosity readings. Such bias could have arisen from user error and variations in the parallelism of opposite surfaces, thereby contributing to measurement uncertainties. Furthermore, the relatively smaller dimensions of the IP conductivity specimens meant that any localised measurement errors or material defects were likely to have a proportionally greater influence on the overall readings, compared to the larger PPS laminate specimens where such effects would be more effectively averaged out. In the calculations for porosity or void fraction, it was also assumed that variation of nominal prepreg density provided by manufacturers was negligible and that the nominal porosity for the prepreg was also assumed as zero and treated as the theoretical limit of the prepreg. This is in line with standard practice as outlined in ASTM 2734.

Overall, PPS-090 emerges as the most consistent and well-consolidated laminate configuration, while the high variance observed in the PA12 data warrants cautious interpretation and further investigation. The results presented are broadly consistent with the observations derived from the UT C-scans, though the contribution of macrovoids was found to be somewhat overestimated in the case of PPS-000. While more advanced methodologies, such as acid digestion and detailed μ CT volume scans, are employed in research for porosity characterisation, the approach presented here was adopted as it provided a practical and cost-effective alternative for estimating the void content within the laminate configurations studied.

4.3.2 Flexural Stiffness and Flexural Strength

The three-point bending flexural tests yielded results for flexural stiffness (E_B) and maximum flexural strength (σ_B), which were calculated in correspondence with ASTM D790. To calculate this, flexural strain (ε_B) was found using Eq. (11); maximum σ_B was then found using the error-corrected formula for high span-to-thickness ratios, seen in Eq. (12). The corrected version of the equation including additional terms helped ensure validity of the results by accounting for large deflections and compliance within the fixture. The maximum was automatically selected and manually inspected to obtain σ_B . Here, D is the deflection measured by the machine, t is the thickness of the material, L represents the support span and P represents the force applied by the loading nose.

$$\varepsilon_B = \frac{6Dt}{L^2} \quad (11)$$

$$\sigma_B = \frac{3PL}{2wt^2} \left(1 + 6 \left(\frac{L}{S} \right)^2 - 4 \left(\frac{Lt}{S^2} \right) \right) \quad (12)$$

Following this, E_B was calculated using the chord modulus methodology seen in Eq. (13). This approach uses the gradient of the force-displacement slope from the range between $\varepsilon_B = 0.1\%$ and $\varepsilon_B = 0.3\%$; corresponding to force values, $P_{0.3\%}$ and $P_{0.1\%}$ and displacement values $\Delta_{0.3\%}$ and $\Delta_{0.1\%}$. To avoid errors, values were automatically obtained using an automated index and match approach in MS Excel, the force-displacement region selected was then inspected and ensured to be linear.

$$E_B = \frac{L^3}{4wt^3} \cdot \frac{P_{0.3\%} - P_{0.1\%}}{\Delta_{0.3\%} - \Delta_{0.1\%}} \quad (13)$$

The stress-strain plots from flexural testing are presented in Figure 28. Averaged results of each configuration are represented in Figure 29 and detailed in Table 8. Inspecting the results, all specimens exhibited stable linear elastic response, followed by non-linearity prior to fracture. The average longitudinal (0°) flexural stiffness of PPS-000 was found to be 107.97 ± 4.30 GPa, with flexural strength of 1570.4 ± 229.0 MPa. In contrast, the average longitudinal flexural strength of PA12-000 was found to be 108.65 ± 4.73 GPa, with flexural strength of 1162.6 ± 110.7 MPa.

While not reported by Toray, the results on PPS-000 are comparable to an equivalent CF/PPS prepreg material from TOPOLOGROUP (76). Using the same ASTM D790 standard, TOPOLOGROUP reported a longitudinal flexural stiffness of 120 GPa with a longitudinal flexural strength of 1500 MPa. The formulation uses T700 carbon fibres from Toray with an unnamed PPS resin matrix and achieves similar fibre volume fractions of $\sim 66\%$, fibre areal weight of $145\text{-}250$ g/m², and density of 1.57 g/cm³. Assuming this was manufactured using the T700G variant, T700 carbon fibres have tensile modulus of 240 GPa, tensile strength of 4900 MPa, and failure strain of 2.0% (77).

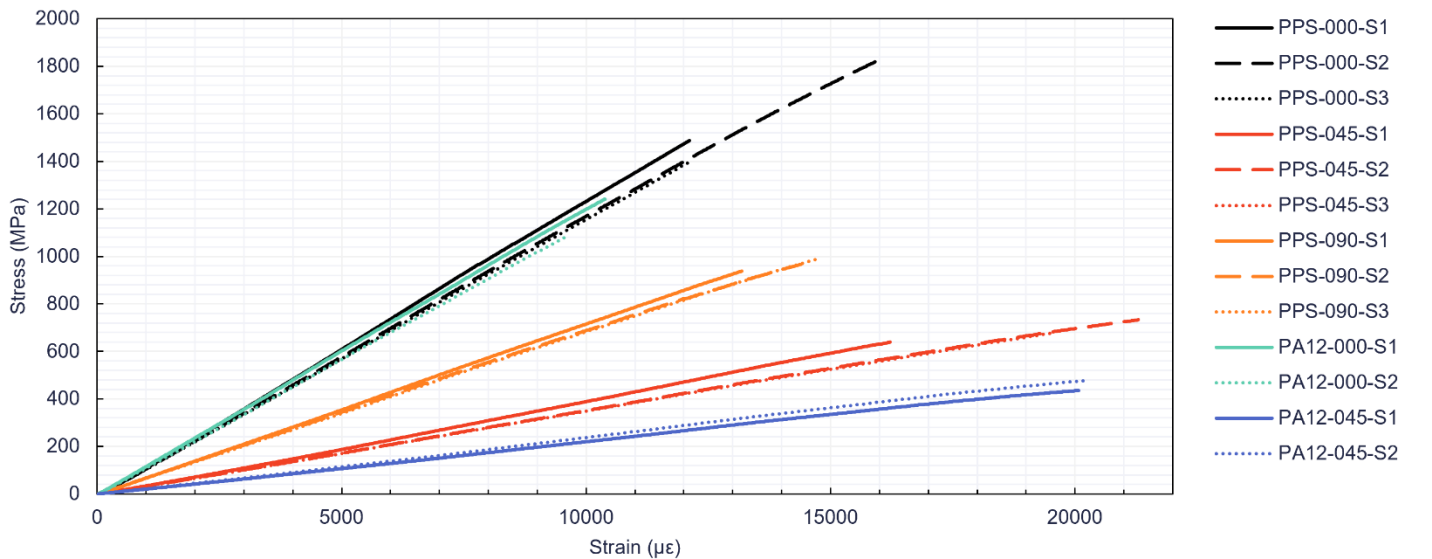


Figure 28. Plots of stress-strain curves flexural tests of each specimen configuration.

In comparison, AS4A carbon fibres used here has a lower tensile modulus of 231 GPa, tensile strength of 4413 MPa, and failure strain of 1.7% (78). It should be also be noted that specimens were tested with a span of 100 mm instead of the specified 120 mm, resulting in fracture occurred near the top surface under compression instead of the bottom surface in tensile. In addition, there was also the clear presence of porosity as noted in previous analysis. However, these three contributing factors did not lower the mechanical properties, instead, they remained comparable despite the differences. There are two possible explanations for this. Firstly, the significant resin squeeze-out observed would have increased fibre volume fraction and required additional force to fracture. Secondly, the slow cooling rate identified could have influenced the crystallinity and therefore mechanical properties. This was confirmed by Oshima et al using the same AS4A/TC1100 CF/PPS prepreg material laid up as $[\pm 45]_{3S}$, they observed an increase of 2.9% in tensile strength when decreasing from a cooling rate of 10 °C/min to 1 °C/min (79).

The results on PA12-000 are higher than reported from Evonik (51). Using the ISO 14125 standard, they reported a longitudinal flexural stiffness of 95 GPa and flexural strength of 900 MPa. The main differences between ISO 14125 and ASTM D790 method is the use of lower span-to-thickness ratio of 40 and lower cross-head speed of 5.3 m/s. The former in failure in compression of the top side of the specimen and viscoelastic effects that can lead to lower stress values. However, as mentioned, the higher mechanical properties of the specimens here could also been due to the slower cooling rate used compared to rates used by Evonik, which may have been up to 20 °C/min (51). In addition, this qualification test only included two specimens, which may have contributed the different values observed here, given the influence of testing standards on results.

Investigating the performance of the other configurations, we can observe that PPS-045 has an average longitudinal flexural stiffness of 32.2 ± 1.43 GPa and flexural strength of 683.2 ± 47.1 MPa, unlike PPS-090 which exhibited higher average longitudinal flexural stiffness of 63.0 ± 1.44 GPa and higher flexural strength of 966.8 ± 27.8 MPa. Conversely, PA12-045 exhibited much lower properties than the two, with longitudinal flexural stiffness of 20.3 ± 0.97 GPa and flexural strength of 456.3 ± 28.6 MPa. These were due to laminate stack effects, which can be demonstrated by calculating the theoretical stiffness values using classical laminate theory (CLT).

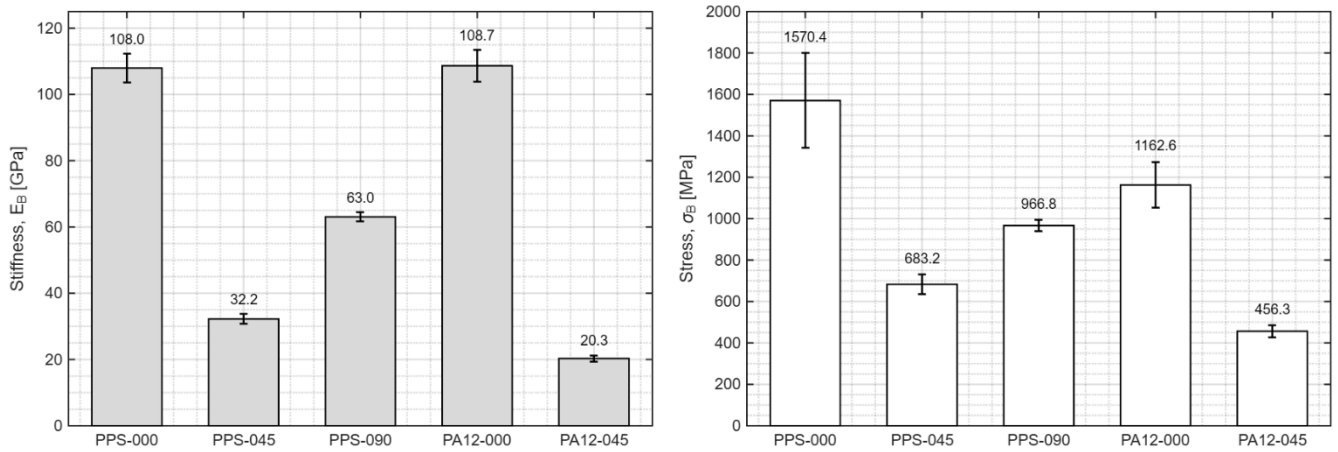


Figure 29. Overview of flexural stiffness and flexural strength of each laminate configuration.

Table 8. Flexural stiffness and flexural strength of each laminate configuration.

Material	$E_{B,theory}$	E_B	SD	COV	σ_B	SD	COV
-	(GPa)	(GPa)	(GPa)	(%)	(MPa)	(MPa)	(%)
PPS-000*	124.37	107.97	4.30	3.99%	1570.4	229.0	14.58%
PPS-045*	38.34	32.23	1.43	4.43%	683.2	47.1	6.90%
PPS-090*	79.87	63.04	1.44	2.28%	966.8	27.8	2.88%
PA12-000	100.00	108.65	4.73	4.35%	1162.6	110.7	9.52%
PA12-045	24.48	20.30	0.97	4.77%	456.3	28.6	6.26%

* These specimens were tested with span = 100 mm.

To calculate the theoretical stiffness, this was achieved using Laminate Analysis Program (LAP) v4.1 from Anaglyph Ltd. The nominal CF/PPS prepreg stiffness values were used, including longitudinal elastic modulus of 124 GPa, transverse elastic modulus of 10 GPa, in-plane shear modulus of 3.5 GPa, and Poisson's ratio of 0.33. On the other hand, the nominal CF/PA12 prepreg stiffness values were used, with longitudinal elastic modulus of 100 GPa, transverse elastic modulus (assumed) of 10 GPa, in-plane shear modulus of 1.4 GPa, and Poisson's ratio of 0.32.

Using the ABD stiffness matrix in Eq. (14), the primary longitudinal bending stiffness element (D_{11}) was then extracted. Here, A_{ij} represents the in-plane extensional stiffness values, B_{ij} represents coupling between in-plane forces and out-of-plane curves, and D_{ij} represents the bending stiffness of the laminate. All laminates are balanced and symmetric, as such B_{ij} is equal to zero.

$$ABD = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \quad (14)$$

From here, $E_{B,theory}$ can then be calculated using Eq. (15). Here, t represents the thickness of the laminate.

$$E_{B,theory} = \frac{D_{11}t^3}{12} \quad (15)$$

Theoretical stress values were also evaluated using CLT, but the predicted maximum bending strength was far lower than the experimental results. This is because three-point bending introduces mixed-mode loading with significant shear, whereas CLT assumes pure bending and cannot capture shear-driven failure. As a result, CLT-based strength values were not included in the final analysis.

4.3.3 Bulk Electrical Conductivity

To calculate the bulk electrical conductivity, volume resistivity (R) was measured and then converted to conductivity (σ) as specified in ASTM D4496. These values were calculated using Eq. (16), where L is the length of the electrical current path between electrodes, R is the measured DC electrical resistance, t is the thickness of the specimen, and w is the width of the specimen. Variables L , w , and t were different depending on the orientation of the specimen. The results are presented in Figure 30 and detailed in Table 9.

$$\sigma = \frac{L}{RWt} \quad (16)$$

Inspecting the longitudinal bulk conductivity values, CF/PPS consistently outperforms CF/PA12 across all three measurement directions, with the performance gap widening as the measurement direction moves away from the fibre axis. In the in-plane longitudinal direction (IP-0), PPS-000 achieves 3522.1 ± 500.7 S/m, as opposed to 1210.0 ± 207.1 S/m for PA12-000. On the other hand, PPS-090, PPS-045, and PA12-045, measured lower values of 2131.0 ± 55.9 S/m, 2588.3 ± 137.7 S/m, and 565.4 ± 135.4 S/m respectively. In all specimens, these results are in the same order of magnitude when done using carbon/epoxy laminates. These results correctly reflect the dominance of direct fibre-to-fibre conduction in this direction, as the CF/PPS was manufactured with a significantly higher fibre volume fraction of 59% compared to 45% used in CF/PA12. There is also the influence of current shortages in between fibres as the distance between fibres is closed, reducing resistance and therefore conductivity; as evidenced by the observed tortuosity observed previously. As such, despite the higher porosity of CF/PPS laminates compared to CF/PA12, voids, there is some evidence here that the macrovoids may have encouraged the fibres to move one another and therefore improved the conductivity of the laminates. There also may be the possibility that this could have been due to variations in carbon fibre type, which can significantly vary between different manufacturers and grades (36).

Inspecting the values in the transverse direction, these values are significantly lower. PPS-000 and PA12 shows a dramatic reduction down to 8.4 ± 0.7 S/m and 0.7 ± 0.1 S/m respectively as the values observed here in essence measures the conductivity of the highly-insulating bulk polymer. It can also be seen that the conductivity drops for PPS-045, PPS-090 and PA12-045, as there are less longitudinal fibres exposed in the measurement direction. However, at 876.9 ± 52.6 S/m this outlier is likely an effect of aspect ratio as the in this orientation, transverse fibres can successfully remain in contact from one-side to the other due to the flat geometry.

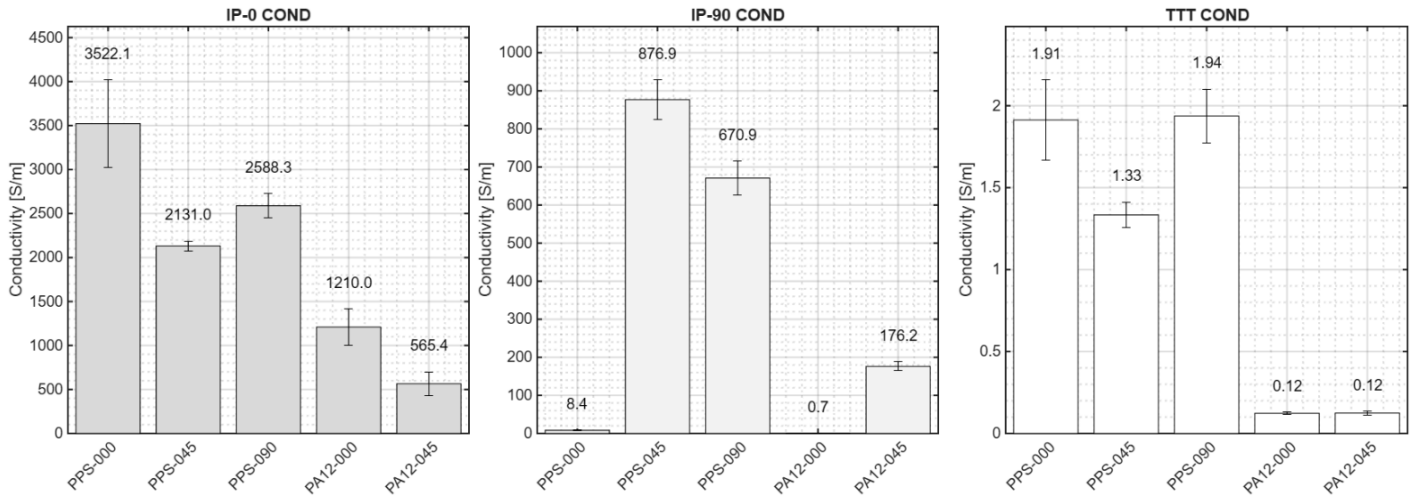


Figure 30. Overview of bulk electrical conductivity of each laminate configuration.

Table 9. Bulk electrical conductivity of each laminate configuration.

Material	$\sigma_{ }$	SD	COV	$\sigma_{\perp}$	SD	COV	σ_{tt}	SD	COV
-	(S/m)	(S/m)	(%)	(S/m)	(S/m)	(%)	(S/m)	(S/m)	(%)
PPS-000	3522.1	500.7	14.22%	8.4	0.7	8.69%	1.91	0.25	12.89%
PPS-045	2131.0	55.9	2.63%	876.9	52.6	6.00%	1.33	0.08	5.84%
PPS-090	2588.3	137.7	5.32%	670.9	44.6	6.65%	1.94	0.16	8.47%
PA12-000	1210.0	207.1	17.12%	0.7	0.1	20.00%	0.12	0.01	6.65%
PA12-045	565.4	134.4	23.78%	176.2	11.8	6.71%	0.12	0.01	8.51%

There is also the effect of field lines whereby multi-direction layups generally have an impact on electric field lines. Ideally parallel within the geometry, current paths should be parallel, however layups result in a redistribution of these pathways into elliptical patterns between two electrodes (27,80). With this same characteristic, Mikinka (27) observed that with an increase in size of the specimen used, this reduces the impact of this effect, as such, size of measurements may be made with larger samples, with high-aspect ratio laminates oriented with the longer dimension between electrodes.

Pivoting towards through-thickness results, it can be seen PPS-000, PPS-045, and PPS-090 all centre around similar values, which are about an order of magnitude larger than PA12-000 and PA12-045. In this direction, inter-ply polymer layers play a major role in defining the actual conductivity due to the gaps in between the two electrodes. However, the aforementioned interactions between fibres likely surpass this impact, which may be why CF/PPS results are significantly higher in contrast to CF/PA12 specimens, which are in-line to literature in the order of 0.1 S/m in through-thickness values reported for most conventional thermoset CFRP laminates in literature (81,82). However, in one study, Mikinka was able to measure up to over 20 S/m (27).

Exploring the systematic level of the approach used here, one crucial parameter of influence is the impact of Holm's general electrical contact resistance theory. As illustrated in Figure 31, the theory describes that when two or more electrically conductive materials pressed together, contact resistance arises at the interface of the two as current passes through (83). This arises due to the presence of surface irregularities in the form of microscopic peaks and troughs, resulting in contact only happening at discrete locations (also referred to as asperity spots or a-spots) that restrict current flow. For UD composite materials, research has noted that these a-spots play a significant role in measuring electrical conductivity in anisotropic carbon/epoxy composites (27,39).

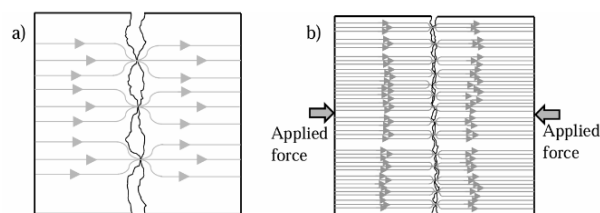


Figure 31. Influence of pressure over number of asperities in contact. (a) no pressure applied, (b) pressure applied. Reproduced from (27)

4.4 Electromagnetic Shielding Effectiveness and Wider Perspective

4.4.1 Electromagnetic Shielding Effectiveness

During the post-processing of data on the electromagnetic shielding effectiveness (SE), especially when measuring materials with high-shielding effectiveness, it is common to observe high-frequency noise due to low-measured power (84). To address this, studies apply an appropriate Savitzky-Golay (S-Golay) filter to eliminate high-frequency noise during acquisition, allowing improved identification of trends or modes (84–86). For example, in the study by Evers et al, their study applied a 10-point S-Golay filter on SE values for CFRP composite laminates that were achieving 80–120 dB (84). However, it is important to acknowledge that the S-Golay filter relies on the ordinary least squares equation (87) that operates on assumptions of linear parameters. This assumption implies that the S-Golay filter should be applied to linear magnitude S-parameters prior to conversion to SE values to prevent bias.

Considering this, the final algorithm adopted was a third-order seven-point ($m = 3, fl = 7$) S-Golay filter applied to the S_{11} and S_{21} linear magnitudes calculated from S-parameters, before converting the filtered data to SE values using Eq. (5-7). Final selection of this approach involved developing two algorithms and comparing the two approaches to test the hypothesis of potential bias. Through this comparison, it was found that applying the same S-Golay filter directly to SE results generally resulted in average SE_T values being biased upwards by up to +1 dB.

To demonstrate the approach used, the algorithm was applied to the aluminium reference. Figure 32 presents the results of the S-Golay filter when applied to S-parameters and compares it to the alternative approach mentioned previously. Focusing on the comparison between the two approaches (center and right), both implementations removed high-frequency noise in the form of extreme sharp peaks. However, the final approach appears to have reduced sharp peaks more than the alternative algorithm, as seen near 2.25 GHz and 2.5 GHz. This likely played a critical role in affecting the average response of SE_T which was more conservative at 75.188 dB in comparison to 76.264 dB in the alternative approach. For reference, the average SE_T of the raw data was found to be 76.261 dB.

The preliminary evidence presented here suggest the final approach reduced noise, suppresses extreme peaks well, returning conservative results. As such, the results from the final approach will be reported. However, future work is necessary to further validate this approach by conducting further studies on other materials, frequency ranges and standards.

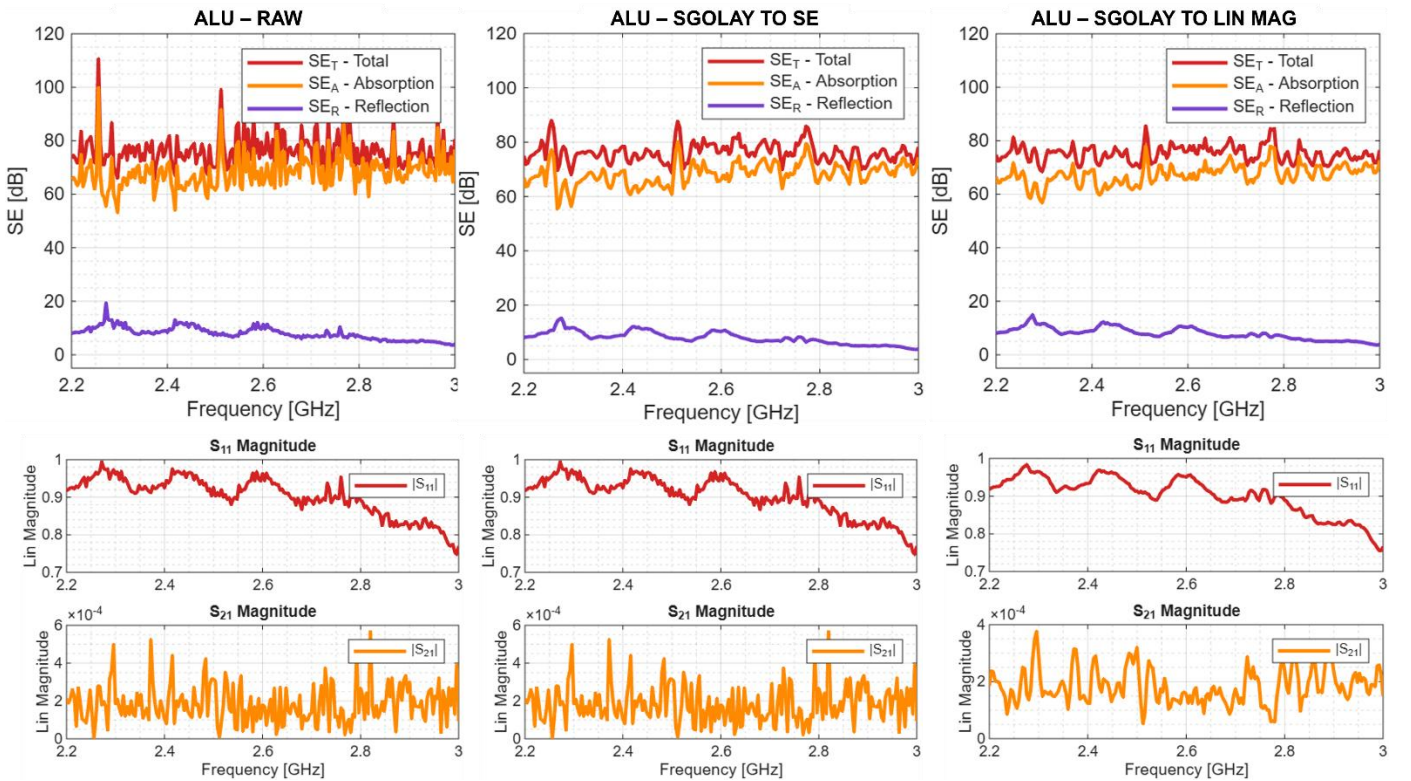


Figure 32. Electromagnetic shielding effectiveness data processing approaches.

From left to right: raw data; alternative approach; and final approach.

The rest of this section focuses on the electromagnetic shielding effectiveness of the laminate configurations investigated in this study. Figure 33 presents the spectrums recorded from each configuration, reflecting the different materials used in each study. Figure 34 presents the averaged shielding effectiveness results across each configuration. Likewise, Table 10 shows the corresponding standard deviation and variance across the configurations investigated. R&S ZVL3 VNA dynamic range measures up to 115 dB for 20 MHz to 3 GHz frequencies measured (88), as such the values here are valid.

Analysing the results presented, the average total shielding effectiveness ($SE_{T,avg}$) of PPS-000 was found to be 17.076 ± 0.275 dB, with an average absorbed sound energy ($SE_{A,avg}$) of 11.892 ± 0.274 dB and an average reflected sound energy ($SE_{R,avg}$) of 5.184 ± 0.036 dB. Similarly, PA12-000 recorded a $SE_{T,avg}$ of 12.703 ± 0.436 dB, with $SE_{A,avg}$ of 9.428 ± 0.499 dB and $SE_{R,avg}$ of 3.276 ± 0.456 dB. Both of these laminates had high in-plane longitudinal electrical conductivity and theoretically should have correlated to high shielding effectiveness results, however the plies here are oriented in the x-direction, resulting in a ‘polarisation’ effect. As noted in Section 2.2.1, this ultimately leads to weaker shielding effectiveness. Given the electric fields are oriented in the y-direction, this means that the higher transverse electrical conductivity of PPS-000 increased the shielding effectiveness. The higher number of plies in PPS-000 may also have played a role in delivering this higher shielding effectiveness. Higher number of plies here may mean more opportunities for electrons to be conducted away from the transverse electric field.

Moving onto the other configurations, PPS-045 showed a $SE_{T,avg}$ of 73.812 ± 1.474 dB, with $SE_{A,avg}$ of 62.886 ± 1.927 dB and $SE_{R,avg}$ of 10.926 ± 0.452 dB. Comparatively, PPS-090 exhibited a $SE_{T,avg}$ of 74.158 ± 1.814 dB, $SE_{A,avg}$ of 64.057 ± 1.886 dB, and $SE_{R,avg}$ of 10.101 ± 0.208 dB. In contrast, PA12-045 yielded a $SE_{T,avg}$ of 65.576 ± 0.319 dB, $SE_{A,avg}$ of 57.055 ± 2.988 dB, and $SE_{R,avg}$ of 8.521 ± 2.798 dB. The three laminates here exhibited significantly improved higher shielding effectiveness values compared to the unidirectional configurations. These values are similar to the shielding effectiveness observed by numerous studies for the same thickness (16,36). The effect of the electric fields interacting with PPS-090 more than PPS-045 due to the parallel lines appears to be true here, which matches the findings from literature (16,38). Similarly, the measured lower conductivity of PA12-045 is evidence of the decrease of 8.58 dB from numbers found for PPS-045.

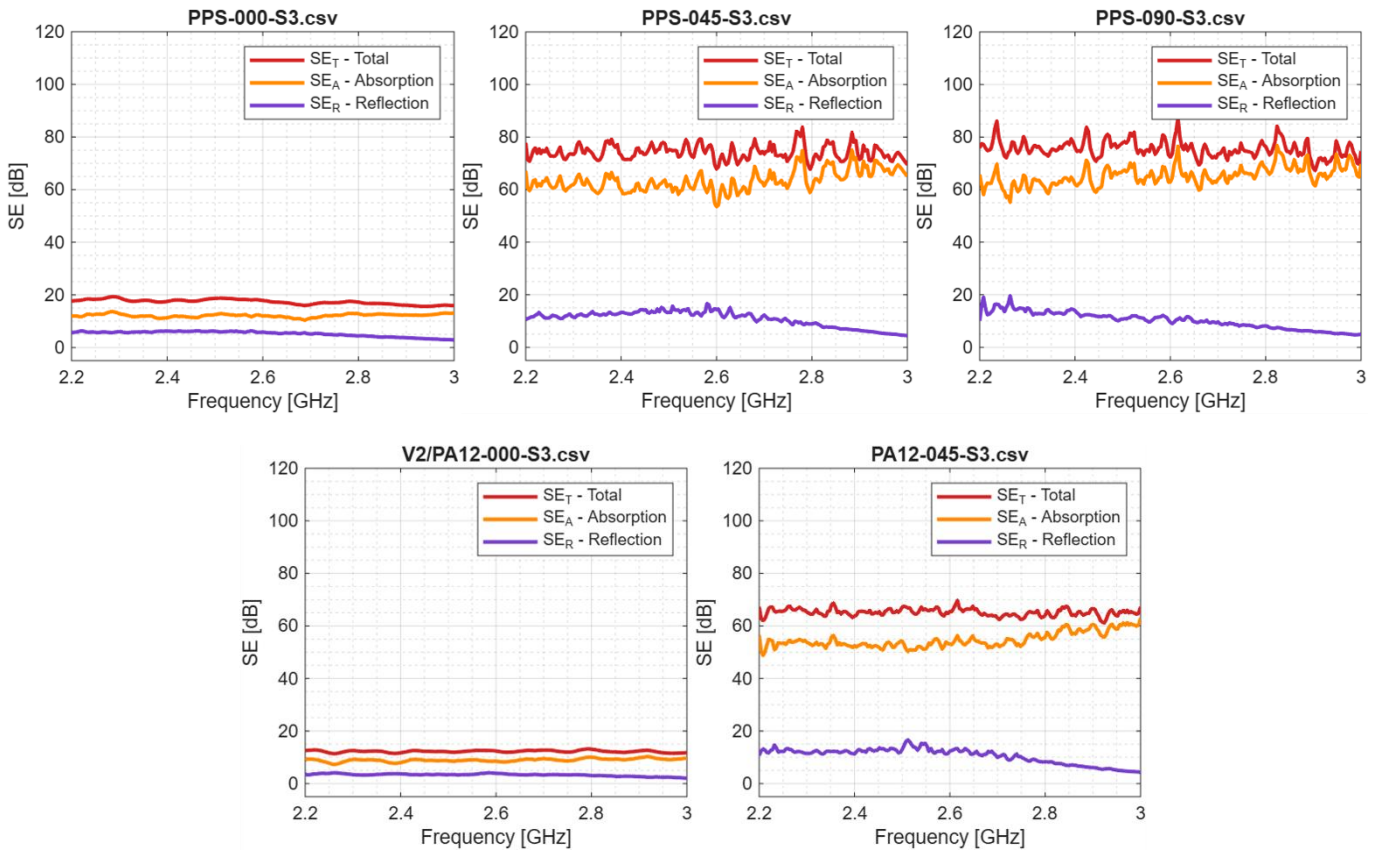


Figure 33. Comparison of electromagnetic shielding effectiveness spectrums of each laminate configuration.

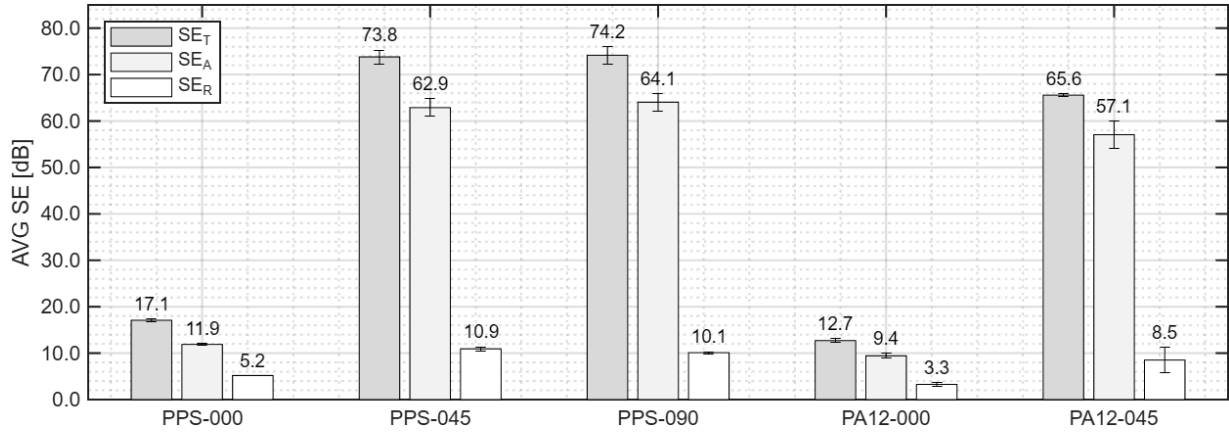


Figure 34. Overview of averaged total electromagnetic shielding effectiveness (2.2 – 3 GHz) of each laminate configuration.

Table 10. Electromagnetic shielding effectiveness (2.2 – 3GHz) of each laminate configuration.

Material	$SE_{T,avg}$	SD	COV	$SE_{A,avg}$	SD	COV	$SE_{R,avg}$	SD	COV
-	(dB)	(dB)	(dB)	(dB)	(dB)	(dB)	(dB)	(dB)	(dB)
PPS-000	17.076	0.275	1.61%	11.892	0.274	2.31%	5.184	0.036	0.70%
PPS-045	73.812	1.474	2.00%	62.886	1.927	3.06%	10.926	0.452	4.14%
PPS-090	74.158	1.814	2.45%	64.057	1.886	2.94%	10.101	0.208	2.06%
PA12-000	12.703	0.436	3.43%	9.428	0.499	5.30%	3.276	0.456	13.93%
PA12-045	65.576	0.319	0.49%	57.055	2.988	5.24%	8.521	2.798	32.84%

4.4.2 Discussion on Porosity, Bulk Electrical Conductivity and Electromagnetic Shielding Effectiveness

A summary of the results reported in previous sections can be found in Table 11. We can observe that the higher porosity in PPS-045, may have influenced the shielding effectiveness when compared to PPS-090. These additional air bubbles could have created additional resistance for electrons to flow through the material. Second, observing the pattern of the results from in-plane transverse conductivity, there appears to be a pattern in values similar to those observed in Figure 34. This is further confirmation that the field lines were acting in the y-direction parallel to the short side of the waveguide cross-section.

Considering the density-normalised shielding effectiveness (SSE_T), we can find that PPS-000 equals 11.088 dB cm³/g, PPS-045 equals 48.149 dB cm³/g, PPS-090 equals 48.092 dB cm³/g, PA12-000 equals 9.487 dB cm³/g, and PA12-045 equals 48.937 dB cm³/g. The values found here indicate that PA12-045 provided the best shielding effectiveness to density ratio, followed by PPS-045, PPS-090, PPS-000, and PA12-000.

In contrast, using the results from the aluminium panel investigated tested, which had the same thickness, the average SE_T came to 75.188 dB, with a minimum SE_T of 68.485 dB and maximum SE_T of 78.003 dB. These results are higher than the results measured in SE_T however they are in line with literature. The density-normalised shielding effectiveness for this spectrum was found to be much lower at 27.847 dB cm³/g, much lower than PPS-045, PPS-090, and PA12-045.

Reflecting on the performance of the PPS-045 laminate and comparing it to literature, as observed in Table 2, the results here are lower than the CF/EP system with 16-ply QI layup used by Mikinka and Siwak. The study found that the SE_T to be 93.3 dB, the SE_A as 82.3 dB and SE_R of 11.0 dB (16). Similarly, the performance of the PPS-090 laminate was also lower than the 16-ply CP layup reported, which was found to have SE_T of 93.2 dB, SE_A of 80.6 dB, and SE_R of 12.6 dB. While the layup of the final laminate was different, this probably played a smaller role due to the additional 0° plies being parallel to the x-axis direction. Despite the fibre volume fraction of the tape being 60%, these values were likely higher due to the different experimental setup which was done using ASTM D4935, measurement frequency (30 MHz to 1.5 GHz in comparison to 2.2 GHz to 3 GHz), and larger number of plies used.

Table 11. Summary of experimental results on CF/PPS and CF/PA12 configurations.

Symbol	Unit	PPS-000	PPS-045	PPS-090	PA12-000	PA12-045
ρ	(g/cm ³)	1.540 ± 0.004 (0.26%)	1.533 ± 0.010 (0.64%)	1.542 ± 0.002 (0.13%)	1.339 ± 0.017 (1.25%)	1.350 ± 0.010 (0.72%)
v_v	(%)	3.720 ± 0.253 (6.80%)	4.164 ± 0.612 (14.71%)	3.605 ± 0.126 (3.50%)	1.574 ± 1.228 (78.00%)	0.762 ± 0.713 (93.62%)
E_B	(GPa)	108.0 ± 4.3 (3.99%)	32.2 ± 1.4 (4.43%)	63.0 ± 1.4 (2.28%)	108.6 ± 4.7 (4.35%)	20.3 ± 1.0 (4.77%)
σ_B	(MPa)	1570.4 ± 229.0 (14.58%)	683.2 ± 47.1 (6.90%)	966.8 ± 27.8 (2.88%)	1162.6 ± 110.7 (9.52%)	456.3 ± 28.6 (6.26%)
$\sigma_{ }$	(S/m)	3522.1 ± 500.7 (14.22%)	2131.0 ± 55.9 (2.63%)	2588.3 ± 137.7 (5.32%)	1210.0 ± 207.1 (17.12%)	565.4 ± 134.4 (23.78%)
$\sigma_{\perp}$	(S/m)	8.430 ± 0.732 (8.69%)	876.9 ± 52.6 (6.00%)	670.9 ± 44.6 (6.65%)	0.748 ± 0.150 (20.00%)	176.2 ± 11.8 (6.71%)
σ_{tt}	(S/m)	1.913 ± 0.247 (12.89%)	1.334 ± 0.078 (5.84%)	1.937 ± 0.164 (8.47%)	0.123 ± 0.008 (6.65%)	0.125 ± 0.011 (8.51%)
$SE_{T,avg}$	(dB)	17.076 ± 0.275 (1.61%)	73.812 ± 1.474 (2.00%)	74.158 ± 1.814 (2.45%)	12.703 ± 0.436 (3.43%)	65.576 ± 0.319 (0.49%)
$SE_{A,avg}$	(dB)	11.892 ± 0.274 (2.31%)	62.886 ± 1.927 (3.06%)	64.057 ± 1.886 (2.94%)	9.428 ± 0.499 (5.30%)	57.055 ± 2.988 (5.24%)
$SE_{R,avg}$	(dB)	5.184 ± 0.036 (0.70%)	10.926 ± 0.452 (4.14%)	10.101 ± 0.208 (2.06%)	3.276 ± 0.456 (13.93%)	8.521 ± 2.798 (32.84%)

5.0 Conclusion

This research project was intended to experimentally develop and characterise high-performance electromagnetic shielding for avionics manufactured from thermoplastic unidirectional laminates. The electromagnetic range of interest set out in this study was the 2.2-3 GHz range in the microwave S-Band. Two types of materials were investigated and compared, specifically carbon fibre polyphenylene sulphide (CF/PPS) and carbon fibre polyamide-12 (CF/PA12). Returning to the structure set out at the start of this project:

The objectives of this project were:

1. **Manufacture** carbon fibre thermoplastic composite laminates and evaluate the manufacturing quality of all laminates.
2. **Characterise** the bulk electrical conductivities and flexural performance of the selected thermoplastic laminates.
3. **Measure** the shielding effectiveness of thermoplastic laminates in the S-band region.
4. **Investigate** the relationship between manufacturing quality, bulk electrical conductivity, and shielding performance.
5. **Compare** the shielding effectiveness and mechanical performance to the state-of-the-art, aluminium.

1. Manufacturing

Upon initial analysis of the DSC curves, it was noted that the CF/PPS prepreg may have experienced cold crystallisation. The shifts in DSC peaks were examined in detail and were found to be in line with literature. Once this validation was complete, five configurations of CFRTPL laminates were manufactured and evaluated involving a detailed documentation of the condition of the laminate through a range of approaches involving ultrasonic C-scans to locate porosities. Here, it was found that the CF/PPS laminates were relatively high, up to 4.164%. On the other hand, CF/PA12 laminates performed remarkably well, however while evaluating the porosity, significant variance was found and may have arisen from error while measuring dimensions.

To understand the source of these porosities, a procedure to understand the implications of these results was carried out using computed tomography. By reconstructing the geometry, a phenomenon arising from fibre tortuosity likely impeded voids from escaping, which was determined to possibly have been exacerbated by the low viscosity of PPS polymer. Further inspection, through optical microscopy

of each laminate subsequently confirmed these findings. The results also showed a clearer view of fibre bundling which may have also constricted the flow of the PPS polymer.

The potential issues observed were deduced to be material-specific rather than process-driven, given the same approach was carried out in addition to recommended manufacturing processes by manufacturers. However, further improvements could have been made by exploring a Taguchi method prior to manufacturing so that CF/PPS laminates could be optimised better.

2. Characterise

While characterising the flexural properties of the material, it was found that the results for UD laminate of CF/PPS were comparable to literature, with UD laminates expressing flexural stiffness of 107.97 ± 4.30 GPa, and a flexural strength of 1570.4 ± 229.0 MPa. In contrast, the UD laminates for CF/PA12 were found to be higher than expected with flexural stiffness of 108.65 ± 4.73 GPa, and flexural strength of 1162.6 ± 110.7 MPa. These were attributed to the slow cooling rate of 0.243 °C/min used during manufacturing of the laminates. Using classical laminate theory, the results from the other laminates were also verified to be in-line with expected results.

Investigating the conductivity results, it was found that for the in-plane longitudinal conductivity of UD CF/PPS was 3522.1 ± 500.7 S/m, whereas CF/PA12 was 1210.0 ± 207.1 S/m. This difference was attributed to the difference in fibre volume fraction, in addition to fibre tortuosity. The closer contact of the fibres due to macrovoids moving the fibres closer could have impacted the results. For the in-plane transverse conductivity, UD CF/PPS reported much lower, at 8.4 ± 0.7 S/m; similarly, UD CF/PA12 was found to be 0.7 ± 0.1 S/m. Through-thickness results were low for CF/PPS in general, at 1.91 ± 0.25 S/m whereas CF/PA12 reported 0.12 ± 0.01 S/m. These results in general were in line with results recorded by some articles, despite not using silver paste.

3. Measure, Investigate, Compare

Electromagnetic shielding effectiveness measurements, were processed using a novel Savitzky-Golay filtering approach applied to linear-magnitude S-parameters prior to conversion, where it was revealed that cross-ply and quasi-isotropic configurations achieved average total shielding effectiveness values of 73.8 ± 1.5 dB and 74.2 ± 1.8 dB for CF/PPS, and 65.6 ± 0.3 dB for CF/PA12, while unidirectional configurations were substantially lower at 17.1 ± 0.3 dB and 12.7 ± 0.4 dB respectively. This result was consistent with the polarisation effect of fibres oriented parallel to the propagation direction, however further justification and confirmation of the accuracy of the results were difficult to compare to literature due to differences in setup, frequency and ply number.

Density-normalised analysis confirmed that CF/PA12 in a quasi-isotropic configuration offered the most favourable shielding-to-density ratio at 48.9 dB·cm³/g, marginally outperforming CF/PPS equivalents. Compared to aluminium, which had a total shielding effectiveness of 93.2 dB, the results presented here mark a successful first attempt at achieving equivalent results.

In conclusion, this thesis demonstrates that CFRTP laminates are viable candidates for electromagnetic shielding in avionics enclosures, and that ply orientation, fibre volume fraction, and manufacturing quality are the dominant parameters governing shielding performance; not the thermoplastic matrix system alone. Future recommendations are to:

1. Explore alternative frequencies using a different setup to ensure results here can be validated, for instance, using the X-band waveguide so that further analysis can be established.
- 2.. Experiment with woven ply laminates manufactured from thermoplastic laminates, understanding the difference in performance despite changes in fibre volume fraction is an area that has not been explored.
3. Attempt to forge a more complete relationship between conductivity and shielding effectiveness by carrying out improved conductivity measurements such as, the addition of a torque wrench to monitor torque applied.
4. Determine the influence of carrying out the experiment without the laminate bolted to the flanges, as described by ASTM D5568, as many studies explore this approach.

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APPENDIX A: MATERIALS, MANUFACTURING, CHARACTERISATION DEETAILS

A1 MATERIAL UTILISATION

The cutting scheme developed for this experiment was found to be highly effective. It is possible for laminates containing $\pm 45^\circ$ -degree panels to be manufactured without significant waste material. For comparison, the methodology used by Mikinka et al (27) was set as the benchmark here as shown in Table A1. To the author's best understanding, aspect ratio is the key element to reducing material waste. While this study was done on a UD tape with a width of ~ 100 mm, this relationship should remain true even when scaled up to 300 mm where the most ideal configuration with 300 mm UD tape would be achieved with manufacturing 300 mm x 450 mm laminates.

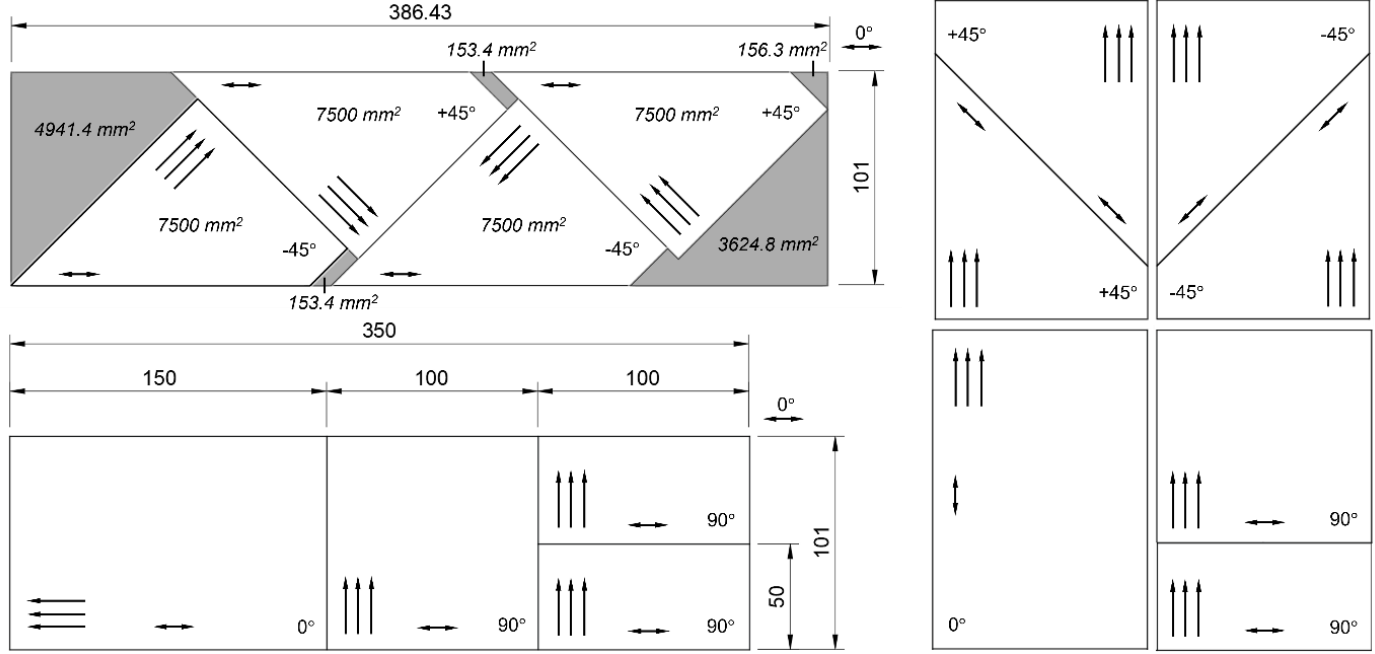


Figure A1. Complete cutting template for CF/PPS.

Table A1. Calculations on effect of aspect ratio (width:length) on material utilisation.

Aspect ratio	A_{total}	A_{used}	A_{waste}	Utilisation factor	Waste factor	Ref.
-	(mm ²)	(mm ²)	(mm ²)	(%)	(%)	-
1:1	127123.1	90000.0	37123.1	70.80%	29.20%	(27)
1:1 (incl. Start Only)	254402.0	180000.0	74402.0	70.75%	29.25%	(27)
1:1 (incl. Start and End)	291680.9	180000.0	111680.9	61.71%	38.29%	(27)
2:3	15153.4	15000.0	153.4	98.99%	1.01%	This study
2:3 (incl. Start Only)	34087.9	30000.0	4087.9	88.01%	11.99%	This study
2:3 (incl. Start and End)	39029.3	30000.0	9029.3	76.87%	23.13%	This study

* Start = the initial piece of offcut required to start a $\pm 45^\circ$ degree ply.

** End = to the final piece of offcut required after two $\pm 45^\circ$ degree plies to start a 0 degree ply.

A2 HEATED PRESS MANUFACTURING

Fabrication of the laminates was done with the picture-frame design as shown below. Three of these picture-frames were fabricated for the simultaneous manufacturing of three laminates per press cycle. Having the frames be individual instead of being a singular steel sheet was useful when ‘pushing-out’ the laminates after the press but also ensured pressure values were exactly as specified.

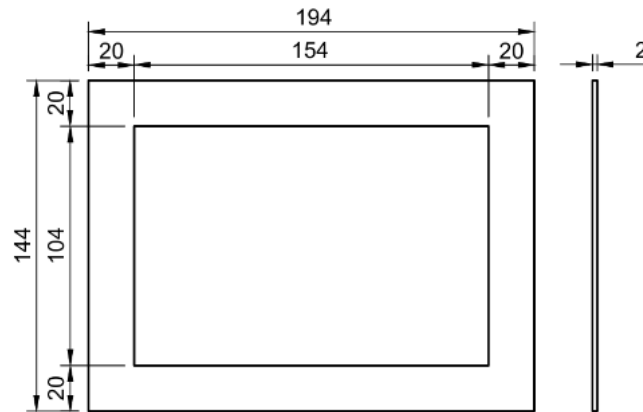


Figure A2: Picture-frame design for the three-element high-temperature mould.

Before starting the press cycle, additional parameters were used. This included setting closing speed to 18mm/s, max pressurising speed to 0.4 mm/s, press minimum closure pressure to 13.0 kN, pressure close position to 40.0 mm, and pressure open position to 150.0 mm. These parameters were kept the same other than pressure close position, which was regularly recorded and checked before using the right input table. After trial and testing, Table A2 was used as the input table for the press cycle for CF/PPS. On the other hand, Table A3 was used as the input table for the press cycle for CF/PA12. The three-stage design was inspired by the work of Patadia et al (42).

The programmes here were saved on the HLP40 (Hofer Presstechnik, Germany) and the logs from each press cycle were saved onto a USB drive. Logs are available upon request and were not included due to length (~9000 rows).

Table A2. Input table for one CF/PPS composite panel.

No.	Time	Duration	Temp	Temp Ramp	Temp Ramp	Pressure	Pressure	Pressure Ramp	Pressure Ramp
	(hr:min)	(min)	(°C)	(°C min ⁻¹)	(sec)	(kPa)	(kN)	(kPa min ⁻¹)	(sec)
1	00:00	0	25 (RT)	-	-	0	0	-	
2	01:15	75	325	+4	4500	500	15*	200	150
3	01:30	15	325	0	0	1000	30*	200	150
4	01:45	15	325	0	0	1200	36*	200	150
5	FINISH	999	25	-10	150	1200	36*	0	0

* Pressure (kN) must be scaled linearly by a scaling factor equal to number of panels under compression at any given time, for example manufacturing three (3) panels would require a scaling factor of 3.

Table A3. Input table for one CF/PA12 composite panel.

No.	Time	Duration	Temp	Temp Ramp	Temp Ramp	Pressure	Pressure	Pressure Ramp	Pressure Ramp
	(hr:min)	(min)	(°C)	(°C min ⁻¹)	(sec)	(kPa)	(kN)	(kPa min ⁻¹)	(sec)
1	00:00	0	25 (RT)	-	-	0	0	-	
2	00:50	50	225	+4	3000	500	15*	200	150
3	00:55	5	225	0	0	1000	30*	200	150
4	FINISH	999	25	-10	150	1000	30*	0	0

* Pressure (kN) must be scaled linearly by a scaling factor equal to number of panels under compression at any given time, for example manufacturing three (3) panels would require a scaling factor of 3.

A3 ALTERNATIVE THROUGH-THICKNESS ELECTRICAL CONDUCTIVITY SETUPS

Two other measurement methods were considered during measurements of the through-thickness conductivity, as seen in Figure A3. Compared to the final decision to go with the clamped methodology, the alternatives were poorer in both their repeatability and accuracy. Data points from each individual specimen under both methodologies is included in Appendix B.

The first method involved directly clamping the Kelvin clip onto the surface of the specimen with the same 3M 1170 conductive aluminium tape and operated under 2-wire DC resistance mode. repeatability was significantly lower, with a COV of 29.1% on average across five readings. The accuracy of the readings wasn't great either when evaluating the readings across the three samples, with an average COV of 23.36%.

The second method considered was extending the two surfaces and having the current run through the tape. However, this also yielded poor repeatability with a COV of 46.3% on average across on five readings. Accuracy was even lower when compared to the direct clamping with an average COV of 28.04%.

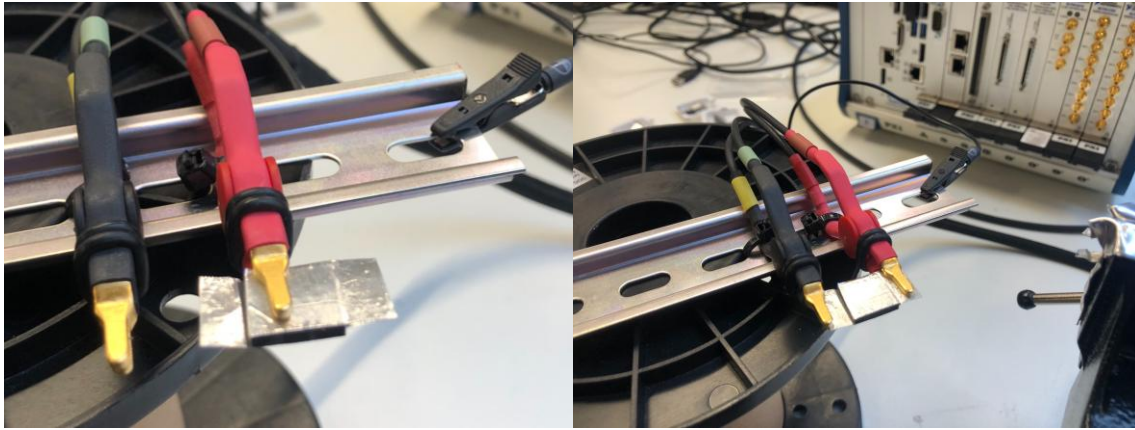


Figure A3. Direct clamping (left) and lead extension method (right).

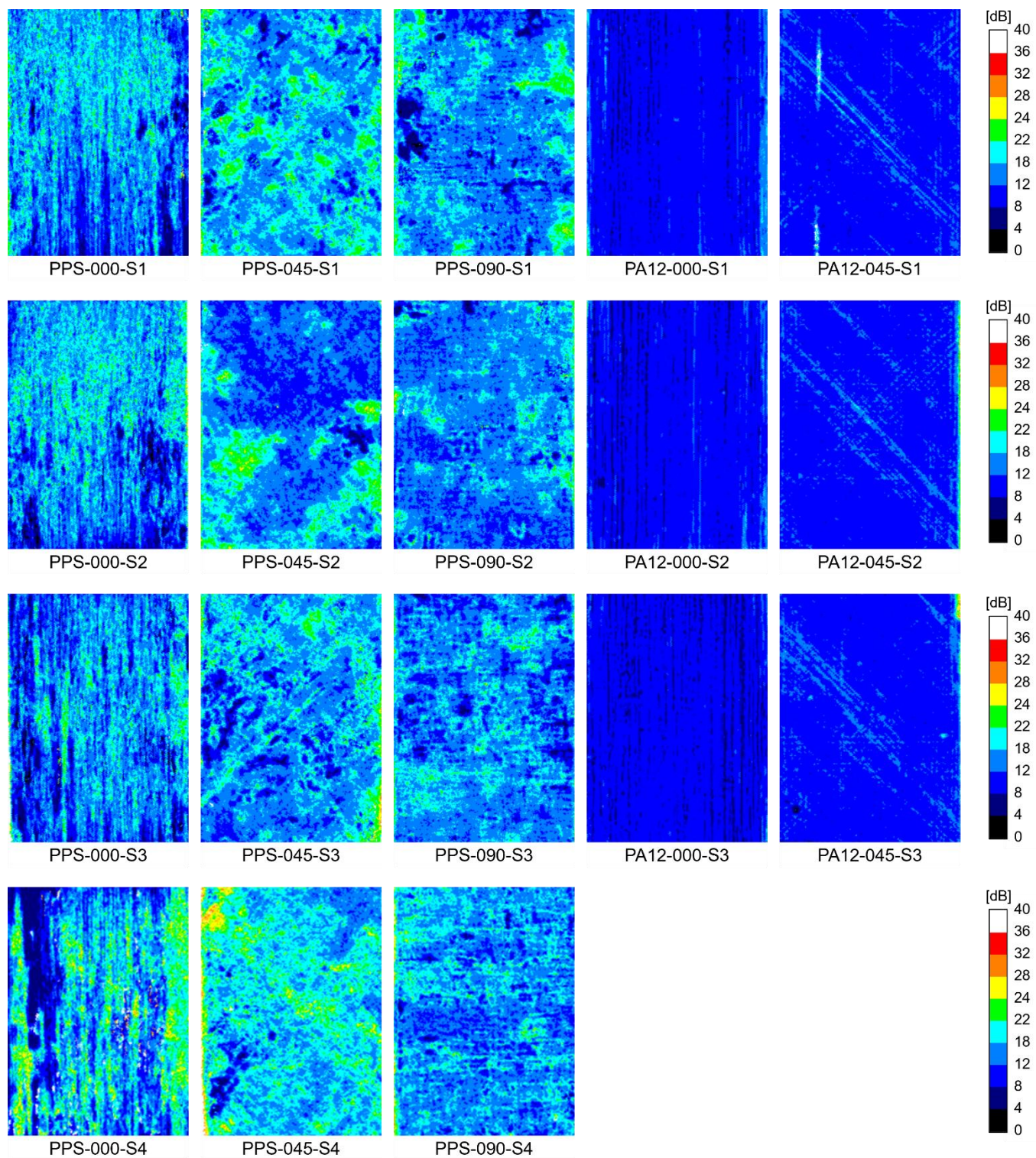
APPENDIX B: CHARACTERISATION DATASETS

B1 DIFFERENTIAL SCANNING CALORIMETRY

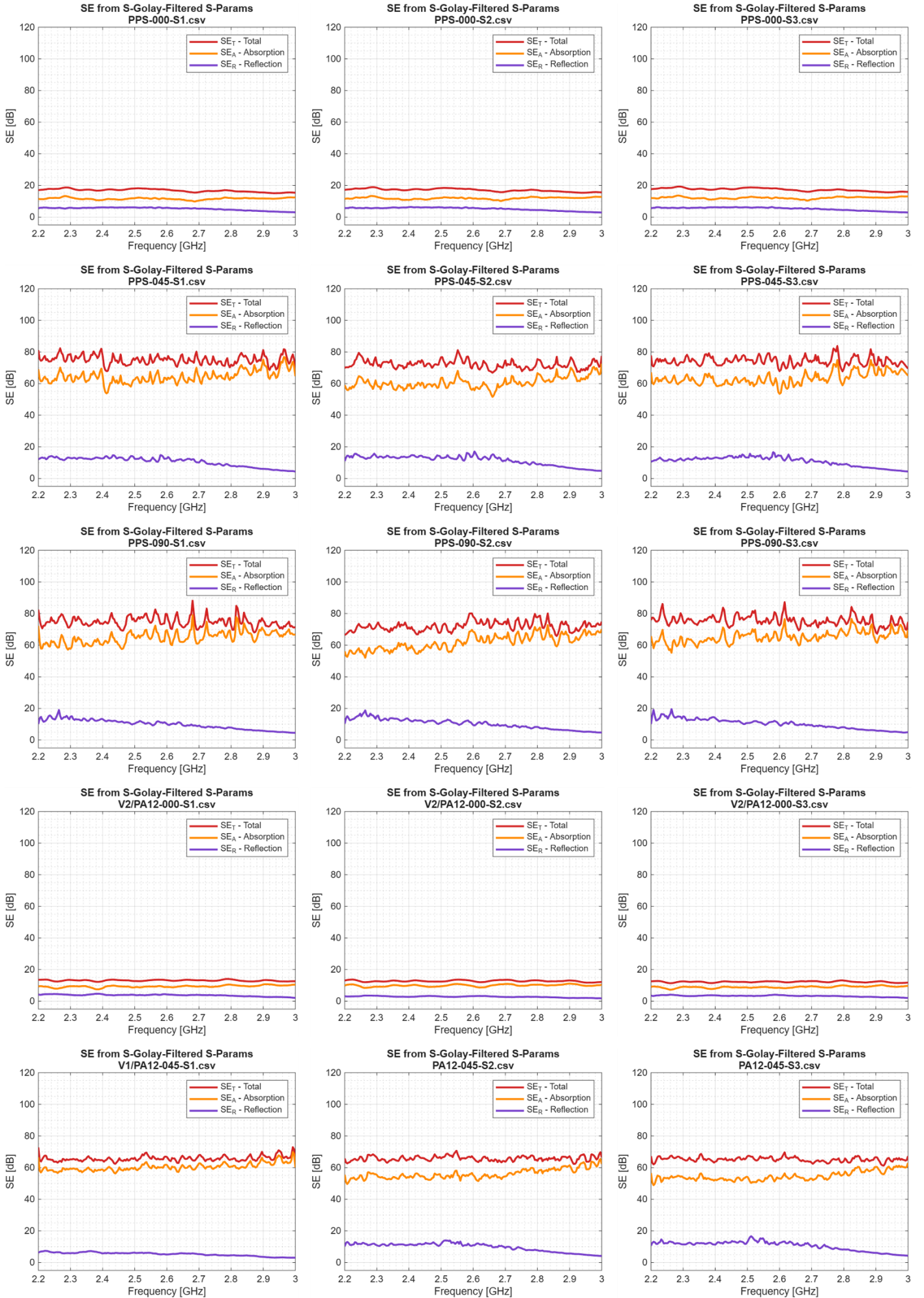
Table A4. Raw data of each measurement from DSC.

Material	Tg	Tm	Tc	Material	Tg	Tm	Tc
-	(°C)	(°C)	(°C)		(°C)	(°C)	(°C)
PPS-S1-R2	83.26	278.34	223.16	PA12-S1-R2	45.53	176.5	159.95
PPS-S1-R3	84.93	279.62	216.54	PA12-S1-R3	55.66	176.46	159.85
PPS-S2-R2	92.71	280.12	222.63	PA12-S2-R2	46.51	176.58	160.13
PPS-S2-R3	87.51	279.21	215.94	PA12-S2-R3	53.75	176.47	160.01
PPS-S3-R2	94.29	280.23	222.84	PA12-S3-R2	49.17	176.58	160.06
PPS-S3-R3	93.24	279.46	216.45	PA12-S3-R3	50.61	176.48	160.02
AVG	89.32	279.50	219.59	AVG	50.21	176.51	160.00
STDEV.S	4.71	0.69	3.61	STDEV.S	3.98	0.05	0.10

B2 ULTRASONIC C-SCANS



B3 ELECTROMAGNETIC SHIELDING EFFECTIVENESS



B4 DENSITY AND POROSITY

Material	Specimen	ρ	ρ_{avg}	ρ_{avg} SD	COV	v_v	$v_{v\ avg}$	$v_{v\ avg}$ SD	COV
-	-	(g/cm ³)	(g/cm ³)	(g/cm ³)	(%)	(%)	(%)	(%)	(%)
PPS-000	PPS-000-S1	1.5413	1.5405	0.0040	0.26%	3.6657	3.7203	0.2528	6.80%
	PPS-000-S2	1.5361				3.9960			
	PPS-000-S3	1.5440				3.4992			
PPS-045	PPS-045-S1	1.5343	1.5334	0.0098	0.64%	4.1044	4.1641	0.6124	14.71%
	PPS-045-S2	1.5427				3.5837			
	PPS-045-S3	1.5231				4.8041			
PPS-090	PPS-090-S1	1.5409	1.5423	0.0020	0.13%	3.6911	3.6050	0.1263	3.50%
	PPS-090-S2	1.5446				3.4600			
	PPS-090-S3	1.5414				3.6640			
PA12-000	PA12-000-S1	1.3578	1.3386	0.0167	1.25%	0.1593	1.5737	1.2275	78.00%
	PA12-000-S2	1.3301				2.1996			
	PA12-000-S3	1.3279				2.3621			
PA12-045	PA12-045-S1	1.3406	1.3496	0.0097	0.72%	1.4234	0.7619	0.7133	93.62%
	PA12-045-S2	1.3484				0.8563			
	PA12-045-S3	1.3599				0.0062			

B5 ELECTROMAGNETIC SHIELDING EFFECTIVENESS – TOTAL

Material	Specimen	AVG SE_T	AVG	SD	COV	MIN SE_T	AVG	SD	COV	MAX SE_T	AVG	SD	COV
-	-	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]
PPS-000	PPS-000-S1	16.814	17.076	0.275	1.61%	15.085	15.354	0.267	1.74%	18.699	18.966	0.297	1.57%
	PPS-000-S2	17.053				15.357				18.914			
	PPS-000-S3	17.362				15.619				19.286			
PPS-045	PPS-045-S1	75.03	73.812	1.474	2.00%	67.875	67.481	0.578	0.86%	82.297	82.412	1.307	1.59%
	PPS-045-S2	72.173				66.818				81.166			
	PPS-045-S3	74.233				67.751				83.773			
PPS-090	PPS-090-S1	74.822	74.158	1.814	2.45%	67.99	67.018	1.112	1.66%	88.188	85.180	4.408	5.17%
	PPS-090-S2	72.105				65.805				80.121			
	PPS-090-S3	75.546				67.26				87.232			
PA12-000	PA12-000-S1*	13.109	12.703	0.436	3.43%	12.244	11.788	0.433	3.67%	14.062	13.651	0.443	3.25%
	PA12-000-S2*	12.758				11.738				13.709			
	PA12-000-S3*	12.243				11.383				13.181			
PA12-045	PA12-045-S1*	65.735	65.576	0.319	0.49%	62.434	61.966	0.686	1.11%	72.905	71.107	1.644	2.31%
	PA12-045-S2	65.784				62.285				70.734			
	PA12-045-S3	65.208				61.178				69.682			

* Data recorded from different session

B6 ELECTROMAGNETIC SHIELDING EFFECTIVENESS – ABSORPTION

Material	Specimen	AVG SE_A	AVG	SD	COV	MIN SE_A	AVG	SD	COV	MAX SE_A	AVG	SD	COV
-	-	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]
PPS-000	PPS-000-S1	11.609	11.892	0.274	2.31%	9.836	10.135	0.277	2.73%	13.325	13.467	0.183	1.36%
	PPS-000-S2	11.911				10.186				13.403			
	PPS-000-S3	12.157				10.383				13.674			
PPS-045	PPS-045-S1	64.482	62.886	1.927	3.06%	53.776	52.969	1.203	2.27%	76.829	74.847	2.093	2.80%
	PPS-045-S2	60.746				51.586				72.658			
	PPS-045-S3	63.431				53.545				75.053			
PPS-090	PPS-090-S1	64.961	64.057	1.886	2.94%	57.16	54.777	2.612	4.77%	78.971	76.171	3.125	4.10%
	PPS-090-S2	61.889				51.985				72.8			
	PPS-090-S3	65.321				55.185				76.743			
PA12-000	PA12-000-S1*	9.386	9.428	0.499	5.30%	7.524	7.748	0.637	8.22%	10.754	10.715	0.429	4.00%
	PA12-000-S2*	9.947				8.467				11.123			
	PA12-000-S3*	8.951				7.253				10.268			
PA12-045	PA12-045-S1*	60.41	57.055	2.988	5.24%	56.295	51.582	4.100	7.95%	69.804	66.089	3.464	5.24%
	PA12-045-S2	56.075				49.61				65.514			
	PA12-045-S3	54.68				48.84				62.948			

* Data recorded from different session

B7 ELECTROMAGNETIC SHIELDING EFFECTIVENESS – REFLECTION

Material	Specimen	AVG SE_R	AVG	SD	COV	MIN SE_R	AVG	SD	COV	MAX SE_R	AVG	SD	COV
-	-	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]
PPS-000	PPS-000-S1	5.204	5.184	0.036	0.70%	2.969	2.917	0.047	1.61%	6.25	6.341	0.079	1.25%
	PPS-000-S2	5.142				2.903				6.386			
	PPS-000-S3	5.205				2.878				6.388			
PPS-045	PPS-045-S1	10.548	10.926	0.452	4.14%	4.352	4.540	0.280	6.17%	14.821	16.182	1.205	7.45%
	PPS-045-S2	11.427				4.862				17.114			
	PPS-045-S3	10.802				4.407				16.611			
PPS-090	PPS-090-S1	9.861	10.101	0.208	2.06%	4.587	4.691	0.094	2.00%	18.988	19.134	0.398	2.08%
	PPS-090-S2	10.216				4.769				18.83			
	PPS-090-S3	10.225				4.716				19.585			
PA12-000	PA12-000-S1*	3.723	3.276	0.456	13.93%	2.129	1.995	0.185	9.28%	4.737	4.145	0.585	14.10%
	PA12-000-S2*	2.811				1.784				3.568			
	PA12-000-S3*	3.293				2.073				4.13			
PA12-045	PA12-045-S1*	5.325	8.521	2.798	32.84%	3.089	3.834	0.648	16.90%	7.404	12.679	4.752	37.48%
	PA12-045-S2	9.709				4.148				14.008			
	PA12-045-S3	10.529				4.265				16.626			

* Data recorded from different session

B8 BULK ELECTRICAL CONDUCTIVITY - IN-PLANE LONGITUDINAL

Material	Specimen	W	T	L	R	σ	σ_{avg}	σ_{avg} SD	COV
-	-	(mm)	(mm)	(mm)	(Ω)	(S/m)	(S/m)	(S/m)	(%)
PPS-000	PPS-000-S1	9.97	1.955	19.90	0.31038	3288.30	3522.07	500.70	14.22%
	PPS-000-S2	10.01	1.955	19.91	0.31978	3181.01			
	PPS-000-S3	10.02	1.952	19.94	0.2488	4096.89			
PPS-045	PPS-045-S1	10.03	1.926	19.94	0.48952	2109.33	2131.02	55.94	2.63%
	PPS-045-S2	9.98	1.921	19.94	0.49784	2089.17			
	PPS-045-S3	9.92	1.923	19.97	0.47702	2194.56			
PPS-090	PPS-090-S1	9.98	2.006	19.88	0.40812	2433.14	2588.27	137.69	5.32%
	PPS-090-S2	10.01	2.005	19.98	0.36944	2696.00			
	PPS-090-S3	10.01	2.006	19.99	0.37796	2635.67			
PA12-000	PA12-000-S1	9.89	1.935	19.93	0.85904	1212.32	1209.96	207.12	17.12%
	PA12-000-S2	9.73	1.958	19.68	1.03146	1001.66			
	PA12-000-S3	9.81	1.939	20.31	0.75462	1415.89			
PA12-045	PA12-045-S1	9.95	1.905	19.85	2.54074	411.97	565.36	134.41	23.78%
	PA12-045-S2	9.92	1.900	19.84	1.58896	662.58			
	PA12-045-S3	9.95	1.926	19.90	1.67134	621.52			

B9 BULK ELECTRICAL CONDUCTIVITY - IN-PLANE TRANSVERSE

Material	Specimen	W	T	L	R	σ	σ_{avg}	σ_{avg} SD	COV
-	-	(mm)	(mm)	(mm)	(Ω)	(S/m)	(S/m)	(S/m)	(%)
PPS-000	PPS-000-S1	9.97	1.955	19.90	32.763	7.8220	8.4297	0.7322	8.69%
	PPS-000-S2	10.01	1.955	19.91	27.819	9.2426			
	PPS-000-S3	10.02	1.952	19.94	31.306	8.2245			
PPS-045	PPS-045-S1	10.03	1.926	19.94	0.27970	933.750	876.908	52.621	6.00%
	PPS-045-S2	9.98	1.921	19.94	0.30038	867.080			
	PPS-045-S3	9.92	1.923	19.97	0.31116	829.894			
PPS-090	PPS-090-S1	9.98	2.006	19.88	0.36812	679.594	670.862	44.592	6.65%
	PPS-090-S2	10.01	2.005	19.98	0.35166	710.442			
	PPS-090-S3	10.01	2.006	19.99	0.40084	622.550			
PA12-000	PA12-000-S1	9.89	1.935	19.93	335.021	0.76548	0.74798	0.14957	20.00%
	PA12-000-S2	9.73	1.958	19.68	427.452	0.59043			
	PA12-000-S3	9.81	1.939	20.31	280.419	0.88803			
PA12-045	PA12-045-S1	9.95	1.905	19.85	1.53742	171.235	176.217	11.831	6.71%
	PA12-045-S2	9.92	1.900	19.84	1.56904	167.691			
	PA12-045-S3	9.95	1.926	19.90	1.36834	189.724			

B10 BULK ELECTRICAL CONDUCTIVITY - THROUGH-THICKNESS - CROCODILE CLAMP

Material	Specimen	W	T	L	R	σ	σ_{avg}	σ_{avg} SD	COV
-	-	(mm)	(mm)	(mm)	(Ω)	(S/m)	(S/m)	(S/m)	(%)
PPS-000	PPS-000-S1	12.57	1.916	12.54	131.947	0.09212	0.10987	0.02169	19.74%
	PPS-000-S2	12.50	1.906	12.56	117.347	0.10344			
	PPS-000-S3	12.44	1.911	12.57	91.236	0.13404			
PPS-045	PPS-045-S1	12.55	1.919	12.54	89.052	0.13696	0.17909	0.04025	22.48%
	PPS-045-S2	12.52	1.914	12.55	56.103	0.21716			
	PPS-045-S3	12.54	1.916	12.54	66.538	0.18314			
PPS-090	PPS-090-S1	12.54	1.963	12.57	75.169	0.16567	0.17749	0.01526	8.60%
	PPS-090-S2	12.56	1.970	12.54	64.241	0.19472			
	PPS-090-S3	12.54	1.976	12.53	73.052	0.17208			
PA12-000	PA12-000-S1	12.51	1.921	12.53	1394.536	0.00878	0.01148	0.00412	35.84%
	PA12-000-S2	12.52	1.924	12.54	1297.343	0.00944			
	PA12-000-S3	12.45	1.918	12.54	757.087	0.01622			
PA12-045	PA12-045-S1	12.47	1.874	12.50	567.666	0.02117	0.01713	0.00517	30.17%
	PA12-045-S2	12.49	1.894	12.52	1070.855	0.01131			
	PA12-045-S3	12.45	1.895	12.52	642.876	0.01891			

B11 BULK ELECTRICAL CONDUCTIVITY - THROUGH-THICKNESS - VICE CLAMP

Material	Specimen	W	T	L	R	σ	σ_{avg}	σ_{avg} SD	COV
-	-	(mm)	(mm)	(mm)	(Ω)	(S/m)	(S/m)	(S/m)	(%)
PPS-000	PPS-000-S1	12.57	1.916	12.54	6.8403	1.7770	1.91296	0.24662	12.89%
	PPS-000-S2	12.50	1.906	12.56	6.8800	1.7643			
	PPS-000-S3	12.44	1.911	12.57	5.5649	2.1976			
PPS-045	PPS-045-S1	12.55	1.919	12.54	9.7521	1.2507	1.33379	0.07792	5.84%
	PPS-045-S2	12.52	1.914	12.55	8.6702	1.4052			
	PPS-045-S3	12.54	1.916	12.54	9.0566	1.3455			
PPS-090	PPS-090-S1	12.54	1.963	12.57	5.9156	2.1052	1.93746	0.16408	8.47%
	PPS-090-S2	12.56	1.970	12.54	7.0382	1.7773			
	PPS-090-S3	12.54	1.976	12.53	6.5135	1.9299			
PA12-000	PA12-000-S1	12.51	1.921	12.53	94.8265	0.1292	0.12348	0.00821	6.65%
	PA12-000-S2	12.52	1.924	12.54	107.4171	0.1141			
	PA12-000-S3	12.45	1.918	12.54	96.5486	0.1272			
PA12-045	PA12-045-S1	12.47	1.874	12.50	89.9944	0.1336	0.12454	0.01060	8.51%
	PA12-045-S2	12.49	1.894	12.52	107.2997	0.1129			
	PA12-045-S3	12.45	1.895	12.52	95.5807	0.1272			

B12 BULK ELECTRICAL CONDUCTIVITY - THROUGH-THICKNESS - NO CLAMP

Material	Specimen	W	T	L	R	σ	σ_{avg}	σ_{avg} SD	COV
-	-	(mm)	(mm)	(mm)	(Ω)	(S/m)	(S/m)	(S/m)	(%)
PPS-000	PPS-000-S1	12.57	1.916	12.54	487.23	0.024947	0.032140	0.006233	19.39%
	PPS-000-S2	12.50	1.906	12.56	337.48	0.035967			
	PPS-000-S3	12.44	1.911	12.57	344.44	0.035506			
PPS-045	PPS-045-S1	12.55	1.919	12.54	354.12	0.034443	0.040327	0.005098	12.64%
	PPS-045-S2	12.52	1.914	12.55	282.43	0.043138			
	PPS-045-S3	12.54	1.916	12.54	280.76	0.043401			
PPS-090	PPS-090-S1	12.54	1.963	12.57	199.97	0.062276	0.048982	0.019903	40.63%
	PPS-090-S2	12.56	1.970	12.54	213.58	0.058569			
	PPS-090-S3	12.54	1.976	12.53	481.63	0.026100			
PA12-000	PA12-000-S1	12.51	1.921	12.53	4842.72	0.002530	0.003191	0.000649	20.32%
	PA12-000-S2	12.52	1.924	12.54	3202.57	0.003826			
	PA12-000-S3	12.45	1.918	12.54	3815.11	0.003219			
PA12-045	PA12-045-S1	12.47	1.874	12.50	3391.66	0.003544	0.003974	0.001876	47.21%
	PA12-045-S2	12.49	1.894	12.52	5153.68	0.002350			
	PA12-045-S3	12.45	1.895	12.52	2017.23	0.006027			

APPENDIX C: ELECTROMAGNETIC SHIELDING EFFECTIVENESS CODE

```
1 clear;
2 clc;
3
4 % =====
5 % 1. USER OPTIONS
6 % =====
7 % Option 1: running without averaging (false)
8 % Option 2: running with averaging across 3 files (true)
9 use_averaging = false;
10
11 % Option to easily disable the S-Parameter Linear Magnitude plots
12 enable_se_plots = true;
13 enable_sparam_plots = true;
14
15 % Define filenames (Update these with your actual file names)
16 filename1 = 'V1/ALU_cal_260727.csv';
17 filename2 = 'PPS-045-S2.csv'; % Only used if use_averaging = true
18 filename3 = 'PPS-045-S3.csv'; % Only used if use_averaging = true
19
20 % =====
21 % 2. DATA IMPORT & AVERAGING
22 % =====
23 opts = detectImportOptions(filename1, 'CommentStyle', '#');
24 opts.VariableNamesLine = 7;
25 opts.DataLines = [8 Inf];
26
27 if ~use_averaging
28     % --- Option 1: Single Reading ---
29     data = readtable(filename1, opts);
30     disp('Data successfully loaded (Single run). First 5 rows:');
31     disp(head(data, 5));
32
33     freq = data{:,1} ./ 1e9;
34     S11 = data{:,2};
35     S21 = data{:,4};
36 else
37     % --- Option 2: 3-Sweep Linear Averaging ---
38     data1 = readtable(filename1, opts);
39     data2 = readtable(filename2, opts);
40     data3 = readtable(filename3, opts);
41
42     disp('Data successfully loaded (3 runs averaged). First 5 rows of Run 1:');
43     disp(head(data1, 5));
44
45     freq = data1{:,1} ./ 1e9; % Assuming frequency vector is identical across runs
46
47     % Linear scale averaging of S-parameters before converting to dB
48     S11 = (data1{:,2} + data2{:,2} + data3{:,2}) / 3;
49     S21 = (data1{:,4} + data2{:,4} + data3{:,4}) / 3;
50 end
51
52 % =====
53 % 3. FILTERING PIPELINE (ON S-PARAMETERS)
54 % =====
55 % Step A: Median Filter (removes sharp VNA/manual spikes)
56 med_win = 1; % Window size for median filter
57 S11_med = medfilt1(S11, med_win);
58 S21_med = medfilt1(S21, med_win);
59
60 % Step B: Savitzky-Golay Filter (smoothes residual noise)
61 m = 3;
62 framelen = 7;
63 S11_final = sgolayfilt(S11_med, m, framelen);
```

```

64 S21_final = sgolayfilt(S21_med, m, framelen);
65
66 % =====
67 % X. SHIFTING METHOD
68 % =====
69
70 % % finds >1 and substitutes it for 0.99999
71 % scout = 0;
72 % offset = 0.0001;
73 % for i = 1:length(S11_final)
74 %     if S11(i) >= 1
75 %         S11(i) = 1 - offset;
76 %     end
77 %     if S11_med(i) >= 1
78 %         S11_med(i) = 1 - offset;
79 %     end
80 %     if S11_final(i) >= 1
81 %         S11_final(i) = 1 - offset;
82 %         scout = scout + 1;
83 %     end
84 % end
85 %
86 % fprintf("Number of S11 points fixed: %.8f\n\n",scout)
87
88
89 % counts how many are off
90 pcount = 0;
91 for i = 1:length(S11_final)
92     if S11_final(i) >= 1
93         pcount = pcount + 1;
94     end
95 end
96
97 % count = 0;
98 % offset = 0.0001;
99 % while any(S11_final >= 1)
100 %     S11 = S11 - offset;
101 %     S11_med = S11_med - offset;
102 %     S11_final = S11_final - offset;
103 %     count = count + 1;
104 % end
105 %
106 %
107 % fprintf("S11 linear magnitude shifted by: %.8f\n\n",-count*offset)
108
109 % =====
110 % 4. CALCULATE SE [dB]
111 % =====
112 % Calculate SE from RAW S-parameters
113 SET_raw = -10*log10(abs(S21).^2);
114 SER_raw = -10*log10(1-abs(S11).^2);
115 SEA_raw = -10*log10(abs(S21).^2./(1-abs(S11).^2));
116
117 % Calculate SE from MEDIAN-FILTERED S-parameters
118 SET_med = -10*log10(abs(S21_med).^2);
119 SER_med = -10*log10(1-abs(S11_med).^2);
120 SEA_med = -10*log10(abs(S21_med).^2./(1-abs(S11_med).^2));
121
122 % Calculate SE from SGOLAY-FILTERED (FINAL) S-parameters
123 SET_final = -10*log10(abs(S21_final).^2);
124 SER_final = -10*log10(1-abs(S11_final).^2);
125 SEA_final = -10*log10(abs(S21_final).^2./(1-abs(S11_final).^2));
126
127 % =====
128 % 5. EXTRACT AND PRINT SUMMARY METRICS

```

```

129 % =====
130 disp('=====');
131 disp('          S-PARAMETERS SUMMARY METRICS          ');
132 disp('=====');
133
134 % Print metrics using fprintf for the final smoothed data
135 fprintf('Raw S11          | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
136         mean(S11), min(S11), max(S11));
137 fprintf('Raw S21          | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
138         mean(S21), min(S21), max(S21));
139 fprintf('Med S11          | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
140         mean(S11_med), min(S11_med), max(S11_med));
141 fprintf('Med S21          | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
142         mean(S21_med), min(S21_med), max(S21_med));
143 fprintf('Fin S11          | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
144         mean(S11_final), min(S11_final), max(S11_final));
145 fprintf('Fin S21          | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
146         mean(S21_final), min(S21_final), max(S21_final));
147
148
149 disp('=====');
150 disp('  SHIELDING EFFECTIVENESS (SE) SUMMARY METRICS  ');
151 disp('=====');
152
153 % Print metrics using fprintf for the final smoothed data
154 fprintf('Tot SE (SE_T)      | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
155         mean(SET_final), min(SET_final), max(SET_final));
156 fprintf('Abs SE (SE_A)      | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
157         mean(SEA_final), min(SEA_final), max(SEA_final));
158 fprintf('Ref SE (SE_R)      | Avg = %.3f dB | Min = %.3f dB | Max = %.3f dB\n', ...
159         mean(SER_final), min(SER_final), max(SER_final));
160
161 disp('=====');
162 % prepare values
163 vals = [mean(SET_final), min(SET_final), max(SET_final), ...
164         mean(SEA_final), min(SEA_final), max(SEA_final), ...
165         mean(SER_final), min(SER_final), max(SER_final)];
166
167 % print CSV row to console
168 fprintf('%.3f,%.3f,%.3f,%.3f,%.3f,%.3f,%.3f,%.3f,%.3f\n', vals);
169 disp('=====');
170
171 newcolors = [0.83 0.14 0.14
172             1.00 0.54 0.00
173             0.47 0.25 0.80
174             0.25 0.80 0.54];
175
176 % =====
177 % 6. PLOTTING SE RESULTS (3 Separate Figures)
178 % =====
179 % Group the datasets to loop through and plot easily
180 if enable_se_plots
181
182     datasets = {{SET_raw, SEA_raw, SER_raw}, ...
183               {SET_med, SEA_med, SER_med}, ...
184               {SET_final, SEA_final, SER_final}};
185
186     plot_titles = {'SE from Raw S-Params', ...
187                  'SE from Median-Filtered S-Params', ...
188                  'SE from S-Golay-Filtered S-Params'};
189
190     for i = 1:3
191         current_data = datasets{i};
192         c_SET = current_data{1};
193         c_SEA = current_data{2};

```

```

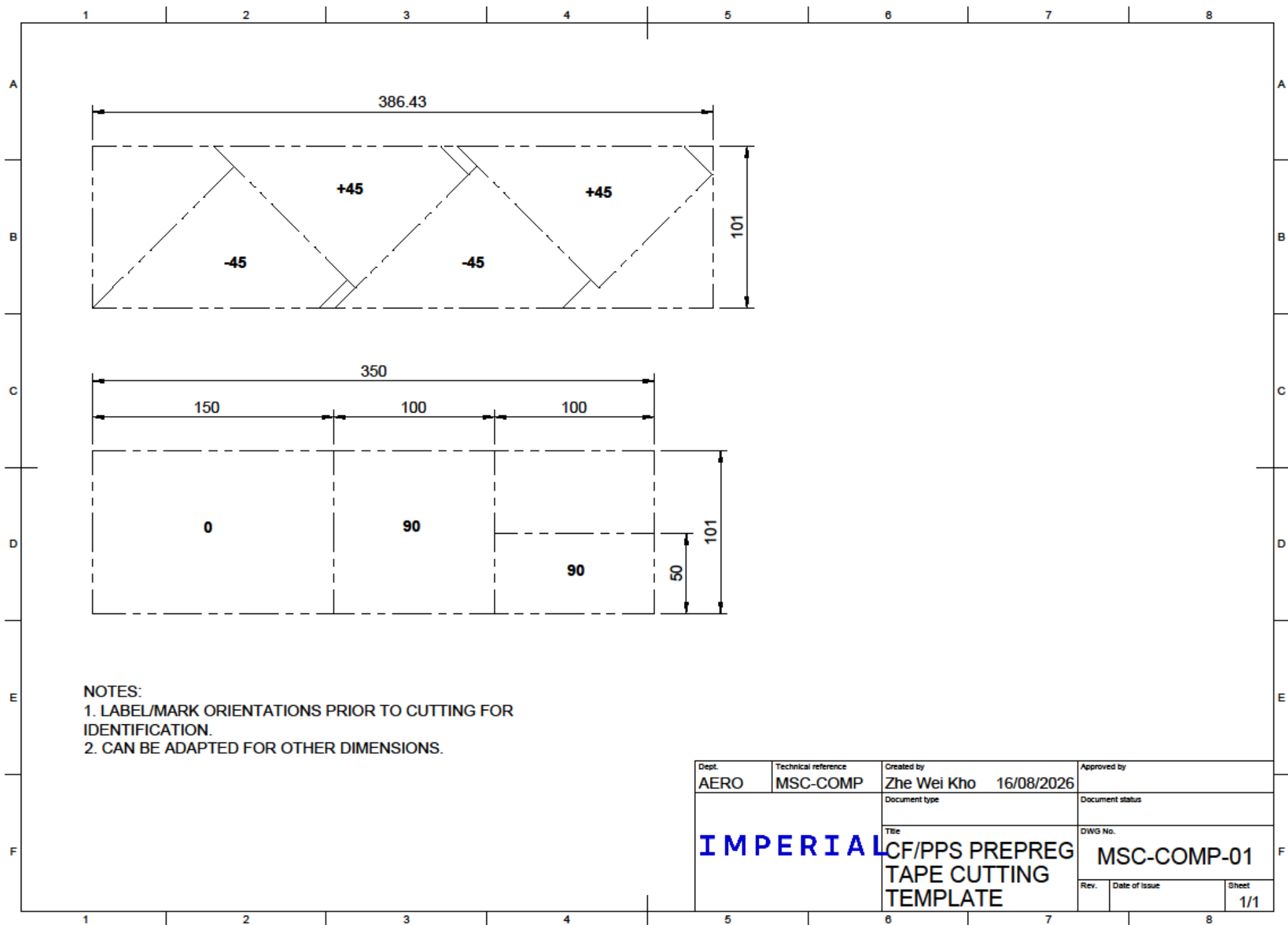
194     c_SER = current_data{3};
195
196     f = figure('Name', sprintf("EMI - %s", plot_titles{i}),...
197             'NumberTitle', 'off', 'Color', 'w',...
198             'Units', 'normalized');
199
200     colororder(newcolors)
201     set(f, 'defaultLineLineWidth', 2)
202
203     plot(freq, c_SET, 'DisplayName', 'SE_T - Total')
204     hold on
205     plot(freq, c_SEA, 'DisplayName', 'SE_A - Absorption')
206     plot(freq, c_SER, 'DisplayName', 'SE_R - Reflection')
207     hold off
208
209     ylabel('SE [dB]')
210     ylim([-5,120])
211
212     if use_averaging
213         title(sprintf('%s\n(Averaged across 3 runs of %s)',
214             plot_titles{i},filename1), 'Interpreter', 'none');
215     else
216         title(sprintf('%s\n%s', plot_titles{i}, filename1), 'Interpreter', 'none');
217     end
218
219     xlabel('Frequency [GHz]');
220     xlim([2.2, 3.0])
221     grid on
222     grid minor
223     legend()
224 end
225 % =====
226 % 7. PLOTTING S-PARAMETERS (LIN MAGNITUDE)
227 % =====
228 if enable_sparam_plots
229
230     % Group the S-parameter datasets for the loop
231     sp_datasets = {{S11, S21}, ...
232                 {S11_med, S21_med}, ...
233                 {S11_final, S21_final}};
234
235     sp_plot_titles = {'S-Params 1: Raw (After Averaging)', ...
236                     'S-Params 2: After Median Filter', ...
237                     'S-Params 3: After S-Golay Filter (Final)'};
238
239     for i = 1:3
240         current_sp = sp_datasets{i};
241         c_S11 = abs(current_sp{1}); % Convert to Linear Magnitude
242         c_S21 = abs(current_sp{2}); % Convert to Linear Magnitude
243
244         f_sp = figure('Name', sprintf("S-Params - %s", sp_plot_titles{i}),...
245                 'NumberTitle', 'off', 'Color', 'w',...
246                 'Units', 'normalized');
247
248         % Use tiledlayout for stacked subplots
249         t = tiledlayout(2, 1, 'TileSpacing', 'compact', 'Padding', 'compact');
250
251         % --- Top Tile: S11 ---
252         nexttile
253         set(f_sp, 'defaultLineLineWidth', 2)
254         plot(freq, c_S11, 'Color', newcolors(1,:), 'DisplayName', '|S_{11}|')
255         ylabel('Lin Magnitude')
256         xlim([2.2, 3.0])
257         title('S_{11} Magnitude')

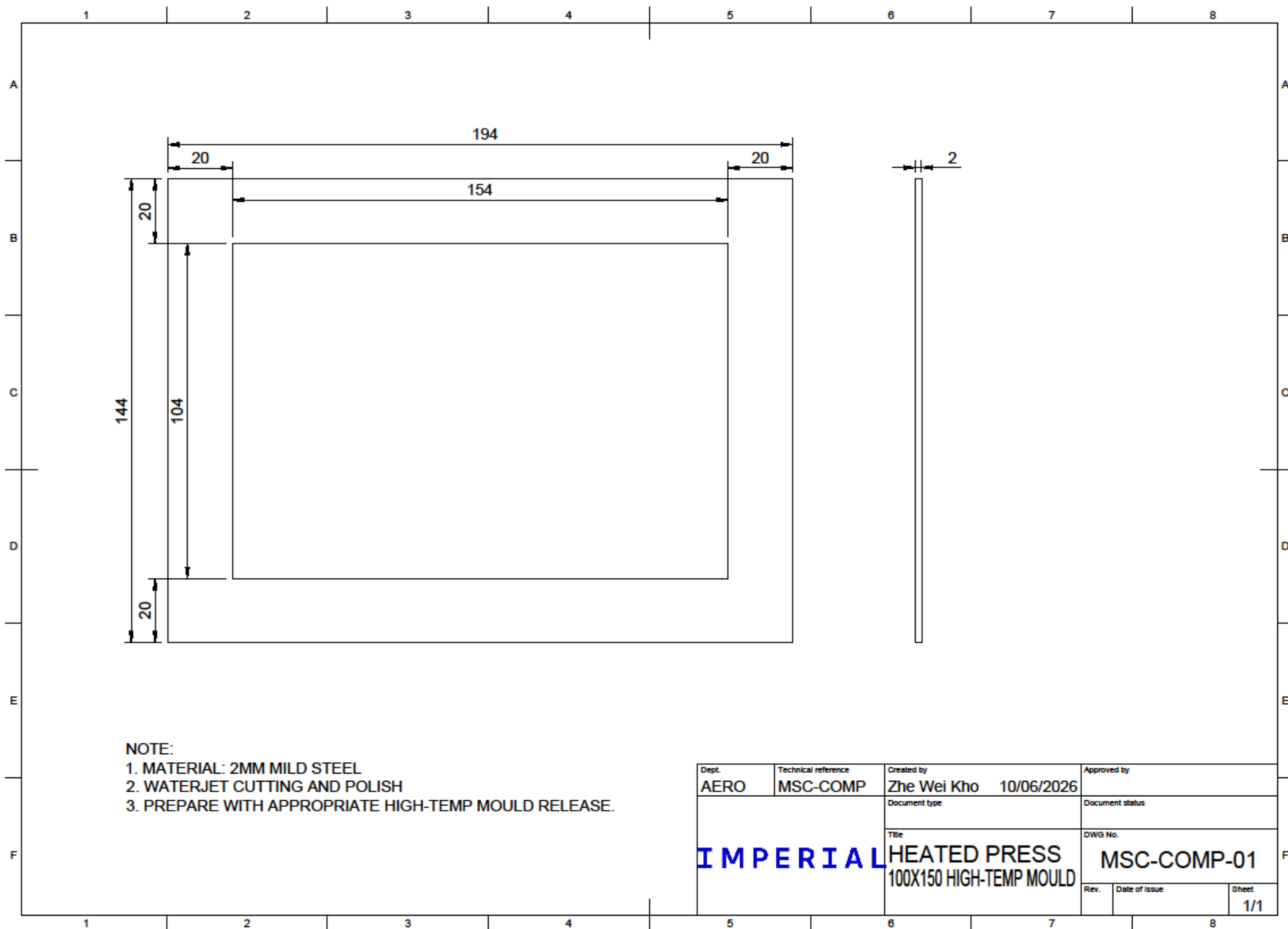
```

```

258         grid on
259         grid minor
260         legend()
261
262         % --- Bottom Tile: S21 ---
263         nexttile
264         set(f_sp, 'defaultLineLineWidth', 2)
265         plot(freq, c_S21, 'Color', newcolors(2,:), 'DisplayName', '|S_{21}|')
266         ylabel('Lin Magnitude')
267         xlabel('Frequency [GHz]')
268         xlim([2.2, 3.0])
269         title('S_{21} Magnitude')
270         grid on
271         grid minor
272         legend()
273
274         % Overall Figure Title
275         if use_averaging
276             title(t, sprintf('%s\n(Averaged across 3 runs of %s)', sp_plot_titles{i},
filename1), 'Interpreter', 'none');
277         else
278             title(t, sprintf('%s\nof %s', sp_plot_titles{i}, filename1), 'Interpreter',
'none');
279         end
280     end
281 end
282
283 fprintf("Number of S11 points outside bounds: %.8f\n\n",pcount)

```





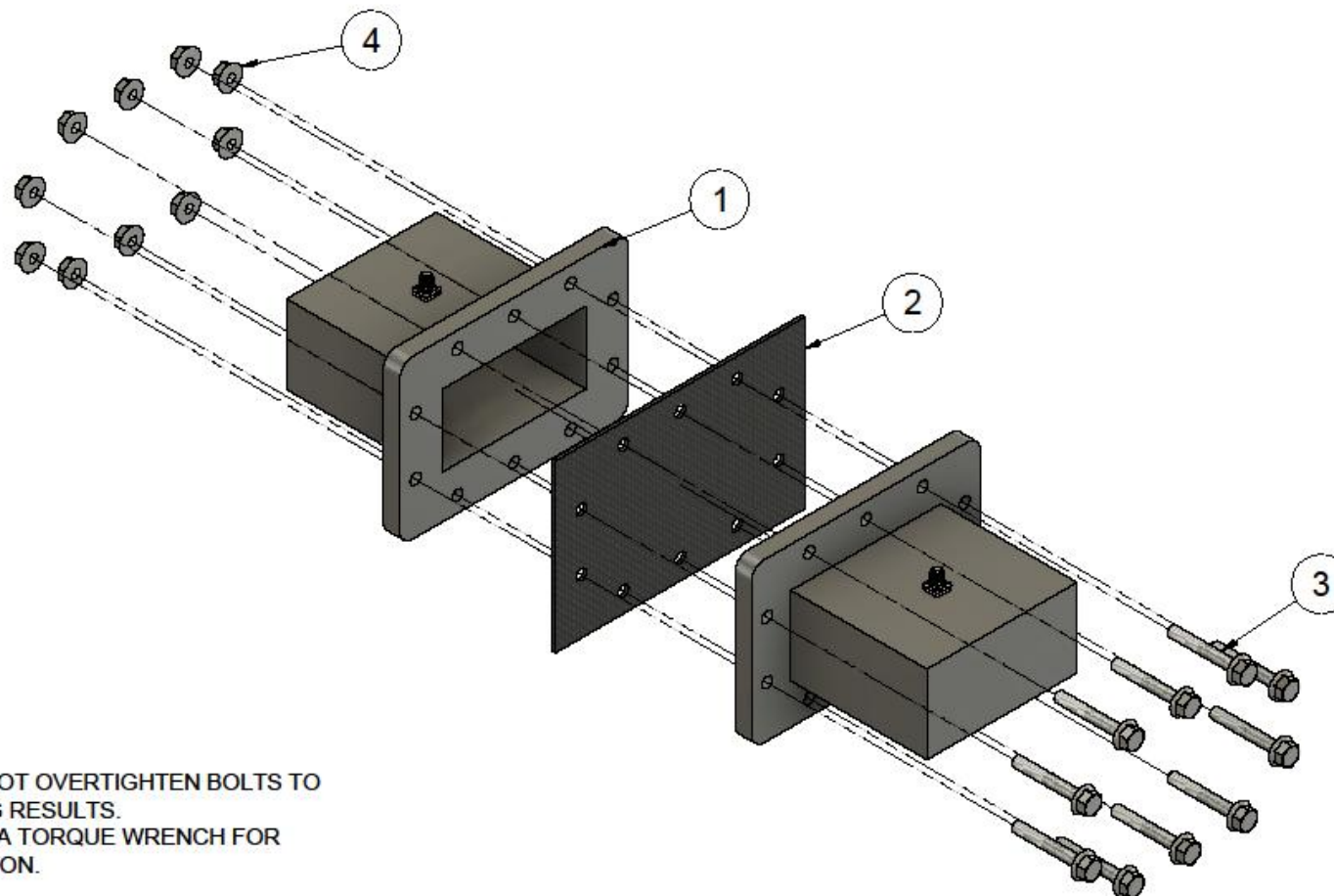
NOTE:

1. MATERIAL: 2MM MILD STEEL
2. WATERJET CUTTING AND POLISH
3. PREPARE WITH APPROPRIATE HIGH-TEMP MOULD RELEASE.

Dept. AERO	Technical reference MSC-COMP	Created by Zhe Wei Kho 10/06/2026	Approved by
IMPERIAL		Document type	Document status
		Title HEATED PRESS 100X150 HIGH-TEMP MOULD	DWG No. MSC-COMP-01
		Rev.	Date of issue
		Sheet 1/1	

NOTE:

1. ENSURE TO NOT OVERTIGHTEN BOLTS TO AVOID SKEWING RESULTS.
2. IDEALLY USE A TORQUE WRENCH FOR STANDARDISATION.



4	10	M6 HEX NUTS	STEEL
3	10	M6 X 40 HEX BOLTS	STEEL
2	1	EM SHIELDING SPECIMEN	COMPOSITE
1	2	A-INFO W340CAS-FDP26 ADAPTER TO SMA FEMALE	ALUMINIUM
ITEM	QTY	PART	MATERIAL
PARTS LIST			

Dept. AERO	Technical reference MSC-COMP	Created by Zhe Wei Kho 19/08/2026	Approved by
IMPERIAL		Document type	Document status
		Title EM SHIELDING ASSEMBLY	DWG No. MSC-COMP-01
		Rev.	Date of issue
		Sheet 1/2	

